HCIA-Security Comprehensive Exam Question Bank

V3.0 Certification Preparation

Total Questions: 1070

Question Types: Single Choice, Multiple Choice, True/False

Coverage: All HCIA-Security V3.0 Exam Topics

Generated for HCIA-Security exam preparation

Contents

PART I

Single-Answer Questions

1 Network Security Concepts

Question 1 **Single Choice | Topic: Network Security Concepts**

In a broad sense, network security refers to:

A. Physical security of devices only
B. Cybersecurity / cyberspace security at a national level
C. Only firewall deployment
D. Only antivirus protection

Correct Answer: B

Concept: Broadly, network security is synonymous with cybersecurity, covering national laws, regulations, and processes to protect cyberspace.

Question 2 **Single Choice | Topic: Network Security Concepts**

In a narrow sense, network security focuses on:

A. National cyberspace policies only
B. Vendor devices and solutions that ensure secure running of enterprise networks
C. Only social engineering defense
D. Only data-center cooling

Correct Answer: B

Concept: In a narrow sense, network security refers to measures such as firewalls, IPS, VPNs, and policies that protect enterprise networks.

Question 3 **Single Choice | Topic: Network Security Concepts**

Which period of security history mainly used stream ciphers and physical protection?

A. Information security period
B. Communication security period
C. Cyberspace security period
D. Information assurance period

Correct Answer: B

Concept: The communication security period (1940s) relied on physical security and cryptographic communication, mainly stream ciphers.

Question 4 **Single Choice | Topic: Network Security Concepts**

During the information security period, which three goals were the main focus?

A. Confidentiality, integrity, availability
B. Speed, bandwidth, latency
C. Power, cooling, cabling
D. Routing, switching, NAT

Correct Answer: A

Concept: The classic CIA triad — confidentiality, integrity, and availability — became the core objectives of information security.

Question 5 **Single Choice | Topic: Network Security Concepts**

Which two properties were added to the CIA triad during the information assurance period?

A. Controllability and non-repudiation
B. Speed and redundancy
C. Encryption and hashing
D. NAT and VLAN

Correct Answer: A

Concept: In the information assurance period, controllability (monitoring and control) and non-repudiation (proof of actions) joined the CIA triad.

Question 6 **Single Choice | Topic: Network Security Concepts**

The cyberspace security period aims to protect:

A. Only web servers
B. Only user passwords
C. Facilities, data, users, and operations across cyberspace
D. Only email gateways

Correct Answer: C

Concept: Cyberspace security expands protection to the entire digital ecosystem: facilities, data, users, and operations.

Question 7 **Single Choice | Topic: Network Security Concepts**

Which term describes the practice of protecting networks and data from unauthorized access?

A. Cybersecurity
B. Marketing
C. Accounting
D. Logistics

Correct Answer: A

Concept: Cybersecurity protects networks, devices, and data from unauthorized access or attack.

Question 8 **Single Choice | Topic: Network Security Concepts**

Which of the following best defines a vulnerability?

A. A weakness that can be exploited
B. A security patch
C. A firewall rule
D. A network diagram

Correct Answer: A

Concept: A vulnerability is a weakness in a system that a threat actor can exploit.

Question 9 **Single Choice | Topic: Network Security Concepts**

What is an exploit?

A. A software patch
B. A method to take advantage of a vulnerability
C. A type of firewall
D. A routing protocol

Correct Answer: B

Concept: An exploit is a technique or code that leverages a vulnerability.

Question 10 **Single Choice | Topic: Network Security Concepts**

Which term describes a potential danger to an asset?

A. Risk
B. Threat
C. Vulnerability
D. Control

Correct Answer: B

Concept: A threat is any potential danger that could exploit a vulnerability.

Question 11 **Single Choice | Topic: Network Security Concepts**

Which term describes the likelihood and impact of a threat exploiting a vulnerability?

A. Threat
B. Risk
C. Asset
D. Control

Correct Answer: B

Concept: Risk measures the probability and impact of a security incident.

Question 12 | **Single Choice | Topic: Network Security Concepts**

Future security architecture is moving toward:

A. Single perimeter defense
B. Zero-trust and dynamic trust models
C. Open anonymous access
D. No logging

Correct Answer: B

Concept: Zero trust assumes no implicit trust and verifies every access request dynamically.

Question 13 | **Single Choice | Topic: Network Security Concepts**

Which technology is increasingly used to detect unknown threats?

A. Signature-only antivirus
B. Artificial intelligence and machine learning
C. Hub-based networks
D. Plaintext protocols

Correct Answer: B

Concept: AI/ML helps analyze behavior and detect anomalies and unknown threats.

Question 14 | **Single Choice | Topic: Network Security Concepts**

What does confidentiality ensure?

A. Only speed
B. Information is accessible only to authorized users
C. Unlimited access
D. Data duplication

Correct Answer: B

Concept: Confidentiality ensures that information is disclosed only to authorized individuals.

Question 15 | **Single Choice | Topic: Network Security Concepts**

What does integrity ensure?

A. Fast transmission
B. Information is not tampered with during transmission/storage
C. Free access
D. Unencrypted storage

Correct Answer: B

Concept: Integrity ensures data is not altered or destroyed in an unauthorized manner.

Question 16 **Single Choice | Topic: Network Security Concepts**

What does availability ensure?

A. Authorized users can access information when needed
B. All users can access
C. Systems are offline
D. Data is hidden

Correct Answer: A

Concept: Availability guarantees reliable and timely access to information for authorized users.

Question 17 **Single Choice | Topic: Network Security Concepts**

What does controllability mean in information security?

A. No monitoring
B. Ability to monitor and control information/systems
C. Unlimited downloads
D. No access controls

Correct Answer: B

Concept: Controllability enables organizations to monitor, manage, and protect information and systems.

Question 18 **Single Choice | Topic: Network Security Concepts**

What does non-repudiation prevent?

A. Fast data transfer
B. Senders or receivers from denying their actions
C. Encryption
D. Backups

Correct Answer: B

Concept: Non-repudiation provides proof of origin and delivery so parties cannot deny participation.

Question 19 **Single Choice | Topic: Network Security Concepts**

The broad definition of network security is closest to:

A. Cybersecurity
B. Firewall configuration
C. Antivirus deployment
D. Physical security

Correct Answer: A

Concept: Broadly, network security equates to cybersecurity covering national and societal cyberspace protection.

Question 20 **Single Choice | Topic: Network Security Concepts**

The narrow definition of network security focuses on:

A. National laws only
B. Enterprise network protection solutions
C. Social media policies
D. Hardware warranties

Correct Answer: B

Concept: Narrowly, network security means vendor solutions and measures that keep enterprise networks secure.

Question 21 **Single Choice | Topic: Network Security Concepts**

During which period did communication cipher security emerge?

A. Cyberspace security period
B. Communication security period
C. Information assurance period
D. Cloud security period

Correct Answer: B

Concept: Stream ciphers and communication confidentiality appeared during the communication security period.

Question 22 **Single Choice | Topic: Network Security Concepts**

The information security period added which goal to the earlier communication security focus?

A. Availability
B. Confidentiality, integrity, and availability
C. Non-repudiation only
D. Controllability only

Correct Answer: B

Concept: The information security period formalized confidentiality, integrity, and availability as core goals.

Question 23 **Single Choice | Topic: Network Security Concepts**

Controllability and non-repudiation became prominent during:

A. Communication security period
B. Information security period
C. Information assurance period
D. Cloud computing period

Correct Answer: C

Concept: The information assurance period expanded security to include monitoring and proof of actions.

Question 24 **Single Choice | Topic: Network Security Concepts**

Which concept is described as: is synonymous with national cyberspace security?

A. Availability
B. Non-repudiation
C. Broad network security
D. AI and machine learning

Correct Answer: C

Concept: Broad network security refers to the concept where is synonymous with national cyberspace security.

Question 25 **Single Choice | Topic: Network Security Concepts**

Which concept is described as: focuses on enterprise network protection solutions?

A. Future security trend
B. Narrow network security
C. AI and machine learning
D. Controllability

Correct Answer: B

Concept: Narrow network security refers to the concept where focuses on enterprise network protection solutions.

Question 26 **Single Choice | Topic: Network Security Concepts**

Which concept is described as: consists of confidentiality, integrity, and availability?

A. CIA triad
B. Cyberspace security
C. Exploit
D. AI and machine learning

Correct Answer: A

Concept: CIA triad refers to the concept where consists of confidentiality, integrity, and availability.

Question 27 **Single Choice | Topic: Network Security Concepts**

Which concept is described as: added controllability and non-repudiation?

A. Information assurance period
B. Threat
C. Confidentiality
D. Broad network security

Correct Answer: A

Concept: Information assurance period refers to the concept where added controllability and non-repudiation.

Question 28 **Single Choice | Topic: Network Security Concepts**

Which concept is described as: protects facilities, data, users, and operations?

A. Exploit
B. Broad network security
C. Cyberspace security
D. AI and machine learning

Correct Answer: C

Concept: Cyberspace security refers to the concept where protects facilities, data, users, and operations.

Question 29 **Single Choice | Topic: Network Security Concepts**

Which concept is described as: ensures information is accessible only to authorized users?

A. Broad network security
B. AI and machine learning
C. Confidentiality
D. Risk

Correct Answer: C

Concept: Confidentiality refers to the concept where ensures information is accessible only to authorized users.

Question 30 **Single Choice | Topic: Network Security Concepts**

Which concept is described as: ensures data is not tampered with during transmission or storage?

A. Non-repudiation
B. Threat
C. Integrity
D. CIA triad

Correct Answer: C

Concept: Integrity refers to the concept where ensures data is not tampered with during transmission or storage.

Question 31 **Single Choice | Topic: Network Security Concepts**

Which concept is described as: ensures authorized users can access information when needed?

A. Availability
B. Threat
C. Risk
D. Vulnerability

Correct Answer: A

Concept: Availability refers to the concept where ensures authorized users can access information when needed.

Question 32 **Single Choice | Topic: Network Security Concepts**

Which concept is described as: enables monitoring and control of information and systems?

A. Controllability
B. Exploit
C. AI and machine learning
D. Confidentiality

Correct Answer: A

Concept: Controllability refers to the concept where enables monitoring and control of information and systems.

Question 33 **Single Choice | Topic: Network Security Concepts**

Which concept is described as: prevents senders or receivers from denying actions?

A. Non-repudiation
B. Broad network security
C. Vulnerability
D. Narrow network security

Correct Answer: A

Concept: Non-repudiation refers to the concept where prevents senders or receivers from denying actions.

Question 34 **Single Choice | Topic: Network Security Concepts**

Which concept is described as: is the combination of threat likelihood and impact?

A. Availability
B. Risk
C. Narrow network security
D. Broad network security

Correct Answer: B

Concept: Risk refers to the concept where is the combination of threat likelihood and impact.

Question 35 **Single Choice | Topic: Network Security Concepts**

Which concept is described as: is a weakness that can be exploited by threats?

A. Information assurance period
B. Threat
C. Vulnerability
D. Narrow network security

Correct Answer: C

Concept: Vulnerability refers to the concept where is a weakness that can be exploited by threats.

Question 36 **Single Choice | Topic: Network Security Concepts**

Which concept is described as: is a potential danger to an asset?

A. Confidentiality
B. Threat
C. CIA triad
D. Availability

Correct Answer: B

Concept: Threat refers to the concept where is a potential danger to an asset.

Question 37 **Single Choice | Topic: Network Security Concepts**

Which concept is described as: is a method that takes advantage of a vulnerability?

A. Confidentiality
B. Exploit
C. Availability
D. Threat

Correct Answer: B

Concept: Exploit refers to the concept where is a method that takes advantage of a vulnerability.

Question 38 **Single Choice | Topic: Network Security Concepts**

Which concept is described as: includes zero-trust architecture?

A. Non-repudiation
B. Future security trend
C. Confidentiality
D. Exploit

Correct Answer: B

Concept: Future security trend refers to the concept where includes zero-trust architecture.

Question 39 **Single Choice | Topic: Network Security Concepts**

Which concept is described as: help detect unknown threats and anomalies?

A. Information assurance period
B. Broad network security
C. AI and machine learning
D. Vulnerability

Correct Answer: C

Concept: AI and machine learning refers to the concept where help detect unknown threats and anomalies.

2 Network Security Standards

Question 1 Single Choice | Topic: Network Security Standards

ISO 27001 specifies requirements for:

A. An information security management system (ISMS)
B. A firewall hardware model
C. A routing protocol
D. A programming language

Correct Answer: A

Concept: ISO/IEC 27001 is the international standard for establishing, implementing, maintaining, and continually improving an ISMS.

Question 2 Single Choice | Topic: Network Security Standards

ISO 27002 is best described as:

A. A certification standard
B. A code of practice with security controls
C. A firewall configuration guide
D. A wireless protocol

Correct Answer: B

Concept: ISO/IEC 27002 provides best-practice security controls, while ISO 27001 is the certifiable requirements standard.

Question 3 Single Choice | Topic: Network Security Standards

Cybersecurity Classified Protection 2.0 is issued by:

A. Huawei
B. China (as a national standard/policy)
C. IEEE
D. IETF

Correct Answer: B

Concept: Classified Protection 2.0 is a Chinese national cybersecurity baseline and compliance framework.

Question 4 Single Choice | Topic: Network Security Standards

Classified Protection 2.0 uses a protection-level scale from:

A. Level 1 to Level 5
B. Level 0 to Level 4
C. Level A to Level E
D. Level I to Level V

Correct Answer: A

Concept: The Chinese classified protection system defines five levels, with Level 5 being the highest.

Question 5 **Single Choice | Topic: Network Security Standards**

Which of the following is NOT a typical Classified Protection 2.0 lifecycle phase?

A. Classification
B. Filing
C. Construction and rectification
D. Marketing and sales

Correct Answer: D

Concept: The Classified Protection lifecycle includes classification, filing, construction/rectification, level assessment, and supervision/inspection.

Question 6 **Single Choice | Topic: Network Security Standards**

Which ISO standard family addresses information security management?

A. ISO 9000
B. ISO 27000
C. ISO 14000
D. ISO 20000

Correct Answer: B

Concept: The ISO/IEC 27000 family covers information security management systems and controls.

Question 7 **Single Choice | Topic: Network Security Standards**

Which Chinese regulation is the basis for Classified Protection?

A. Cybersecurity Law
B. GDPR
C. HIPAA
D. PCI DSS

Correct Answer: A

Concept: China Cybersecurity Law underpins the Classified Protection compliance framework.

Question 8 **Single Choice | Topic: Network Security Standards**

ISO 27001 uses which methodology for risk management?

A. Ignore risks
B. Plan-Do-Check-Act (PDCA)
C. Random patching
D. No reviews

Correct Answer: B

Concept: ISO 27001 follows the PDCA cycle for continuous improvement of the ISMS.

Question 9 Single Choice | Topic: Network Security Standards

Which document in ISO 27000 family provides implementation guidance for controls?

A. ISO 27001
B. ISO 27002
C. ISO 27003
D. ISO 27005

Correct Answer: B

Concept: ISO 27002 lists security controls and implementation guidance.

Question 10 Single Choice | Topic: Network Security Standards

Which is a key principle of Classified Protection 2.0?

A. Security synchronization with construction
B. Security as an afterthought
C. No auditing
D. Anonymous access

Correct Answer: A

Concept: Classified Protection requires security to be planned, constructed, and used simultaneously with information systems.

Question 11 Single Choice | Topic: Network Security Standards

In Classified Protection 2.0, the security protection level is preliminarily determined by the network operator, but when the level reaches which level or above, expert review and approval are required?

A. Level 1
B. Level 2
C. Level 3
D. Level 4

Correct Answer: B

Concept: For protection Level 2 and above, the preliminary determination must be reviewed and approved; Level 1 can be determined by the operator.

Question 12 Single Choice | Topic: Network Security Standards

Which standard is used to certify an ISMS?

A. ISO 27001
B. ISO 27002
C. ISO 9001
D. ISO 14001

Correct Answer: A

Concept: ISO 27001 contains the certifiable requirements for an ISMS.

Question 13 Single Choice | Topic: Network Security Standards

ISO 27002 is primarily a:

A. Certification standard
B. Code of practice for security controls
C. Risk assessment methodology
D. Networking protocol

Correct Answer: B

Concept: ISO 27002 provides best-practice controls but is not itself a certification standard.

Question 14 Single Choice | Topic: Network Security Standards

Classified Protection 2.0 compliance is primarily relevant in:

A. China
B. United States only
C. European Union only
D. Every country equally

Correct Answer: A

Concept: Classified Protection 2.0 is a Chinese national cybersecurity compliance framework.

Question 15 Single Choice | Topic: Network Security Standards

Which is the highest protection level in Classified Protection 2.0?

A. Level 1
B. Level 3
C. Level 5
D. Level 10

Correct Answer: C

Concept: Classified Protection defines five levels, with Level 5 being the most stringent.

Question 16 Single Choice | Topic: Network Security Standards

Expert review is required for Classified Protection levels starting at:

A. Level 1
B. Level 2
C. Level 3
D. Level 4

Correct Answer: B

Concept: For Level 2 and above, expert review and approval are required.

Question 17 **Single Choice | Topic: Network Security Standards**

Which concept is described as: specifies certifiable ISMS requirements?

A. Cybersecurity Law
B. ISO 27001
C. ISO 27001:2022
D. Classified Protection lifecycle

Correct Answer: B

Concept: ISO 27001 refers to the concept where specifies certifiable ISMS requirements.

Question 18 **Single Choice | Topic: Network Security Standards**

Which concept is described as: provides a code of practice for security controls?

A. ISO 27001:2022
B. Classified Protection lifecycle
C. Level 2 and above
D. ISO 27002

Correct Answer: D

Concept: ISO 27002 refers to the concept where provides a code of practice for security controls.

Question 19 **Single Choice | Topic: Network Security Standards**

Which concept is described as: is the continuous improvement methodology used in ISO 27001?

A. Classified Protection 2.0
B. PDCA
C. Security synchronization
D. Level 2 and above

Correct Answer: B

Concept: PDCA refers to the concept where is the continuous improvement methodology used in ISO 27001.

Question 20 **Single Choice | Topic: Network Security Standards**

Which concept is described as: is a Chinese national cybersecurity framework?

A. ISO 27002
B. Security synchronization
C. ISO 27001
D. Classified Protection 2.0

Correct Answer: D

Concept: Classified Protection 2.0 refers to the concept where is a Chinese national cybersecurity framework.

Question 21 **Single Choice | Topic: Network Security Standards**

Which concept is described as: range from Level 1 to Level 5?

A. ISO 27002
B. Classified Protection levels
C. Classified Protection 2.0
D. ISO 27001

Correct Answer: B

Concept: Classified Protection levels refers to the concept where range from Level 1 to Level 5.

Question 22 **Single Choice | Topic: Network Security Standards**

Which concept is described as: require expert review and approval?

A. Security synchronization
B. ISO 27001
C. Level 2 and above
D. PDCA

Correct Answer: C

Concept: Level 2 and above refers to the concept where require expert review and approval.

Question 23 **Single Choice | Topic: Network Security Standards**

Which concept is described as: includes classification, filing, construction, assessment, and supervision?

A. ISO 27002
B. Level 2 and above
C. Classified Protection 2.0
D. Classified Protection lifecycle

Correct Answer: D

Concept: Classified Protection lifecycle refers to the concept where includes classification, filing, construction, assessment, and supervision.

Question 24 **Single Choice | Topic: Network Security Standards**

Which concept is described as: means planning security with information system construction?

A. Security synchronization
B. Level 2 and above
C. PDCA
D. Classified Protection levels

Correct Answer: A

Concept: Security synchronization refers to the concept where means planning security with information system construction.

Question 25 **Single Choice | Topic: Network Security Standards**

Which concept is described as: organizes controls into four themes?

A. Classified Protection lifecycle
B. ISO 27001:2022
C. ISO 27002
D. Level 2 and above

Correct Answer: B

Concept: ISO 27001:2022 refers to the concept where organizes controls into four themes.

Question 26 **Single Choice | Topic: Network Security Standards**

Which concept is described as: underpins Classified Protection in China?

A. Level 2 and above
B. PDCA
C. Cybersecurity Law
D. Classified Protection levels

Correct Answer: C

Concept: Cybersecurity Law refers to the concept where underpins Classified Protection in China.

3 Network Fundamentals

Question 1 **Single Choice | Topic: Network Fundamentals**

What is the default subnet mask for a Class C IPv4 address?

A. 255.0.0.0
B. 255.255.0.0
C. 255.255.255.0
D. 255.255.255.255

Correct Answer: C

Concept: Class C networks use a default /24 subnet mask: 255.255.255.0.

Question 2 **Single Choice | Topic: Network Fundamentals**

Which protocol is used to automatically assign IPv6 addresses using MAC addresses?

A. DHCPv6
B. SLAAC
C. ARP
D. DNS

Correct Answer: B

Concept: SLAAC can generate IPv6 addresses from MAC addresses.

Question 3 **Single Choice | Topic: Network Fundamentals**

Which layer-2 protocol prevents loops in redundant Ethernet topologies?

A. VTP
B. STP
C. RSTP
D. Both STP and RSTP

Correct Answer: D

Concept: Spanning Tree Protocol and Rapid STP prevent Layer 2 loops.

Question 4 **Single Choice | Topic: Network Fundamentals**

What does VLAN stand for?

A. Virtual Local Area Network
B. Virtual Large Area Network
C. Verified Local Access Node
D. Virtual Link Aggregation Network

Correct Answer: A

Concept: VLANs logically segment a switch into separate broadcast domains.

Question 5 **Single Choice | Topic: Network Fundamentals**

Which device interconnects VLANs at Layer 3?

A. Layer 2 switch
B. Router or Layer 3 switch
C. Hub
D. Bridge

Correct Answer: B

Concept: Routers or Layer 3 switches perform inter-VLAN routing.

Question 6 **Single Choice | Topic: Network Fundamentals**

Which OSI layer is responsible for reliable data transfer between hosts?

A. Network
B. Transport
C. Data link
D. Physical

Correct Answer: B

Concept: The transport layer manages end-to-end reliability, flow control, and error recovery.

Question 7 — Single Choice | Topic: Network Fundamentals

In the TCP/IP model, which layer corresponds to the OSI network layer?

A. Internet layer
B. Network interface layer
C. Transport layer
D. Application layer

Correct Answer: A

Concept: The TCP/IP internet layer maps to the OSI network layer and handles IP routing.

Question 8 — Single Choice | Topic: Network Fundamentals

Which layer handles encryption and data format conversion?

A. Application
B. Presentation
C. Session
D. Transport

Correct Answer: B

Concept: The presentation layer handles syntax, encryption, compression, and character encoding.

Question 9 — Single Choice | Topic: Network Fundamentals

Which layer establishes, manages, and terminates application dialogs?

A. Application
B. Presentation
C. Session
D. Transport

Correct Answer: C

Concept: The session layer controls dialog establishment, maintenance, and termination.

Question 10 — Single Choice | Topic: Network Fundamentals

Which protocol is used to send email between servers?

A. HTTP
B. SMTP
C. POP3
D. IMAP

Correct Answer: B

Concept: SMTP is used to send and relay email.

Question 11 | **Single Choice | Topic: Network Fundamentals**

Which protocol is used by clients to retrieve email from a server while keeping messages on the server?

A. POP3
B. IMAP
C. SMTP
D. SNMP

Correct Answer: B

Concept: IMAP allows clients to access and manage email while messages remain on the server.

Question 12 | **Single Choice | Topic: Network Fundamentals**

Which protocol dynamically assigns IP addresses to hosts?

A. DNS
B. DHCP
C. ARP
D. FTP

Correct Answer: B

Concept: DHCP dynamically assigns IP addresses and network configuration parameters.

Question 13 | **Single Choice | Topic: Network Fundamentals**

Which protocol maps IP addresses to MAC addresses on a local network?

A. DNS
B. ARP
C. DHCP
D. ICMP

Correct Answer: B

Concept: ARP resolves IP addresses to MAC addresses.

Question 14 | **Single Choice | Topic: Network Fundamentals**

Which protocol is used for network management and monitoring?

A. SNMP
B. SMTP
C. FTP
D. Telnet

Correct Answer: A

Concept: SNMP manages and monitors network devices.

Question 15 — Single Choice | Topic: Network Fundamentals

What does ICMP primarily provide?

A. File transfer
B. Error reporting and diagnostics
C. Email delivery
D. Web browsing

Correct Answer: B

Concept: ICMP reports errors and provides diagnostic functions such as ping and traceroute.

Question 16 — Single Choice | Topic: Network Fundamentals

Which port does HTTP use by default?

A. 20
B. 21
C. 80
D. 443

Correct Answer: C

Concept: HTTP uses TCP port 80 by default.

Question 17 — Single Choice | Topic: Network Fundamentals

Which port does HTTPS use by default?

A. 80
B. 443
C. 22
D. 25

Correct Answer: B

Concept: HTTPS uses TCP port 443 by default, providing encrypted web traffic.

Question 18 — Single Choice | Topic: Network Fundamentals

Which port does SSH use by default?

A. 21
B. 22
C. 23
D. 25

Correct Answer: B

Concept: SSH uses TCP port 22 for secure remote access.

Question 19 Single Choice | Topic: Network Fundamentals

Telnet uses which port and is considered insecure because it transmits data in plaintext?

A. 21
B. 22
C. 23
D. 25

Correct Answer: C

Concept: Telnet uses TCP port 23 and sends data, including credentials, in plaintext.

Question 20 Single Choice | Topic: Network Fundamentals

Which address is a Layer 2 address?

A. IP address
B. MAC address
C. Port number
D. Domain name

Correct Answer: B

Concept: A MAC address is a hardware address used at the data link layer.

Question 21 Single Choice | Topic: Network Fundamentals

A router forwards packets based on:

A. MAC address
B. IP address
C. Port number
D. VLAN ID

Correct Answer: B

Concept: Routers use IP addresses to make forwarding decisions.

Question 22 Single Choice | Topic: Network Fundamentals

What is a subnet mask used for?

A. MAC address
B. IP address class
C. Network and host portions of an IP address
D. DNS server

Correct Answer: C

Concept: A subnet mask divides an IP address into network and host portions.

Question 23 **Single Choice | Topic: Network Fundamentals**

The OSI model layer closest to the physical medium is:

A. Data link
B. Physical
C. Network
D. Transport

Correct Answer: B

Concept: The physical layer deals with bits, cables, and signals.

Question 24 **Single Choice | Topic: Network Fundamentals**

TCP is considered reliable because it uses:

A. Sequence numbers and acknowledgments
B. Broadcasting
C. No connection setup
D. Fixed ports only

Correct Answer: A

Concept: TCP uses sequence numbers, acknowledgments, and retransmissions for reliability.

Question 25 **Single Choice | Topic: Network Fundamentals**

UDP is preferred for:

A. Reliable file transfer
B. Real-time streaming and low-latency applications
C. Email delivery
D. Web browsing requiring guaranteed delivery

Correct Answer: B

Concept: UDP low overhead and lack of retransmission make it suitable for real-time traffic.

Question 26 **Single Choice | Topic: Network Fundamentals**

DNS primarily uses:

A. TCP port 53
B. UDP port 53
C. TCP port 80
D. UDP port 161

Correct Answer: B

Concept: DNS queries primarily use UDP port 53.

Question 27 **Single Choice | Topic: Network Fundamentals**

A Layer 3 switch combines functions of a:

A. Hub and repeater
B. Switch and router
C. Firewall and IDS
D. Modem and gateway

Correct Answer: B

Concept: Layer 3 switches perform both Layer 2 switching and Layer 3 routing.

Question 28 **Single Choice | Topic: Network Fundamentals**

Which concept is described as: has seven layers?

A. Data link layer
B. SMTP
C. DHCP
D. OSI model

Correct Answer: D

Concept: OSI model refers to the concept where has seven layers.

Question 29 **Single Choice | Topic: Network Fundamentals**

Which concept is described as: has four layers?

A. TCP/IP standard model
B. Hub
C. ARP
D. TCP/IP equivalent model

Correct Answer: A

Concept: TCP/IP standard model refers to the concept where has four layers.

Question 30 **Single Choice | Topic: Network Fundamentals**

Which concept is described as: has five layers?

A. STP
B. MAC address
C. HTTPS
D. TCP/IP equivalent model

Correct Answer: D

Concept: TCP/IP equivalent model refers to the concept where has five layers.

Question 31 **Single Choice | Topic: Network Fundamentals**

Which concept is described as: transmits raw bits over a medium?

A. TCP
B. Switch
C. Router
D. Physical layer

Correct Answer: D

Concept: Physical layer refers to the concept where transmits raw bits over a medium.

Question 32 **Single Choice | Topic: Network Fundamentals**

Which concept is described as: handles framing and MAC addressing?

A. FTP
B. Data link layer
C. Session layer
D. TCP

Correct Answer: B

Concept: Data link layer refers to the concept where handles framing and MAC addressing.

Question 33 **Single Choice | Topic: Network Fundamentals**

Which concept is described as: handles logical addressing and routing?

A. OSI model
B. Network layer
C. TCP/IP standard model
D. HTTPS

Correct Answer: B

Concept: Network layer refers to the concept where handles logical addressing and routing.

Question 34 **Single Choice | Topic: Network Fundamentals**

Which concept is described as: provides end-to-end communication and reliability?

A. Data link layer
B. Transport layer
C. HTTPS
D. Switch

Correct Answer: B

Concept: Transport layer refers to the concept where provides end-to-end communication and reliability.

Question 35 **Single Choice | Topic: Network Fundamentals**

Which concept is described as: manages application dialogs?

A. Session layer
B. IPv6 address
C. Switch
D. ICMP

Correct Answer: A

Concept: Session layer refers to the concept where manages application dialogs.

Question 36 **Single Choice | Topic: Network Fundamentals**

Which concept is described as: handles encryption and data format conversion?

A. Network layer
B. TCP
C. IPv4 address
D. Presentation layer

Correct Answer: D

Concept: Presentation layer refers to the concept where handles encryption and data format conversion.

Question 37 **Single Choice | Topic: Network Fundamentals**

Which concept is described as: provides network services to applications?

A. Physical layer
B. ICMP
C. Application layer
D. DNS

Correct Answer: C

Concept: Application layer refers to the concept where provides network services to applications.

Question 38 **Single Choice | Topic: Network Fundamentals**

Which concept is described as: is connection-oriented and reliable?

A. TCP
B. STP
C. SSH
D. IPv4 address

Correct Answer: A

Concept: TCP refers to the concept where is connection-oriented and reliable.

Question 39 **Single Choice | Topic: Network Fundamentals**

Which concept is described as: is connectionless and has low overhead?

A. DHCP
B. UDP
C. STP
D. IPv6 address

Correct Answer: B

Concept: UDP refers to the concept where is connectionless and has low overhead.

Question 40 **Single Choice | Topic: Network Fundamentals**

Which concept is described as: uses TCP ports 20 and 21?

A. IPv4 address
B. FTP
C. IPv6 address
D. RFC 1918

Correct Answer: B

Concept: FTP refers to the concept where uses TCP ports 20 and 21.

Question 41 **Single Choice | Topic: Network Fundamentals**

Which concept is described as: uses TCP port 80?

A. TCP/IP standard model
B. Telnet
C. TCP/IP equivalent model
D. HTTP

Correct Answer: D

Concept: HTTP refers to the concept where uses TCP port 80.

Question 42 **Single Choice | Topic: Network Fundamentals**

Which concept is described as: uses TCP port 443?

A. HTTPS
B. Hub
C. DNS
D. RFC 1918

Correct Answer: A

Concept: HTTPS refers to the concept where uses TCP port 443.

Question 43 Single Choice | Topic: Network Fundamentals

Which concept is described as: uses TCP port 22?

A. UDP
B. FTP
C. VLAN
D. SSH

Correct Answer: D

Concept: SSH refers to the concept where uses TCP port 22.

Question 44 Single Choice | Topic: Network Fundamentals

Which concept is described as: uses TCP port 23 and is insecure?

A. Transport layer
B. TCP/IP equivalent model
C. Router
D. Telnet

Correct Answer: D

Concept: Telnet refers to the concept where uses TCP port 23 and is insecure.

Question 45 Single Choice | Topic: Network Fundamentals

Which concept is described as: primarily uses UDP port 53?

A. Physical layer
B. DNS
C. STP
D. IPv4 address

Correct Answer: B

Concept: DNS refers to the concept where primarily uses UDP port 53.

Question 46 Single Choice | Topic: Network Fundamentals

Which concept is described as: uses TCP port 25?

A. DHCP
B. ICMP
C. Hub
D. SMTP

Correct Answer: D

Concept: SMTP refers to the concept where uses TCP port 25.

Question 47 **Single Choice | Topic: Network Fundamentals**

Which concept is described as: dynamically assigns IP addresses?

A. DHCP
B. Presentation layer
C. Hub
D. HTTPS

Correct Answer: A

Concept: DHCP refers to the concept where dynamically assigns IP addresses.

Question 48 **Single Choice | Topic: Network Fundamentals**

Which concept is described as: maps IP addresses to MAC addresses?

A. TCP
B. ARP
C. TCP/IP equivalent model
D. DNS

Correct Answer: B

Concept: ARP refers to the concept where maps IP addresses to MAC addresses.

Question 49 **Single Choice | Topic: Network Fundamentals**

Which concept is described as: provides error reporting and diagnostics?

A. STP
B. IPv4 address
C. Switch
D. ICMP

Correct Answer: D

Concept: ICMP refers to the concept where provides error reporting and diagnostics.

Question 50 **Single Choice | Topic: Network Fundamentals**

Which concept is described as: is 32 bits long?

A. TCP/IP equivalent model
B. Hub
C. IPv4 address
D. FTP

Correct Answer: C

Concept: IPv4 address refers to the concept where is 32 bits long.

Question 51 — Single Choice | Topic: Network Fundamentals

Which concept is described as: is 128 bits long?

- **A.** Session layer
- **B.** IPv6 address
- **C.** HTTPS
- **D.** TCP/IP standard model

Correct Answer: B

Concept: IPv6 address refers to the concept where is 128 bits long.

Question 52 — Single Choice | Topic: Network Fundamentals

Which concept is described as: is a Layer 2 hardware address?

- **A.** RFC 1918
- **B.** ICMP
- **C.** MAC address
- **D.** SMTP

Correct Answer: C

Concept: MAC address refers to the concept where is a Layer 2 hardware address.

Question 53 — Single Choice | Topic: Network Fundamentals

Which concept is described as: operates at Layer 2 and forwards frames by MAC?

- **A.** Router
- **B.** Switch
- **C.** STP
- **D.** FTP

Correct Answer: B

Concept: Switch refers to the concept where operates at Layer 2 and forwards frames by MAC.

Question 54 — Single Choice | Topic: Network Fundamentals

Which concept is described as: operates at Layer 3 and forwards packets by IP?

- **A.** STP
- **B.** UDP
- **C.** FTP
- **D.** Router

Correct Answer: D

Concept: Router refers to the concept where operates at Layer 3 and forwards packets by IP.

Question 55 **Single Choice | Topic: Network Fundamentals**

Which concept is described as: operates at Layer 1 and repeats signals?

A. HTTPS
B. FTP
C. Presentation layer
D. Hub

Correct Answer: D

Concept: Hub refers to the concept where operates at Layer 1 and repeats signals.

Question 56 **Single Choice | Topic: Network Fundamentals**

Which concept is described as: logically segments a network into broadcast domains?

A. FTP
B. Data link layer
C. VLAN
D. Presentation layer

Correct Answer: C

Concept: VLAN refers to the concept where logically segments a network into broadcast domains.

Question 57 **Single Choice | Topic: Network Fundamentals**

Which concept is described as: prevents Layer 2 loops?

A. TCP/IP equivalent model
B. Physical layer
C. SSH
D. STP

Correct Answer: D

Concept: STP refers to the concept where prevents Layer 2 loops.

Question 58 **Single Choice | Topic: Network Fundamentals**

Which concept is described as: defines private IPv4 address ranges?

A. ARP
B. SMTP
C. RFC 1918
D. ICMP

Correct Answer: C

Concept: RFC 1918 refers to the concept where defines private IPv4 address ranges.

4 OSI and TCP/IP Models

Question 1 — Single Choice | Topic: OSI and TCP/IP Models

How many layers does the OSI reference model have?

A. 4
B. 5
C. 7
D. 9

Correct Answer: C

Concept: The OSI model has seven layers: physical, data link, network, transport, session, presentation, and application.

Question 2 — Single Choice | Topic: OSI and TCP/IP Models

How many layers does the TCP/IP standard model have?

A. 4
B. 5
C. 6
D. 7

Correct Answer: A

Concept: The TCP/IP standard model has four layers: network interface, internet, transport, and application.

Question 3 — Single Choice | Topic: OSI and TCP/IP Models

Which layer is responsible for end-to-end communication and flow control?

A. Network layer
B. Transport layer
C. Data link layer
D. Physical layer

Correct Answer: B

Concept: The transport layer provides end-to-end communication, reliability, and flow control (e.g., TCP).

Question 4 — Single Choice | Topic: OSI and TCP/IP Models

Which layer handles logical addressing and routing?

A. Transport layer
B. Network layer
C. Session layer
D. Presentation layer

Correct Answer: B

Concept: The network layer uses logical addresses (IP) and routes packets between networks.

Question 5 **Single Choice | Topic: OSI and TCP/IP Models**

Which layer is responsible for framing, MAC addressing, and error detection on a local link?

A. Network layer
B. Transport layer
C. Data link layer
D. Application layer

Correct Answer: C

Concept: The data link layer handles framing, MAC addresses, and local link error detection.

5 Network Protocols

Question 1 **Single Choice | Topic: Network Protocols**

Which protocol uses ports 20 and 21 for file transfer?

A. HTTP
B. FTP
C. SMTP
D. DNS

Correct Answer: B

Concept: FTP uses TCP port 21 for control and port 20 for data transfer.

Question 2 **Single Choice | Topic: Network Protocols**

Which protocol resolves domain names to IP addresses?

A. DHCP
B. DNS
C. SNMP
D. NTP

Correct Answer: B

Concept: DNS (Domain Name System) maps human-readable domain names to IP addresses.

Question 3 **Single Choice | Topic: Network Protocols**

Which transport protocol is connection-oriented and reliable?

A. UDP
B. TCP
C. ICMP
D. IP

Correct Answer: B

Concept: TCP provides connection-oriented, reliable, ordered delivery with flow and congestion control.

Question 4 Single Choice | Topic: Network Protocols

Which transport protocol is connectionless and has low overhead?

A. TCP
B. UDP
C. HTTP
D. FTP

Correct Answer: B

Concept: UDP is connectionless, offers low latency and overhead, but does not guarantee delivery.

6 Network Devices

Question 1 Single Choice | Topic: Network Devices

Which device operates at Layer 2 and forwards frames based on MAC addresses?

A. Router
B. Switch
C. Hub
D. Firewall

Correct Answer: B

Concept: A switch operates at the data link layer and forwards frames using MAC address tables.

Question 2 Single Choice | Topic: Network Devices

Which device operates at Layer 3 and forwards packets based on IP addresses?

A. Switch
B. Router
C. Hub
D. Bridge

Correct Answer: B

Concept: A router operates at the network layer and makes forwarding decisions based on IP routing tables.

Question 3 Single Choice | Topic: Network Devices

Which device simply repeats signals to all connected ports?

A. Switch
B. Router
C. Hub
D. Firewall

Correct Answer: C

Concept: A hub is a Layer 1 device that broadcasts signals to all connected devices.

7 Enterprise Security Threats

Question 1 **Single Choice | Topic: Enterprise Security Threats**

Which attack sends an overwhelming amount of traffic from many sources?

A. DoS
B. DDoS
C. Phishing
D. Spoofing

Correct Answer: B

Concept: DDoS uses multiple distributed sources to flood a target.

Question 2 **Single Choice | Topic: Enterprise Security Threats**

Which attack falsifies the source address of packets?

A. Phishing
B. Spoofing
C. Sniffing
D. Spamming

Correct Answer: B

Concept: Spoofing impersonates a trusted source by falsifying addresses or identities.

Question 3 **Single Choice | Topic: Enterprise Security Threats**

Which type of malware replicates itself across networks without user action?

A. Virus
B. Worm
C. Trojan
D. Spyware

Correct Answer: B

Concept: Worms can self-replicate and spread across networks, often without user interaction.

Question 4 **Single Choice | Topic: Enterprise Security Threats**

Which malware disguises itself as legitimate software?

A. Virus
B. Worm
C. Trojan
D. Adware

Correct Answer: C

Concept: A Trojan horse masquerades as legitimate software to trick users into executing it.

Question 5 **Single Choice | Topic: Enterprise Security Threats**

Ransomware primarily does what?

A. Steals credentials
B. Encrypts data and demands payment
C. Deletes logs
D. Monitors traffic

Correct Answer: B

Concept: Ransomware encrypts files and demands a ransom for decryption.

Question 6 **Single Choice | Topic: Enterprise Security Threats**

Which attack intercepts communications between two parties without their knowledge?

A. Man-in-the-middle
B. DoS
C. Phishing
D. Tailgating

Correct Answer: A

Concept: A man-in-the-middle attack intercepts and possibly alters communication between parties.

Question 7 **Single Choice | Topic: Enterprise Security Threats**

Which protocol can help prevent man-in-the-middle attacks by verifying server identity?

A. HTTP
B. HTTPS
C. FTP
D. Telnet

Correct Answer: B

Concept: HTTPS uses TLS/SSL to encrypt traffic and authenticate the server.

Question 8 **Single Choice | Topic: Enterprise Security Threats**

Which wireless security protocol is more secure than WEP?

A. WPA2/WPA3
B. WEP
C. WPA (TKIP only)
D. Open network

Correct Answer: A

Concept: WPA2 and WPA3 provide stronger encryption and authentication than WEP.

Question 9 **Single Choice | Topic: Enterprise Security Threats**

What is a DMZ typically used for?

A. Hosting internal databases only
B. Hosting public-facing servers
C. User workstations
D. Printer sharing

Correct Answer: B

Concept: A DMZ hosts public-facing services between the Internet and internal network.

Question 10 **Single Choice | Topic: Enterprise Security Threats**

Which security zone usually has the highest priority in Huawei firewalls?

A. Untrust
B. Trust
C. DMZ
D. Local

Correct Answer: D

Concept: The Local zone (firewall itself) has priority 100, the highest default priority.

Question 11 **Single Choice | Topic: Enterprise Security Threats**

Which practice limits the damage from compromised accounts?

A. Least privilege
B. Sharing admin passwords
C. Disabling logs
D. Open shares

Correct Answer: A

Concept: Least privilege gives users only the access they need, limiting blast radius.

Question 12 **Single Choice | Topic: Enterprise Security Threats**

What is the purpose of host-based intrusion prevention?

A. Route packets
B. Detect and block malicious host activity
C. Assign IP addresses
D. Synchronize time

Correct Answer: B

Concept: Host-based IPS monitors and blocks suspicious activity on an endpoint.

Question 13 **Single Choice | Topic: Enterprise Security Threats**

Which technology collects and analyzes security logs centrally?

A. SIEM
B. SNMP
C. DHCP
D. ARP

Correct Answer: A

Concept: SIEM centralizes log collection and analysis.

Question 14 **Single Choice | Topic: Enterprise Security Threats**

What is a key benefit of centralized security management?

A. Reduced visibility
B. Unified policy and monitoring
C. More complex passwords
D. Faster malware spread

Correct Answer: B

Concept: Centralized management provides unified visibility, policy control, and monitoring.

Question 15 **Single Choice | Topic: Enterprise Security Threats**

What is an insider threat?

A. An attack from outside the network
B. A security risk originating from within the organization
C. A hardware vendor
D. A natural disaster

Correct Answer: B

Concept: Insider threats come from employees, contractors, or partners with internal access.

Question 16 **Single Choice | Topic: Enterprise Security Threats**

Which attack sends oversized ICMP packets to crash a target?

A. Ping of death
B. Ping sweep
C. ARP spoofing
D. DNS hijacking

Correct Answer: A

Concept: Ping of death sends ICMP packets larger than the maximum allowed size.

Question 17 **Single Choice | Topic: Enterprise Security Threats**

Which attack uses broadcast pings to amplify traffic?

A. Smurf attack
B. SYN flood
C. Fraggle
D. Session hijacking

Correct Answer: A

Concept: Smurf attacks send ICMP echo requests to broadcast addresses with a spoofed source.

Question 18 **Single Choice | Topic: Enterprise Security Threats**

Which attack intercepts ARP messages to redirect traffic?

A. ARP spoofing
B. DNS spoofing
C. IP spoofing
D. MAC flooding

Correct Answer: A

Concept: ARP spoofing sends forged ARP replies to associate an attacker MAC with another IP.

Question 19 **Single Choice | Topic: Enterprise Security Threats**

A SYN flood is a type of:

A. Denial of Service attack
B. Phishing attack
C. Data exfiltration
D. Physical attack

Correct Answer: A

Concept: SYN floods exhaust server connection resources with half-open TCP connections.

Question 20 **Single Choice | Topic: Enterprise Security Threats**

ARP spoofing is used to:

A. Intercept traffic on a local network
B. Encrypt files
C. Speed up routing
D. Authenticate users

Correct Answer: A

Concept: Attackers use forged ARP messages to redirect local traffic through themselves.

Question 21 Single Choice | Topic: Enterprise Security Threats

The principle of least privilege aims to:

A. Give everyone admin rights
B. Limit user access to the minimum necessary
C. Disable all logging
D. Remove firewalls

Correct Answer: B

Concept: Least privilege reduces risk by restricting access to only what is needed.

Question 22 Single Choice | Topic: Enterprise Security Threats

A SIEM system is primarily used for:

A. Centralized log analysis and threat detection
B. Assigning IP addresses
C. Switching frames
D. Manufacturing cables

Correct Answer: A

Concept: SIEM aggregates and analyzes security events and logs.

Question 23 Single Choice | Topic: Enterprise Security Threats

Which concept is described as: monitors traffic without altering it?

A. Passive attack
B. DDoS attack
C. Social engineering
D. Ping of death

Correct Answer: A

Concept: Passive attack refers to the concept where monitors traffic without altering it.

Question 24 Single Choice | Topic: Enterprise Security Threats

Which concept is described as: modifies or disrupts systems or data?

A. Active attack
B. Least privilege
C. DDoS attack
D. SIEM

Correct Answer: A

Concept: Active attack refers to the concept where modifies or disrupts systems or data.

Question 25 **Single Choice | Topic: Enterprise Security Threats**

Which concept is described as: includes viruses, worms, Trojans, and ransomware?

A. Virus
B. Social engineering
C. Insider threat
D. Malware

Correct Answer: D

Concept: Malware refers to the concept where includes viruses, worms, Trojans, and ransomware.

Question 26 **Single Choice | Topic: Enterprise Security Threats**

Which concept is described as: requires a host program to spread?

A. SYN flood
B. Virus
C. DDoS attack
D. DMZ

Correct Answer: B

Concept: Virus refers to the concept where requires a host program to spread.

Question 27 **Single Choice | Topic: Enterprise Security Threats**

Which concept is described as: can self-replicate across networks?

A. Patch management
B. Man-in-the-middle attack
C. SYN flood
D. Worm

Correct Answer: D

Concept: Worm refers to the concept where can self-replicate across networks.

Question 28 **Single Choice | Topic: Enterprise Security Threats**

Which concept is described as: disguises itself as legitimate software?

A. Malware
B. Ping of death
C. Trojan
D. DMZ

Correct Answer: C

Concept: Trojan refers to the concept where disguises itself as legitimate software.

Question 29 **Single Choice | Topic: Enterprise Security Threats**

Which concept is described as: encrypts data and demands payment?

A. DoS attack
B. Ransomware
C. DMZ
D. SYN flood

Correct Answer: B

Concept: Ransomware refers to the concept where encrypts data and demands payment.

Question 30 **Single Choice | Topic: Enterprise Security Threats**

Which concept is described as: uses deception to steal information?

A. Ping of death
B. Smurf attack
C. Social engineering
D. Phishing

Correct Answer: D

Concept: Phishing refers to the concept where uses deception to steal information.

Question 31 **Single Choice | Topic: Enterprise Security Threats**

Which concept is described as: manipulates people into breaking security rules?

A. Social engineering
B. Worm
C. Virus
D. Least privilege

Correct Answer: A

Concept: Social engineering refers to the concept where manipulates people into breaking security rules.

Question 32 **Single Choice | Topic: Enterprise Security Threats**

Which concept is described as: aims to make a service unavailable?

A. ARP spoofing
B. DDoS attack
C. Trojan
D. DoS attack

Correct Answer: D

Concept: DoS attack refers to the concept where aims to make a service unavailable.

Question 33 **Single Choice | Topic: Enterprise Security Threats**

Which concept is described as: uses many distributed sources to flood a target?

A. ARP spoofing
B. DDoS attack
C. Insider threat
D. Active attack

Correct Answer: B

Concept: DDoS attack refers to the concept where uses many distributed sources to flood a target.

Question 34 **Single Choice | Topic: Enterprise Security Threats**

Which concept is described as: intercepts communication between two parties?

A. Passive attack
B. Worm
C. Man-in-the-middle attack
D. Smurf attack

Correct Answer: C

Concept: Man-in-the-middle attack refers to the concept where intercepts communication between two parties.

Question 35 **Single Choice | Topic: Enterprise Security Threats**

Which concept is described as: forges ARP replies to redirect traffic?

A. Phishing
B. ARP spoofing
C. SIEM
D. Ping of death

Correct Answer: B

Concept: ARP spoofing refers to the concept where forges ARP replies to redirect traffic.

Question 36 **Single Choice | Topic: Enterprise Security Threats**

Which concept is described as: sends many half-open TCP connections?

A. Least privilege
B. Malware
C. SYN flood
D. Worm

Correct Answer: C

Concept: SYN flood refers to the concept where sends many half-open TCP connections.

Question 37 **Single Choice | Topic: Enterprise Security Threats**

Which concept is described as: sends oversized ICMP packets?

A. Trojan
B. Ping of death
C. Passive attack
D. Worm

Correct Answer: B

Concept: Ping of death refers to the concept where sends oversized ICMP packets.

Question 38 **Single Choice | Topic: Enterprise Security Threats**

Which concept is described as: uses broadcast pings to amplify traffic?

A. Insider threat
B. Passive attack
C. Trojan
D. Smurf attack

Correct Answer: D

Concept: Smurf attack refers to the concept where uses broadcast pings to amplify traffic.

Question 39 **Single Choice | Topic: Enterprise Security Threats**

Which concept is described as: limits user access to the minimum necessary?

A. Least privilege
B. Ransomware
C. DDoS attack
D. Insider threat

Correct Answer: A

Concept: Least privilege refers to the concept where limits user access to the minimum necessary.

Question 40 **Single Choice | Topic: Enterprise Security Threats**

Which concept is described as: eliminates known vulnerabilities?

A. Patch management
B. Ping of death
C. SYN flood
D. ARP spoofing

Correct Answer: A

Concept: Patch management refers to the concept where eliminates known vulnerabilities.

Question 41 **Single Choice | Topic: Enterprise Security Threats**

Which concept is described as: centralizes log collection and analysis?

A. Worm
B. Malware
C. SIEM
D. Smurf attack

Correct Answer: C

Concept: SIEM refers to the concept where centralizes log collection and analysis.

Question 42 **Single Choice | Topic: Enterprise Security Threats**

Which concept is described as: hosts public-facing servers between Internet and internal network?

A. Passive attack
B. SYN flood
C. DMZ
D. DDoS attack

Correct Answer: C

Concept: DMZ refers to the concept where hosts public-facing servers between Internet and internal network.

Question 43 **Single Choice | Topic: Enterprise Security Threats**

Which concept is described as: originates from within the organization?

A. Active attack
B. Insider threat
C. SIEM
D. Social engineering

Correct Answer: B

Concept: Insider threat refers to the concept where originates from within the organization.

8 Communication Network Security

Question 1 **Single Choice | Topic: Communication Network Security**

Which threat involves intercepting data traveling across a network?

A. Eavesdropping
B. DoS
C. Phishing
D. Tailgating

Correct Answer: A

Concept: Eavesdropping intercepts network communications to steal information.

9 Zone Border Security

Question 1 — **Single Choice | Topic: Zone Border Security**

What is the main purpose of a security zone?

A. Increase Internet speed
B. Group interfaces with similar security levels
C. Replace antivirus
D. Disable routing

Correct Answer: B

Concept: Security zones group network interfaces with similar trust levels to enforce inter-zone policies.

Question 2 — **Single Choice | Topic: Zone Border Security**

Which device is typically deployed at a zone border to enforce access control?

A. Hub
B. Firewall
C. Repeater
D. Bridge

Correct Answer: B

Concept: Firewalls are deployed at network boundaries to enforce security policies between zones.

10 Computing Environment Security

Question 1 — **Single Choice | Topic: Computing Environment Security**

Which control helps protect endpoints from malware?

A. Antivirus/EDR
B. Physical locks only
C. Hub replacement
D. Disabling DNS

Correct Answer: A

Concept: Antivirus and endpoint detection and response (EDR) protect hosts from malware and suspicious behavior.

11 Firewall Basics

Question 1 **Single Choice | Topic: Firewall Basics**

Which firewall feature inspects application-layer data to detect attacks?

A. Packet filtering
B. Application inspection
C. Hub forwarding
D. NAT

Correct Answer: B

Concept: Application inspection examines payload content for malicious patterns.

Question 2 **Single Choice | Topic: Firewall Basics**

Which Huawei firewall command mode is used for system-level configuration?

A. User view
B. System view
C. Interface view
D. Policy view

Correct Answer: B

Concept: System view allows global-level configuration on Huawei devices.

Question 3 **Single Choice | Topic: Firewall Basics**

Which view is entered after the system-view command?

A. User view
B. System view
C. Interface view
D. None

Correct Answer: B

Concept: The system-view command enters system view from user view.

Question 4 **Single Choice | Topic: Firewall Basics**

What is the purpose of a security profile in NGFW?

A. Physical installation
B. Applying advanced security features such as IPS/AV/URL filtering
C. Layer 1 forwarding
D. DHCP relay

Correct Answer: B

Concept: Security profiles group advanced inspection features applied to policies.

Question 5 — **Single Choice | Topic: Firewall Basics**

Which firewall generation combines traditional firewall functions with application awareness and intrusion prevention?

A. Packet-filtering firewall
B. Next-generation firewall
C. Stateful firewall
D. Proxy firewall

Correct Answer: B

Concept: Next-generation firewalls integrate application awareness, IPS, antivirus, and user identity.

Question 6 — **Single Choice | Topic: Firewall Basics**

What is the purpose of a default security policy deny action?

A. Improve performance
B. Ensure least-privilege access
C. Enable logging
D. Allow management

Correct Answer: B

Concept: Default deny enforces least privilege by blocking unspecified traffic.

Question 7 — **Single Choice | Topic: Firewall Basics**

Which zone would typically contain public web servers?

A. Trust
B. Untrust
C. DMZ
D. Local

Correct Answer: C

Concept: Public-facing servers are placed in the DMZ for isolation.

Question 8 — **Single Choice | Topic: Firewall Basics**

What does policy hit counting help with?

A. Routing optimization
B. Verifying policy usage and troubleshooting
C. MAC learning
D. DHCP assignment

Correct Answer: B

Concept: Hit counts show which policies are triggered, aiding auditing and troubleshooting.

Question 9 | **Single Choice | Topic: Firewall Basics**

Which of the following is NOT a firewall deployment mode?

A. Routing mode
B. Transparent mode
C. Hybrid mode
D. Broadcast mode

Correct Answer: D

Concept: Firewalls deploy in routing, transparent (bridge), or hybrid modes; broadcast mode is not standard.

Question 10 | **Single Choice | Topic: Firewall Basics**

What happens when traffic moves from a higher-priority zone to a lower-priority zone?

A. Inbound traffic
B. Outbound traffic
C. No traffic allowed
D. Always dropped

Correct Answer: B

Concept: Traffic from a higher-priority zone to a lower-priority zone is considered outbound.

Question 11 | **Single Choice | Topic: Firewall Basics**

What is the priority range for custom security zones on Huawei firewalls?

A. 0-50
B. 1-100
C. 1-99 excluding defaults
D. 0-100

Correct Answer: C

Concept: Custom zone priorities range from 1 to 99, avoiding reserved default values.

Question 12 | **Single Choice | Topic: Firewall Basics**

Which action in a security policy can log permitted or denied traffic?

A. Permit
B. Deny
C. Both permit and deny
D. Neither

Correct Answer: C

Concept: Security policies can enable logging for both permit and deny actions.

Question 13 — Single Choice | Topic: Firewall Basics

Which security policy action is typically applied to suspicious traffic for further inspection?

A. Permit
B. Deny
C. Block
D. Redirect

Correct Answer: B

Concept: Suspicious traffic is commonly denied or blocked by policy.

Question 14 — Single Choice | Topic: Firewall Basics

When a return packet arrives, what does a stateful firewall check first?

A. Routing table
B. Session table
C. MAC table
D. ARP table

Correct Answer: B

Concept: A stateful firewall first checks the session table to find a matching connection.

Question 15 — Single Choice | Topic: Firewall Basics

What happens if a return packet does not match any session table entry?

A. Forwarded normally
B. Dropped or inspected against policies
C. Always permitted
D. Sent to DMZ

Correct Answer: B

Concept: Without a matching session, the firewall treats the packet as a new connection and applies policies.

Question 16 — Single Choice | Topic: Firewall Basics

Which protocol commonly requires ASPF support due to dynamic port negotiation?

A. HTTP
B. FTP
C. DNS
D. NTP

Correct Answer: B

Concept: FTP negotiates dynamic data ports in its control channel, requiring ASPF.

Question 17 — Single Choice | Topic: Firewall Basics

ASPF primarily inspects traffic at which layer?

A. Network layer only
B. Application layer
C. Physical layer
D. Data link layer

Correct Answer: B

Concept: ASPF performs application-layer inspection to identify negotiated connections.

Question 18 — Single Choice | Topic: Firewall Basics

In Huawei firewalls, the default priority of Trust is:

A. 5
B. 50
C. 85
D. 100

Correct Answer: C

Concept: Trust zone default priority is 85.

Question 19 — Single Choice | Topic: Firewall Basics

Traffic from a low-priority zone to a high-priority zone is considered:

A. Outbound
B. Inbound
C. Blocked by default
D. Always permitted

Correct Answer: B

Concept: Traffic entering a higher-priority zone from a lower-priority zone is inbound.

Question 20 — Single Choice | Topic: Firewall Basics

The default security policy action is:

A. Permit
B. Deny
C. Log
D. Redirect

Correct Answer: B

Concept: Huawei firewalls deny traffic by default unless explicitly permitted.

Question 21 Single Choice | Topic: Firewall Basics

Stateful inspection relies on:

A. Session tables
B. MAC tables only
C. DNS records
D. DHCP leases

Correct Answer: A

Concept: Session tables track connection state for stateful inspection.

Question 22 Single Choice | Topic: Firewall Basics

ASPF helps firewalls handle:

A. Multi-channel protocols
B. Static routing
C. MAC learning
D. DHCP relay

Correct Answer: A

Concept: ASPF inspects protocols that negotiate dynamic secondary channels.

Question 23 Single Choice | Topic: Firewall Basics

Which concept is described as: is 85 by default?

A. Trust zone priority
B. Stateful firewall
C. Routing mode
D. Local zone priority

Correct Answer: A

Concept: Trust zone priority refers to the concept where is 85 by default.

Question 24 Single Choice | Topic: Firewall Basics

Which concept is described as: is 50 by default?

A. Security policy logging
B. Stateful firewall
C. Policy hit count
D. DMZ zone priority

Correct Answer: D

Concept: DMZ zone priority refers to the concept where is 50 by default.

Question 25 Single Choice | Topic: Firewall Basics

Which concept is described as: is 5 by default?

A. Hybrid mode
B. Untrust zone priority
C. Stateful firewall
D. ASPF

Correct Answer: B

Concept: Untrust zone priority refers to the concept where is 5 by default.

Question 26 Single Choice | Topic: Firewall Basics

Which concept is described as: is 100 by default?

A. Local zone priority
B. Transparent mode
C. Routing mode
D. Policy hit count

Correct Answer: A

Concept: Local zone priority refers to the concept where is 100 by default.

Question 27 Single Choice | Topic: Firewall Basics

Which concept is described as: is deny?

A. Security zones
B. Routing mode
C. Policy hit count
D. Default security policy action

Correct Answer: D

Concept: Default security policy action refers to the concept where is deny.

Question 28 Single Choice | Topic: Firewall Basics

Which concept is described as: tracks connections in a session table?

A. Security policy logging
B. Stateful firewall
C. Packet-filtering firewall
D. Local zone priority

Correct Answer: B

Concept: Stateful firewall refers to the concept where tracks connections in a session table.

Question 29 **Single Choice | Topic: Firewall Basics**

Which concept is described as: inspects packets individually without state?

A. Packet-filtering firewall
B. Security policy logging
C. Local zone priority
D. Intra-zone traffic

Correct Answer: A

Concept: Packet-filtering firewall refers to the concept where inspects packets individually without state.

Question 30 **Single Choice | Topic: Firewall Basics**

Which concept is described as: inspects multi-channel protocols like FTP?

A. ASPF
B. Stateful firewall
C. Next-generation firewall
D. Security policy logging

Correct Answer: A

Concept: ASPF refers to the concept where inspects multi-channel protocols like FTP.

Question 31 **Single Choice | Topic: Firewall Basics**

Which concept is described as: includes application awareness and IPS?

A. Policy hit count
B. Next-generation firewall
C. Security policy logging
D. Untrust zone priority

Correct Answer: B

Concept: Next-generation firewall refers to the concept where includes application awareness and IPS.

Question 32 **Single Choice | Topic: Firewall Basics**

Which concept is described as: group interfaces with similar trust levels?

A. Untrust zone priority
B. Security zones
C. Hybrid mode
D. DMZ zone priority

Correct Answer: B

Concept: Security zones refers to the concept where group interfaces with similar trust levels.

Question 33 **Single Choice | Topic: Firewall Basics**

Which concept is described as: can be controlled by policies?

A. Intra-zone traffic
B. Policy hit count
C. Local zone priority
D. Security policy logging

Correct Answer: A

Concept: Intra-zone traffic refers to the concept where can be controlled by policies.

Question 34 **Single Choice | Topic: Firewall Basics**

Which concept is described as: helps verify which rules are used?

A. Routing mode
B. Hybrid mode
C. Policy hit count
D. Security zones

Correct Answer: C

Concept: Policy hit count refers to the concept where helps verify which rules are used.

Question 35 **Single Choice | Topic: Firewall Basics**

Which concept is described as: can be enabled for permit and deny actions?

A. Security policy logging
B. Next-generation firewall
C. ASPF
D. Policy hit count

Correct Answer: A

Concept: Security policy logging refers to the concept where can be enabled for permit and deny actions.

Question 36 **Single Choice | Topic: Firewall Basics**

Which concept is described as: is a firewall deployment mode?

A. Transparent mode
B. ASPF
C. Next-generation firewall
D. Routing mode

Correct Answer: D

Concept: Routing mode refers to the concept where is a firewall deployment mode.

Question 37 — **Single Choice | Topic: Firewall Basics**

Which concept is described as: is a firewall deployment mode?

A. Transparent mode
B. Security zones
C. Hybrid mode
D. Local zone priority

Correct Answer: A

Concept: Transparent mode refers to the concept where is a firewall deployment mode.

Question 38 — **Single Choice | Topic: Firewall Basics**

Which concept is described as: is a firewall deployment mode?

A. Security policy logging
B. Local zone priority
C. Hybrid mode
D. Trust zone priority

Correct Answer: C

Concept: Hybrid mode refers to the concept where is a firewall deployment mode.

12 Security Zones

Question 1 — **Single Choice | Topic: Security Zones**

What is the default priority of the Trust zone on a Huawei firewall?

A. 5
B. 50
C. 85
D. 100

Correct Answer: C

Concept: The Trust zone has a default priority of 85.

Question 2 — **Single Choice | Topic: Security Zones**

What is the default priority of the DMZ zone on a Huawei firewall?

A. 5
B. 50
C. 85
D. 100

Correct Answer: B

Concept: The DMZ zone has a default priority of 50.

Question 3 — **Single Choice | Topic: Security Zones**

What is the default priority of the Untrust zone on a Huawei firewall?

A. 5
B. 50
C. 85
D. 100

Correct Answer: A

Concept: The Untrust zone has a default priority of 5.

Question 4 — **Single Choice | Topic: Security Zones**

What is the default priority of the Local zone on a Huawei firewall?

A. 5
B. 50
C. 85
D. 100

Correct Answer: D

Concept: The Local zone represents the firewall itself and has the highest default priority of 100.

13 Security Policies

Question 1 — **Single Choice | Topic: Security Policies**

What is the default action of a Huawei firewall security policy if no rule matches?

A. Permit
B. Deny
C. Log only
D. Redirect

Correct Answer: B

Concept: By default, unmatched traffic is denied.

Question 2 — **Single Choice | Topic: Security Policies**

Which components can be used as matching conditions in a security policy? (single best answer)

A. Source/destination zone, address, service, application, user, time
B. Only source IP
C. Only destination port
D. Only protocol

Correct Answer: A

Concept: Security policies can match on zones, addresses, users, services, applications, and time.

14 Stateful Inspection

Question 1 | Single Choice | Topic: Stateful Inspection

What is a session table used for in a stateful firewall?

A. Storing user passwords
B. Tracking established connections
C. Routing packets
D. Encrypting traffic

Correct Answer: B

Concept: The session table tracks connection state so return traffic can be permitted without a new policy lookup.

Question 2 | Single Choice | Topic: Stateful Inspection

Which type of firewall inspects packets individually without connection state?

A. Stateful firewall
B. Packet-filtering firewall
C. Proxy firewall
D. Next-generation firewall

Correct Answer: B

Concept: Packet-filtering firewalls inspect each packet independently based on header information.

15 ASPF

Question 1 | Single Choice | Topic: ASPF

What does ASPF stand for?

A. Advanced Stateful Packet Filtering
B. Application Specific Packet Filtering
C. Automatic Security Policy Forwarding
D. Advanced Stateful Policy Firewall

Correct Answer: B

Concept: ASPF is Application Specific Packet Filtering, used to inspect multi-channel protocols.

Question 2 | Single Choice | Topic: ASPF

Why is ASPF needed for protocols like FTP?

A. FTP only uses one port
B. FTP opens dynamic data channels
C. FTP encrypts all traffic
D. FTP does not use TCP

Correct Answer: B

Concept: FTP opens a dynamic data channel, and ASPF inspects the control channel to permit the negotiated data channel.

16 NAT

Question 1 **Single Choice | Topic: NAT**

Which NAT type is also called many-to-one NAT?

A. Static NAT
B. Dynamic NAT
C. NAPT
D. NAT Server

Correct Answer: C

Concept: NAPT maps multiple private addresses to one public IP using port numbers.

Question 2 **Single Choice | Topic: NAT**

Which NAT type preserves a one-to-one mapping between public and private addresses?

A. Static NAT
B. NAPT
C. Easy-IP
D. Dynamic PAT

Correct Answer: A

Concept: Static NAT maps one public IP to one private IP consistently.

Question 3 **Single Choice | Topic: NAT**

Which NAT type uses the outbound interface IP as the public address?

A. NAPT
B. NAT No-PAT
C. Easy-IP
D. Destination NAT

Correct Answer: C

Concept: Easy-IP uses the firewall egress interface IP for translation.

Question 4 **Single Choice | Topic: NAT**

Which source NAT mode maps one private address to one public address without port translation?

A. NAPT
B. NAT No-PAT
C. Easy-IP
D. NAT Server

Correct Answer: B

Concept: NAT No-PAT performs one-to-one address translation without changing port numbers.

Question 5 — **Single Choice | Topic: NAT**

When many internal hosts access the Internet using one public IP, which NAT is used?

A. NAT No-PAT
B. NAPT
C. NAT Server
D. Static NAT

Correct Answer: B

Concept: NAPT allows many-to-one address translation using port numbers.

Question 6 — **Single Choice | Topic: NAT**

Destination NAT is also known as:

A. Static NAT
B. Port forwarding / virtual IP
C. PAT
D. Proxy ARP

Correct Answer: B

Concept: Destination NAT maps external destinations to internal hosts, commonly called port forwarding or virtual IP.

Question 7 — **Single Choice | Topic: NAT**

Which NAT type maps a public IP/port to a private server IP/port?

A. Source NAT
B. NAT Server
C. NAPT
D. Bidirectional NAT

Correct Answer: B

Concept: NAT Server publishes internal services to external networks.

Question 8 — **Single Choice | Topic: NAT**

Bidirectional NAT is useful when:

A. Only source address translation is needed
B. Both source and destination must be rewritten
C. No NAT is needed
D. Only DNS traffic is involved

Correct Answer: B

Concept: It translates both source and destination addresses in a single flow.

Question 9 **Single Choice | Topic: NAT**

Which problem can occur if NAT ALG is disabled for FTP?

A. FTP control channel fails
B. FTP data channel may fail because dynamic ports are not opened
C. FTP becomes encrypted
D. FTP speeds up

Correct Answer: B

Concept: Without ALG, the firewall cannot open the negotiated FTP data channel.

Question 10 **Single Choice | Topic: NAT**

Which address type is preserved by NAT No-PAT?

A. Source port
B. Destination port
C. Both source and destination ports
D. No ports

Correct Answer: C

Concept: NAT No-PAT translates only IP addresses, preserving port numbers.

Question 11 **Single Choice | Topic: NAT**

What is a major benefit of NAPT?

A. One-to-one address mapping
B. Conserves public IP addresses
C. Full protocol transparency
D. No ALG needed

Correct Answer: B

Concept: NAPT conserves public IPs by sharing one address across many internal hosts.

Question 12 **Single Choice | Topic: NAT**

A NAT Server entry is typically:

A. Bidirectional and persistent
B. Temporary and random
C. Encrypted
D. Only for outbound traffic

Correct Answer: A

Concept: NAT Server entries are configured statically and persist for inbound access.

Question 13 — Single Choice | Topic: NAT

NAT ALG is often enabled together with:

A. ASPF
B. Routing
C. DHCP
D. DNS

Correct Answer: A

Concept: ALG and ASPF work together to inspect dynamic application channels.

Question 14 — Single Choice | Topic: NAT

Which RFC defines private IPv4 address space?

A. RFC 1918
B. RFC 2328
C. RFC 793
D. RFC 959

Correct Answer: A

Concept: RFC 1918 defines the private IPv4 address ranges.

Question 15 — Single Choice | Topic: NAT

Hairpin NAT allows internal users to access an internal server using:

A. The server private IP only
B. The server public NAT address
C. A VPN tunnel
D. Broadcast address

Correct Answer: B

Concept: Hairpin NAT lets internal hosts reach an internal server via its public NAT address.

Question 16 — Single Choice | Topic: NAT

NAPT is also known as:

A. PAT
B. Static NAT
C. NAT Server
D. Bridge NAT

Correct Answer: A

Concept: Port Address Translation (PAT) is another name for NAPT.

Question 17 — Single Choice | Topic: NAT

Static NAT is most often used when:

A. A server must always use the same public IP
B. Many hosts share one IP
C. Ports must be preserved
D. Outbound only traffic exists

Correct Answer: A

Concept: Static NAT provides a consistent one-to-one public-to-private mapping.

Question 18 — Single Choice | Topic: NAT

Destination NAT is commonly used to:

A. Allow internal hosts to browse the web
B. Publish internal services to external users
C. Encrypt VPN traffic
D. Assign IP addresses

Correct Answer: B

Concept: Destination NAT maps external destinations to internal servers.

Question 19 — Single Choice | Topic: NAT

Which feature rewrites IP addresses inside application payloads?

A. NAT ALG
B. NAPT
C. Static NAT
D. Route lookup

Correct Answer: A

Concept: NAT ALG handles protocols that embed address information in payloads.

Question 20 — Single Choice | Topic: NAT

Which concept is described as: translates IP addresses between private and public?

A. Easy-IP
B. NAT ALG
C. NAT
D. NAT Server

Correct Answer: C

Concept: NAT refers to the concept where translates IP addresses between private and public.

Question 21 **Single Choice | Topic: NAT**

Which concept is described as: translates the source IP of outbound packets?

A. NAT No-PAT
B. NAT
C. Source NAT
D. Destination NAT

Correct Answer: C

Concept: Source NAT refers to the concept where translates the source IP of outbound packets.

Question 22 **Single Choice | Topic: NAT**

Which concept is described as: translates the destination IP of inbound packets?

A. NAT
B. Static NAT
C. NAT Server
D. Destination NAT

Correct Answer: D

Concept: Destination NAT refers to the concept where translates the destination IP of inbound packets.

Question 23 **Single Choice | Topic: NAT**

Which concept is described as: publishes internal services to the Internet?

A. Destination NAT
B. NAT No-PAT
C. NAPT
D. NAT Server

Correct Answer: D

Concept: NAT Server refers to the concept where publishes internal services to the Internet.

Question 24 **Single Choice | Topic: NAT**

Which concept is described as: translates both source and destination addresses?

A. Bidirectional NAT
B. Hairpin NAT
C. NAPT
D. Source NAT

Correct Answer: A

Concept: Bidirectional NAT refers to the concept where translates both source and destination addresses.

Question 25 **Single Choice | Topic: NAT**

Which concept is described as: translates many private addresses to one public IP using ports?

A. NAPT
B. Destination NAT
C. NAT
D. Static NAT

Correct Answer: A

Concept: NAPT refers to the concept where translates many private addresses to one public IP using ports.

Question 26 **Single Choice | Topic: NAT**

Which concept is described as: performs one-to-one address translation preserving ports?

A. NAT No-PAT
B. Hairpin NAT
C. NAT
D. Destination NAT

Correct Answer: A

Concept: NAT No-PAT refers to the concept where performs one-to-one address translation preserving ports.

Question 27 **Single Choice | Topic: NAT**

Which concept is described as: uses the outbound interface IP for translation?

A. NAT
B. NAT ALG
C. Easy-IP
D. NAPT

Correct Answer: C

Concept: Easy-IP refers to the concept where uses the outbound interface IP for translation.

Question 28 **Single Choice | Topic: NAT**

Which concept is described as: rewrites addresses inside application payloads?

A. Static NAT
B. NAPT
C. NAT ALG
D. NAT No-PAT

Correct Answer: C

Concept: NAT ALG refers to the concept where rewrites addresses inside application payloads.

Question 29 **Single Choice | Topic: NAT**

Which concept is described as: defines private IPv4 address ranges?

A. Hairpin NAT
B. NAPT
C. RFC 1918
D. Static NAT

Correct Answer: C

Concept: RFC 1918 refers to the concept where defines private IPv4 address ranges.

Question 30 **Single Choice | Topic: NAT**

Which concept is described as: lets internal users access servers via public NAT address?

A. Easy-IP
B. NAT Server
C. Hairpin NAT
D. Source NAT

Correct Answer: C

Concept: Hairpin NAT refers to the concept where lets internal users access servers via public NAT address.

Question 31 **Single Choice | Topic: NAT**

Which concept is described as: provides a persistent one-to-one mapping?

A. Destination NAT
B. Static NAT
C. RFC 1918
D. NAT

Correct Answer: B

Concept: Static NAT refers to the concept where provides a persistent one-to-one mapping.

17 NAT Basics

Question 1 **Single Choice | Topic: NAT Basics**

What does NAT primarily translate?

A. MAC addresses
B. IP addresses
C. Port numbers only
D. Domain names

Correct Answer: B

Concept: NAT translates IP addresses, typically between private and public addresses.

18 Source NAT

Question 1 **Single Choice | Topic: Source NAT**

Which NAT type translates the source IP address of outgoing packets?

A. Source NAT
B. Destination NAT
C. Bidirectional NAT
D. NAT Server

Correct Answer: A

Concept: Source NAT changes the source IP so internal hosts can use public addresses to access the Internet.

Question 2 **Single Choice | Topic: Source NAT**

What is the main purpose of NAPT?

A. Translate many private addresses to one public IP using port multiplexing
B. Hide the destination IP
C. Encrypt traffic
D. Provide VPN access

Correct Answer: A

Concept: NAPT (Network Address and Port Translation) maps multiple private addresses to a single public IP using different ports.

19 Destination NAT

Question 1 **Single Choice | Topic: Destination NAT**

Which NAT type translates the destination IP address of incoming packets?

A. Source NAT
B. Destination NAT
C. Bidirectional NAT
D. NAT Server

Correct Answer: B

Concept: Destination NAT rewrites the destination IP, often to expose internal servers to external users.

20 NAT Server

Question 1 Single Choice | Topic: NAT Server

What is a NAT Server used for?

A. Hiding internal server IP addresses
B. Publishing internal services to the Internet
C. Encrypting server traffic
D. Load balancing only

Correct Answer: B

Concept: NAT Server maps public IPs/ports to private server IPs/ports so external users can access internal services.

21 NAT ALG

Question 1 Single Choice | Topic: NAT ALG

What is the purpose of NAT ALG?

A. Encrypt NAT traffic
B. Translate application-layer addresses embedded in protocol payloads
C. Replace firewalls
D. Assign IP addresses

Correct Answer: B

Concept: NAT Application Level Gateway inspects and rewrites IP addresses inside application-layer protocols (e.g., FTP, SIP).

22 Firewall Hot Standby

Question 1 Single Choice | Topic: Firewall Hot Standby

In active/standby mode, how many firewalls forward traffic at the same time?

A. One
B. Two
C. All
D. None

Correct Answer: A

Concept: In active/standby mode, one firewall actively forwards while the other remains standby.

Question 2 **Single Choice | Topic: Firewall Hot Standby**

In load-sharing mode, how many firewalls forward traffic?

A. One
B. Two or more
C. None
D. Only the backup

Correct Answer: B

Concept: Load-sharing mode allows multiple firewalls to forward traffic simultaneously.

Question 3 **Single Choice | Topic: Firewall Hot Standby**

What is the default VRRP advertisement interval?

A. 1 second
B. 3 seconds
C. 10 seconds
D. 30 seconds

Correct Answer: A

Concept: VRRP master sends advertisements approximately every 1 second by default.

Question 4 **Single Choice | Topic: Firewall Hot Standby**

Which VRRP priority value is highest configurable?

A. 1
B. 100
C. 254
D. 255

Correct Answer: C

Concept: VRRP priority ranges from 1 to 254 configurable; 255 is reserved for the IP owner.

Question 5 **Single Choice | Topic: Firewall Hot Standby**

What happens if a backup router stops receiving VRRP advertisements?

A. It remains backup
B. It preempts and becomes master
C. It shuts down
D. It sends ARP requests

Correct Answer: B

Concept: Missing advertisements trigger a backup to assume the master role.

Question 6 **Single Choice | Topic: Firewall Hot Standby**

What does VGMP group priority determine?

A. NAT pool size
B. Which firewall becomes active
C. Number of interfaces
D. DHCP lease time

Correct Answer: B

Concept: VGMP priority determines the active/standby role among redundant firewalls.

Question 7 **Single Choice | Topic: Firewall Hot Standby**

Which HRP state indicates the firewall is currently forwarding traffic?

A. Standby
B. Active
C. Initialize
D. Backup

Correct Answer: B

Concept: The active HRP peer forwards traffic and synchronizes state to the standby.

Question 8 **Single Choice | Topic: Firewall Hot Standby**

Which HRP command enables automatic configuration synchronization?

A. hrp enable
B. hrp auto-sync config
C. hrp mirror session
D. hrp vrrp enable

Correct Answer: B

Concept: hrp auto-sync config enables automatic configuration synchronization between peers.

Question 9 **Single Choice | Topic: Firewall Hot Standby**

Which deployment places both firewalls inline with the same traffic path?

A. Dual-hot standby
B. Active/standby inline
C. Route backup only
D. Transparent bridge

Correct Answer: B

Concept: Inline active/standby deployment places firewalls in the same forwarding path.

Question 10 — Single Choice | Topic: Firewall Hot Standby

What is a common requirement for HRP heartbeat links?

A. Use the same interface as user traffic
B. Use a dedicated high-reliability link
C. Use wireless link only
D. No link required

Correct Answer: B

Concept: Heartbeat links should be dedicated and reliable to avoid false failovers.

Question 11 — Single Choice | Topic: Firewall Hot Standby

Which mechanism reduces unnecessary VRRP state changes when interface flaps briefly?

A. Preempt delay
B. Track interface
C. Authentication
D. Priority 255

Correct Answer: A

Concept: Preempt delay prevents rapid master/backup flapping.

Question 12 — Single Choice | Topic: Firewall Hot Standby

Which HRP feature ensures sessions survive a failover?

A. Session backup
B. Route backup
C. NAT ALG
D. URL filtering

Correct Answer: A

Concept: HRP session backup keeps session state synchronized so connections continue after failover.

Question 13 — Single Choice | Topic: Firewall Hot Standby

VRRP master election is based primarily on:

A. Priority value
B. Router hostname
C. Number of interfaces
D. Uptime only

Correct Answer: A

Concept: The router with the highest priority becomes the VRRP master.

Question 14 Single Choice | Topic: Firewall Hot Standby

VGMP group management prevents:

A. Split-brain VRRP states
B. NAT exhaustion
C. DNS failures
D. DHCP conflicts

Correct Answer: A

Concept: VGMP ensures VRRP groups fail over consistently.

Question 15 Single Choice | Topic: Firewall Hot Standby

HRP is used to synchronize:

A. Session and configuration data
B. Email messages
C. Web pages
D. Printer drivers

Correct Answer: A

Concept: HRP synchronizes sessions, NAT, server-map entries, and configurations.

Question 16 Single Choice | Topic: Firewall Hot Standby

In active/standby mode, the standby firewall:

A. Does not forward traffic until failover
B. Forwards half the traffic
C. Processes all traffic
D. Acts as a switch

Correct Answer: A

Concept: The standby firewall remains ready but inactive until failover.

Question 17 Single Choice | Topic: Firewall Hot Standby

Which concept is described as: provides gateway redundancy with a virtual IP?

A. VRRP
B. VRRP backup
C. VGMP
D. Load-sharing mode

Correct Answer: A

Concept: VRRP refers to the concept where provides gateway redundancy with a virtual IP.

Question 18 **Single Choice | Topic: Firewall Hot Standby**

Which concept is described as: forwards traffic for the virtual IP?

A. VRRP backup
B. Heartbeat link
C. HRP
D. VRRP master

Correct Answer: D

Concept: VRRP master refers to the concept where forwards traffic for the virtual IP.

Question 19 **Single Choice | Topic: Firewall Hot Standby**

Which concept is described as: takes over if the master fails?

A. HRP session backup
B. VRRP
C. VRRP backup
D. Active/standby mode

Correct Answer: C

Concept: VRRP backup refers to the concept where takes over if the master fails.

Question 20 **Single Choice | Topic: Firewall Hot Standby**

Which concept is described as: manages VRRP group states consistently?

A. HRP
B. Active/standby mode
C. VGMP
D. Load-sharing mode

Correct Answer: C

Concept: VGMP refers to the concept where manages VRRP group states consistently.

Question 21 **Single Choice | Topic: Firewall Hot Standby**

Which concept is described as: synchronizes sessions and configurations?

A. Preempt delay
B. HRP
C. VGMP
D. Load-sharing mode

Correct Answer: B

Concept: HRP refers to the concept where synchronizes sessions and configurations.

Question 22 **Single Choice | Topic: Firewall Hot Standby**

Which concept is described as: has one firewall forwarding traffic?

A. VRRP master
B. Active/standby mode
C. VGMP
D. VRRP backup

Correct Answer: B

Concept: Active/standby mode refers to the concept where has one firewall forwarding traffic.

Question 23 **Single Choice | Topic: Firewall Hot Standby**

Which concept is described as: has multiple firewalls forwarding traffic?

A. HRP
B. Active/standby mode
C. VRRP master
D. Load-sharing mode

Correct Answer: D

Concept: Load-sharing mode refers to the concept where has multiple firewalls forwarding traffic.

Question 24 **Single Choice | Topic: Firewall Hot Standby**

Which concept is described as: should be dedicated and reliable?

A. Heartbeat link
B. VRRP backup
C. VRRP master
D. VRRP

Correct Answer: A

Concept: Heartbeat link refers to the concept where should be dedicated and reliable.

Question 25 **Single Choice | Topic: Firewall Hot Standby**

Which concept is described as: ensures connections survive failover?

A. HRP
B. VRRP backup
C. VRRP master
D. HRP session backup

Correct Answer: D

Concept: HRP session backup refers to the concept where ensures connections survive failover.

Question 26 Single Choice | Topic: Firewall Hot Standby

Which concept is described as: reduces rapid master-backup flapping?

A. HRP
B. Preempt delay
C. VRRP backup
D. HRP session backup

Correct Answer: B

Concept: Preempt delay refers to the concept where reduces rapid master-backup flapping.

23 VRRP

Question 1 Single Choice | Topic: VRRP

What does VRRP stand for?

A. Virtual Router Redundancy Protocol
B. Virtual Routing Resolution Protocol
C. Virtual Redundant Routing Path
D. Virtual Routing Relay Protocol

Correct Answer: A

Concept: VRRP provides gateway redundancy by allowing multiple routers to share a virtual IP address.

Question 2 Single Choice | Topic: VRRP

In VRRP, which router forwards traffic for the virtual IP?

A. Backup router
B. Master router
C. All routers simultaneously
D. Only the physical IP owner

Correct Answer: B

Concept: The VRRP master router owns the virtual IP and forwards traffic; backups take over if it fails.

24 VGMP

Question 1 **Single Choice | Topic: VGMP**

What does VGMP manage on Huawei firewalls?

A. VRRP groups
B. NAT pools
C. Routing tables
D. DHCP scopes

Correct Answer: A

Concept: VGMP (VRRP Group Management Protocol) manages the state of multiple VRRP groups consistently.

25 HRP

Question 1 **Single Choice | Topic: HRP**

What is the primary purpose of HRP?

A. Encrypt traffic
B. Synchronize sessions and configurations between firewalls
C. Assign IP addresses
D. Filter URLs

Correct Answer: B

Concept: HRP (Huawei Redundancy Protocol) synchronizes sessions, configurations, and state between hot-standby firewalls.

26 Intrusion Prevention and Antivirus

Question 1 **Single Choice | Topic: Intrusion Prevention and Antivirus**

Which phase of an attack involves identifying vulnerable services?

A. Exploitation
B. Reconnaissance
C. Installation
D. Exfiltration

Correct Answer: B

Concept: Reconnaissance gathers information about targets before exploitation.

Question 2 Single Choice | Topic: Intrusion Prevention and Antivirus

Which phase involves gaining unauthorized access?

A. Reconnaissance
B. Exploitation
C. Covering tracks
D. Reporting

Correct Answer: B

Concept: Exploitation uses vulnerabilities to gain access to the target.

Question 3 Single Choice | Topic: Intrusion Prevention and Antivirus

Which deployment mode allows IDS to monitor traffic without affecting it?

A. Inline
B. Tap/SPAN
C. Bridge
D. Gateway

Correct Answer: B

Concept: IDS commonly connects to a tap or switch SPAN port for passive monitoring.

Question 4 Single Choice | Topic: Intrusion Prevention and Antivirus

Which signature action drops the offending packet?

A. Alert
B. Block
C. Permit
D. Mirror

Correct Answer: B

Concept: Block actions discard malicious packets.

Question 5 Single Choice | Topic: Intrusion Prevention and Antivirus

What is a false positive in IPS?

A. Attack missed
B. Legitimate traffic blocked as malicious
C. Correct detection
D. Signature update failure

Correct Answer: B

Concept: False positives incorrectly classify legitimate traffic as malicious.

Question 6 **Single Choice | Topic: Intrusion Prevention and Antivirus**

What is a false negative in IPS?

A. Legitimate traffic blocked
B. Attack missed by IPS
C. Correct block
D. Logging error

Correct Answer: B

Concept: False negatives occur when an actual attack is not detected.

Question 7 **Single Choice | Topic: Intrusion Prevention and Antivirus**

Which malware type encrypts user files and demands ransom?

A. Virus
B. Worm
C. Trojan
D. Ransomware

Correct Answer: D

Concept: Ransomware encrypts files and demands a ransom for decryption.

Question 8 **Single Choice | Topic: Intrusion Prevention and Antivirus**

Which approach only allows known-good applications to run?

A. Blacklisting
B. Whitelisting
C. Heuristics
D. Sandboxing

Correct Answer: B

Concept: Whitelisting permits only approved applications.

Question 9 **Single Choice | Topic: Intrusion Prevention and Antivirus**

Which technique detects malware by examining code behavior rather than signatures?

A. Signature scanning
B. Heuristic analysis
C. Whitelisting
D. Patching

Correct Answer: B

Concept: Heuristics analyze code behavior to identify potentially malicious activity.

Question 10 **Single Choice | Topic: Intrusion Prevention and Antivirus**

Which protocol is commonly targeted by brute-force attacks against remote access?

A. HTTP
B. SSH
C. DNS
D. NTP

Correct Answer: B

Concept: SSH and similar remote-access protocols are frequent brute-force targets.

Question 11 **Single Choice | Topic: Intrusion Prevention and Antivirus**

What does an IPS signature database contain?

A. User passwords
B. Known attack patterns
C. Routing tables
D. DNS records

Correct Answer: B

Concept: Signature databases contain patterns used to identify known attacks.

Question 12 **Single Choice | Topic: Intrusion Prevention and Antivirus**

What is the role of the antivirus engine on a firewall?

A. Route packets
B. Scan files and traffic for malware
C. Assign IP addresses
D. Authenticate users

Correct Answer: B

Concept: The antivirus engine inspects files and traffic to detect and block malware.

Question 13 **Single Choice | Topic: Intrusion Prevention and Antivirus**

An IPS differs from an IDS mainly because an IPS can:

A. Block traffic
B. Only log events
C. Operate out-of-band only
D. Never generate alerts

Correct Answer: A

Concept: IPS can take active blocking actions, while IDS primarily detects.

Question 14 **Single Choice | Topic: Intrusion Prevention and Antivirus**

Signature-based detection cannot easily detect:

A. Known attacks
B. Zero-day attacks
C. Old malware
D. Common exploits

Correct Answer: B

Concept: Zero-day attacks have no known signatures.

Question 15 **Single Choice | Topic: Intrusion Prevention and Antivirus**

Sandboxing helps detect malware by:

A. Observing behavior in an isolated environment
B. Matching signatures
C. Comparing file sizes
D. Checking file names

Correct Answer: A

Concept: Sandboxing executes suspicious files safely to observe malicious behavior.

Question 16 **Single Choice | Topic: Intrusion Prevention and Antivirus**

Antivirus engines require regular updates to:

A. Recognize new malware variants
B. Improve routing
C. Increase bandwidth
D. Disable firewalls

Correct Answer: A

Concept: Signature and detection databases must be updated to detect emerging threats.

Question 17 **Single Choice | Topic: Intrusion Prevention and Antivirus**

Which concept is described as: detects intrusions and generates alerts?

A. NIDS
B. Antivirus signatures
C. IDS
D. Exploitation

Correct Answer: C

Concept: IDS refers to the concept where detects intrusions and generates alerts.

Question 18 **Single Choice | Topic: Intrusion Prevention and Antivirus**

Which concept is described as: detects and blocks intrusions inline?

A. False positive
B. Reconnaissance
C. IDS
D. IPS

Correct Answer: D

Concept: IPS refers to the concept where detects and blocks intrusions inline.

Question 19 **Single Choice | Topic: Intrusion Prevention and Antivirus**

Which concept is described as: matches known attack patterns?

A. Whitelisting
B. HIDS
C. Signature detection
D. Heuristic analysis

Correct Answer: C

Concept: Signature detection refers to the concept where matches known attack patterns.

Question 20 **Single Choice | Topic: Intrusion Prevention and Antivirus**

Which concept is described as: flags deviations from normal behavior?

A. Anomaly detection
B. Sandboxing
C. HIDS
D. IDS

Correct Answer: A

Concept: Anomaly detection refers to the concept where flags deviations from normal behavior.

Question 21 **Single Choice | Topic: Intrusion Prevention and Antivirus**

Which concept is described as: incorrectly blocks legitimate traffic?

A. IPS
B. False positive
C. HIDS
D. IDS

Correct Answer: B

Concept: False positive refers to the concept where incorrectly blocks legitimate traffic.

Question 22 **Single Choice | Topic: Intrusion Prevention and Antivirus**

Which concept is described as: misses an actual attack?

A. False negative
B. False positive
C. IPS
D. HIDS

Correct Answer: A

Concept: False negative refers to the concept where misses an actual attack.

Question 23 **Single Choice | Topic: Intrusion Prevention and Antivirus**

Which concept is described as: observes file behavior in an isolated environment?

A. Heuristic analysis
B. False positive
C. IDS
D. Sandboxing

Correct Answer: D

Concept: Sandboxing refers to the concept where observes file behavior in an isolated environment.

Question 24 **Single Choice | Topic: Intrusion Prevention and Antivirus**

Which concept is described as: detects malware by examining behavior?

A. Heuristic analysis
B. Antivirus signatures
C. Signature detection
D. IDS

Correct Answer: A

Concept: Heuristic analysis refers to the concept where detects malware by examining behavior.

Question 25 **Single Choice | Topic: Intrusion Prevention and Antivirus**

Which concept is described as: allows only approved applications?

A. HIDS
B. Reconnaissance
C. Whitelisting
D. Signature detection

Correct Answer: C

Concept: Whitelisting refers to the concept where allows only approved applications.

Question 26 **Single Choice | Topic: Intrusion Prevention and Antivirus**

Which concept is described as: must be updated regularly?

A. Heuristic analysis
B. NIDS
C. HIDS
D. Antivirus signatures

Correct Answer: D

Concept: Antivirus signatures refers to the concept where must be updated regularly.

Question 27 **Single Choice | Topic: Intrusion Prevention and Antivirus**

Which concept is described as: monitors network traffic for intrusions?

A. Whitelisting
B. NIDS
C. HIDS
D. Anomaly detection

Correct Answer: B

Concept: NIDS refers to the concept where monitors network traffic for intrusions.

Question 28 **Single Choice | Topic: Intrusion Prevention and Antivirus**

Which concept is described as: monitors host activity for intrusions?

A. Antivirus signatures
B. HIDS
C. Anomaly detection
D. Exploitation

Correct Answer: B

Concept: HIDS refers to the concept where monitors host activity for intrusions.

Question 29 **Single Choice | Topic: Intrusion Prevention and Antivirus**

Which concept is described as: is an early attack phase?

A. Whitelisting
B. HIDS
C. Exploitation
D. Reconnaissance

Correct Answer: D

Concept: Reconnaissance refers to the concept where is an early attack phase.

Question 30 Single Choice | Topic: Intrusion Prevention and Antivirus

Which concept is described as: gains unauthorized access using vulnerabilities?

A. Anomaly detection
B. Exploitation
C. IPS
D. False positive

Correct Answer: B

Concept: Exploitation refers to the concept where gains unauthorized access using vulnerabilities.

27 Intrusion Overview

Question 1 Single Choice | Topic: Intrusion Overview

What is an intrusion?

A. Authorized access
B. Unauthorized access or attack on a network/system
C. A software patch
D. A backup operation

Correct Answer: B

Concept: An intrusion is any unauthorized attempt to access, manipulate, or harm a system or network.

28 Intrusion Prevention

Question 1 Single Choice | Topic: Intrusion Prevention

What is the main difference between IDS and IPS?

A. IDS only detects; IPS detects and blocks
B. IDS is faster
C. IPS cannot log
D. IDS operates inline

Correct Answer: A

Concept: IDS detects intrusions and alerts; IPS can also take action to block malicious traffic.

Question 2 Single Choice | Topic: Intrusion Prevention

Which detection method compares traffic against known attack signatures?

A. Anomaly detection
B. Signature detection
C. Behavioral analysis
D. Heuristic detection

Correct Answer: B

Concept: Signature-based detection matches traffic patterns against a database of known attack signatures.

Question 3 — Single Choice | Topic: Intrusion Prevention

Which detection method builds a baseline of normal behavior and flags deviations?

A. Signature detection
B. Anomaly detection
C. Static analysis
D. Port scanning

Correct Answer: B

Concept: Anomaly detection identifies traffic that deviates from a learned baseline.

Question 4 — Single Choice | Topic: Intrusion Prevention

Where is an IPS typically deployed?

A. Out-of-band only
B. Inline in the traffic path
C. On user desktops only
D. Inside a printer

Correct Answer: B

Concept: IPS is deployed inline so it can block malicious traffic in real time.

29 Antivirus

Question 1 — Single Choice | Topic: Antivirus

What does antivirus software primarily detect and remove?

A. Routing loops
B. Malware
C. Cable faults
D. DNS errors

Correct Answer: B

Concept: Antivirus detects and removes malware such as viruses, worms, and Trojans.

Question 2 — Single Choice | Topic: Antivirus

Which antivirus technique looks for known byte sequences of malware?

A. Heuristic analysis
B. Signature scanning
C. Sandboxing
D. Whitelisting

Correct Answer: B

Concept: Signature scanning matches files/traffic against known malware signatures.

Question 3 **Single Choice | Topic: Antivirus**

Which method executes suspicious files in an isolated environment to observe behavior?

A. Signature scanning
B. Heuristic analysis
C. Sandboxing
D. Whitelisting

Correct Answer: C

Concept: Sandboxing runs files in an isolated environment to detect malicious behavior.

30 AAA and User Authentication

Question 1 **Single Choice | Topic: AAA and User Authentication**

Which AAA protocol commonly uses UDP ports 1812 and 1813?

A. TACACS+
B. RADIUS
C. LDAP
D. Kerberos

Correct Answer: B

Concept: RADIUS commonly uses UDP ports 1812/1813 (or 1645/1646 legacy).

Question 2 **Single Choice | Topic: AAA and User Authentication**

Which AAA protocol uses TCP port 49 by default?

A. RADIUS
B. TACACS+
C. LDAP
D. DHCP

Correct Answer: B

Concept: TACACS+ uses TCP port 49 by default.

Question 3 **Single Choice | Topic: AAA and User Authentication**

In AAA, which process maps users to privileges?

A. Authentication
B. Authorization
C. Accounting
D. Admission

Correct Answer: B

Concept: Authorization maps authenticated identities to permitted actions/resources.

Question 4 — Single Choice | Topic: AAA and User Authentication

Which authentication type identifies users based on source IP?

- **A.** User authentication
- **B.** Access user authentication
- **C.** Device authentication
- **D.** IP/MAC binding

Correct Answer: B

Concept: Access user authentication can identify users by IP/MAC for transparent access control.

Question 5 — Single Choice | Topic: AAA and User Authentication

Which user type is created directly on the firewall?

- **A.** RADIUS user
- **B.** LDAP user
- **C.** Local user
- **D.** Domain user

Correct Answer: C

Concept: Local users are defined directly in the firewall local user database.

Question 6 — Single Choice | Topic: AAA and User Authentication

Which component is queried when using LDAP authentication?

- **A.** DNS server
- **B.** LDAP directory server
- **C.** NTP server
- **D.** SNMP server

Correct Answer: B

Concept: The firewall queries an LDAP directory server to validate credentials.

Question 7 — Single Choice | Topic: AAA and User Authentication

What is the purpose of a user group?

- **A.** Increase Internet speed
- **B.** Apply policies to multiple users collectively
- **C.** Replace firewalls
- **D.** Encrypt traffic

Correct Answer: B

Concept: User groups simplify policy management by grouping users with similar access needs.

Question 8 **Single Choice | Topic: AAA and User Authentication**

Which authentication method is often used for wired/wireless network access control?

A. Portal
B. 802.1X
C. Local
D. RADIUS accounting

Correct Answer: B

Concept: 802.1X is commonly used for port-based network access control on wired and wireless networks.

Question 9 **Single Choice | Topic: AAA and User Authentication**

What does single sign-on (SSO) provide?

A. Multiple passwords
B. One authentication for multiple services
C. No authentication
D. Local only access

Correct Answer: B

Concept: SSO allows users to authenticate once and access multiple services.

Question 10 **Single Choice | Topic: AAA and User Authentication**

Which user authentication policy condition can match traffic?

A. Source zone
B. Time range
C. Source/destination address
D. All of the above

Correct Answer: D

Concept: Authentication policies can match zones, addresses, time ranges, and other conditions.

Question 11 **Single Choice | Topic: AAA and User Authentication**

The first A in AAA stands for:

A. Authentication
B. Authorization
C. Accounting
D. Auditing

Correct Answer: A

Concept: AAA stands for Authentication, Authorization, and Accounting.

Question 12 **Single Choice | Topic: AAA and User Authentication**

RADIUS commonly runs over:

A. UDP
B. TCP
C. SCTP
D. ICMP

Correct Answer: A

Concept: RADIUS typically uses UDP ports 1812/1813.

Question 13 **Single Choice | Topic: AAA and User Authentication**

TACACS+ commonly runs over:

A. UDP
B. TCP
C. HTTP
D. FTP

Correct Answer: B

Concept: TACACS+ uses TCP port 49.

Question 14 **Single Choice | Topic: AAA and User Authentication**

Portal authentication is typically used for:

A. Web-based guest access
B. Cable modem registration
C. Switch MAC learning
D. DNS resolution

Correct Answer: A

Concept: Portal authentication presents a web login page.

Question 15 **Single Choice | Topic: AAA and User Authentication**

802.1X authentication occurs at:

A. Layer 2 port level
B. Application layer only
C. Physical layer only
D. Transport layer only

Correct Answer: A

Concept: 802.1X provides port-based network access control at Layer 2.

Question 16 **Single Choice | Topic: AAA and User Authentication**

Which concept is described as: verifies user identity?

A. Authentication
B. SSO
C. TACACS+
D. Local authentication

Correct Answer: A

Concept: Authentication refers to the concept where verifies user identity.

Question 17 **Single Choice | Topic: AAA and User Authentication**

Which concept is described as: determines what a user can access?

A. Accounting
B. Authorization
C. TACACS+
D. User group

Correct Answer: B

Concept: Authorization refers to the concept where determines what a user can access.

Question 18 **Single Choice | Topic: AAA and User Authentication**

Which concept is described as: records user activities and resource usage?

A. User group
B. Accounting
C. SSO
D. RADIUS

Correct Answer: B

Concept: Accounting refers to the concept where records user activities and resource usage.

Question 19 **Single Choice | Topic: AAA and User Authentication**

Which concept is described as: uses UDP ports 1812 and 1813?

A. RADIUS
B. Authorization
C. Authentication
D. TACACS+

Correct Answer: A

Concept: RADIUS refers to the concept where uses UDP ports 1812 and 1813.

Question 20 Single Choice | Topic: AAA and User Authentication

Which concept is described as: uses TCP port 49?

A. TACACS+
B. LDAP
C. RADIUS
D. User group

Correct Answer: A

Concept: TACACS+ refers to the concept where uses TCP port 49.

Question 21 Single Choice | Topic: AAA and User Authentication

Which concept is described as: uses a web login page?

A. RADIUS
B. Authentication
C. Portal authentication
D. LDAP

Correct Answer: C

Concept: Portal authentication refers to the concept where uses a web login page.

Question 22 Single Choice | Topic: AAA and User Authentication

Which concept is described as: provides port-based network access control?

A. Accounting
B. 802.1X
C. Authorization
D. TACACS+

Correct Answer: B

Concept: 802.1X refers to the concept where provides port-based network access control.

Question 23 Single Choice | Topic: AAA and User Authentication

Which concept is described as: uses the device user database?

A. Accounting
B. Authorization
C. Local authentication
D. Authentication

Correct Answer: C

Concept: Local authentication refers to the concept where uses the device user database.

Question 24 **Single Choice | Topic: AAA and User Authentication**

Which concept is described as: queries a directory server for authentication?

A. Local authentication
B. SSO
C. LDAP
D. 802.1X

Correct Answer: C

Concept: LDAP refers to the concept where queries a directory server for authentication.

Question 25 **Single Choice | Topic: AAA and User Authentication**

Which concept is described as: applies policies to multiple users collectively?

A. User group
B. TACACS+
C. Portal authentication
D. Local authentication

Correct Answer: A

Concept: User group refers to the concept where applies policies to multiple users collectively.

Question 26 **Single Choice | Topic: AAA and User Authentication**

Which concept is described as: allows one authentication for multiple services?

A. TACACS+
B. SSO
C. Local authentication
D. RADIUS

Correct Answer: B

Concept: SSO refers to the concept where allows one authentication for multiple services.

31 AAA Basics

Question 1 **Single Choice | Topic: AAA Basics**

What does AAA stand for?

A. Authentication, Authorization, Accounting
B. Access, Audit, Alert
C. Attack, Analysis, Action
D. Application, Authorization, Audit

Correct Answer: A

Concept: AAA provides authentication (who are you), authorization (what can you do), and accounting (what did you do).

Question 2 — Single Choice | Topic: AAA Basics

Which AAA component verifies a user identity?

A. Authentication
B. Authorization
C. Accounting
D. Auditing

Correct Answer: A

Concept: Authentication verifies identity, for example, by username/password or certificate.

Question 3 — Single Choice | Topic: AAA Basics

Which AAA component determines what resources a user can access?

A. Authentication
B. Authorization
C. Accounting
D. Auditing

Correct Answer: B

Concept: Authorization defines permissions and access rights after authentication.

Question 4 — Single Choice | Topic: AAA Basics

Which AAA component records user activities for billing or auditing?

A. Authentication
B. Authorization
C. Accounting
D. Auditing

Correct Answer: C

Concept: Accounting logs user activities such as login time, data usage, and commands executed.

32 Firewall User Authentication

Question 1 — Single Choice | Topic: Firewall User Authentication

Which authentication method redirects a user to a web portal for login?

A. Local authentication
B. RADIUS authentication
C. Portal authentication
D. LDAP authentication

Correct Answer: C

Concept: Portal authentication presents a web page to users for credential entry.

Question 2 **Single Choice | Topic: Firewall User Authentication**

Which authentication method uses a client-side agent to authenticate users before network access?

A. Portal authentication
B. 802.1X authentication
C. Local authentication
D. Certificate authentication

Correct Answer: B

Concept: 802.1X uses a supplicant client to authenticate devices before granting network access.

Question 3 **Single Choice | Topic: Firewall User Authentication**

Which protocol is commonly used for centralized authentication and accounting?

A. RADIUS
B. SMTP
C. SNMP
D. NTP

Correct Answer: A

Concept: RADIUS is a widely used AAA protocol for centralized authentication, authorization, and accounting.

Question 4 **Single Choice | Topic: Firewall User Authentication**

Which protocol provides authentication with stronger security and uses TCP?

A. RADIUS
B. TACACS+
C. LDAP
D. DHCP

Correct Answer: B

Concept: TACACS+ uses TCP and separates authentication, authorization, and accounting functions.

33 Cryptography and PKI

Question 1 **Single Choice | Topic: Cryptography and PKI**

Which symmetric algorithm uses variable key lengths of 128, 192, or 256 bits?

A. DES
B. AES
C. RSA
D. MD5

Correct Answer: B

Concept: AES supports 128, 192, and 256-bit keys.

Question 2 — Single Choice | Topic: Cryptography and PKI

How many bits is a DES key?

A. 56
B. 64
C. 128
D. 256

Correct Answer: A

Concept: DES uses a 56-bit effective key length.

Question 3 — Single Choice | Topic: Cryptography and PKI

Which asymmetric algorithm is commonly used for key exchange in TLS?

A. AES
B. RSA
C. SHA-256
D. HMAC

Correct Answer: B

Concept: RSA or Diffie-Hellman are commonly used for TLS key exchange.

Question 4 — Single Choice | Topic: Cryptography and PKI

What is the main disadvantage of symmetric encryption?

A. Slow speed
B. Key distribution challenge
C. Large key sizes
D. No confidentiality

Correct Answer: B

Concept: Distributing the shared secret key securely is the main challenge.

Question 5 — Single Choice | Topic: Cryptography and PKI

Which property ensures a small input change produces a very different hash?

A. Determinism
B. Avalanche effect
C. Fixed output
D. Reversibility

Correct Answer: B

Concept: The avalanche effect means tiny input changes cause large output changes.

Question 6 **Single Choice | Topic: Cryptography and PKI**

Which hash algorithm produces a 128-bit digest?

A. MD5
B. SHA-1
C. SHA-256
D. SHA-512

Correct Answer: A

Concept: MD5 produces a 128-bit digest.

Question 7 **Single Choice | Topic: Cryptography and PKI**

What is HMAC?

A. A hash-based message authentication code
B. A symmetric cipher
C. A key exchange protocol
D. A firewall feature

Correct Answer: A

Concept: HMAC combines a hash function with a secret key to provide message authentication.

Question 8 **Single Choice | Topic: Cryptography and PKI**

Which key is used to create a digital signature?

A. Public key
B. Private key
C. Shared secret
D. Session key

Correct Answer: B

Concept: The signer uses their private key to create a digital signature.

Question 9 **Single Choice | Topic: Cryptography and PKI**

Which key is used to verify a digital signature?

A. Public key
B. Private key
C. Shared secret
D. Master key

Correct Answer: A

Concept: The verifier uses the signer public key to verify the signature.

Question 10 **Single Choice | Topic: Cryptography and PKI**

Which standard defines the format of digital certificates?

A. X.509
B. X.500
C. LDAP
D. DNS

Correct Answer: A

Concept: X.509 is the standard format for public key certificates.

Question 11 **Single Choice | Topic: Cryptography and PKI**

What does OCSP provide?

A. Online certificate status checking
B. Certificate issuance
C. Key generation
D. Email encryption

Correct Answer: A

Concept: OCSP checks certificate revocation status in real time.

Question 12 **Single Choice | Topic: Cryptography and PKI**

In a PKI hierarchy, which certificate is self-signed?

A. End-entity certificate
B. Root CA certificate
C. Intermediate CA certificate
D. Server certificate

Correct Answer: B

Concept: Root CA certificates are self-signed because they are the trust anchor.

Question 13 **Single Choice | Topic: Cryptography and PKI**

Which entity stores issued certificates for retrieval?

A. CA
B. RA
C. Repository
D. CRL

Correct Answer: C

Concept: A repository distributes certificates and CRLs to relying parties.

Question 14 **Single Choice | Topic: Cryptography and PKI**

Which cipher mode provides both confidentiality and authentication?

A. ECB
B. CBC
C. GCM
D. CTR

Correct Answer: C

Concept: Galois/Counter Mode (GCM) provides authenticated encryption.

Question 15 **Single Choice | Topic: Cryptography and PKI**

Which mode should be avoided because identical plaintext blocks produce identical ciphertext?

A. ECB
B. CBC
C. GCM
D. CFB

Correct Answer: A

Concept: ECB reveals patterns in plaintext and is generally insecure.

Question 16 **Single Choice | Topic: Cryptography and PKI**

AES is an example of:

A. Symmetric encryption
B. Asymmetric encryption
C. Hashing
D. Steganography

Correct Answer: A

Concept: AES uses a single shared key for encryption and decryption.

Question 17 **Single Choice | Topic: Cryptography and PKI**

RSA is an example of:

A. Asymmetric encryption
B. Symmetric encryption
C. Hash algorithm
D. Compression algorithm

Correct Answer: A

Concept: RSA uses public and private key pairs.

Question 18 **Single Choice | Topic: Cryptography and PKI**

A hash function produces:

A. A fixed-length digest
B. The original input
C. An encrypted file
D. A random key

Correct Answer: A

Concept: Hash functions map variable input to a fixed-size digest.

Question 19 **Single Choice | Topic: Cryptography and PKI**

MD5 is considered insecure because:

A. Collision vulnerabilities exist
B. It is too slow
C. It produces too long output
D. It requires a public key

Correct Answer: A

Concept: MD5 collision attacks make it unsuitable for security purposes.

Question 20 **Single Choice | Topic: Cryptography and PKI**

A digital certificate binds a public key to:

A. An identity
B. A password
C. A MAC address
D. A VLAN

Correct Answer: A

Concept: Certificates associate a public key with an entity identity.

Question 21 **Single Choice | Topic: Cryptography and PKI**

Certificate revocation status can be checked using:

A. CRL or OCSP
B. DNS only
C. DHCP only
D. SNMP only

Correct Answer: A

Concept: CRLs and OCSP provide revocation status information.

Question 22 **Single Choice | Topic: Cryptography and PKI**

Which concept is described as: uses the same key for encryption and decryption?

A. Private key
B. Symmetric encryption
C. OCSP
D. MD5

Correct Answer: B

Concept: Symmetric encryption refers to the concept where uses the same key for encryption and decryption.

Question 23 **Single Choice | Topic: Cryptography and PKI**

Which concept is described as: uses public and private key pairs?

A. Asymmetric encryption
B. SHA-256
C. CRL
D. Private key

Correct Answer: A

Concept: Asymmetric encryption refers to the concept where uses public and private key pairs.

Question 24 **Single Choice | Topic: Cryptography and PKI**

Which concept is described as: is a symmetric encryption standard?

A. CA
B. AES
C. Symmetric encryption
D. ECB

Correct Answer: B

Concept: AES refers to the concept where is a symmetric encryption standard.

Question 25 **Single Choice | Topic: Cryptography and PKI**

Which concept is described as: is an asymmetric encryption algorithm?

A. Private key
B. RSA
C. X.509
D. OCSP

Correct Answer: B

Concept: RSA refers to the concept where is an asymmetric encryption algorithm.

Question 26 **Single Choice | Topic: Cryptography and PKI**

Which concept is described as: uses a 56-bit effective key?

A. CA
B. Public key
C. RSA
D. DES

Correct Answer: D

Concept: DES refers to the concept where uses a 56-bit effective key.

Question 27 **Single Choice | Topic: Cryptography and PKI**

Which concept is described as: produces a fixed-length digest?

A. Hash function
B. CRL
C. Asymmetric encryption
D. Private key

Correct Answer: A

Concept: Hash function refers to the concept where produces a fixed-length digest.

Question 28 **Single Choice | Topic: Cryptography and PKI**

Which concept is described as: is considered broken due to collision attacks?

A. Hash function
B. Public key
C. RSA
D. MD5

Correct Answer: D

Concept: MD5 refers to the concept where is considered broken due to collision attacks.

Question 29 **Single Choice | Topic: Cryptography and PKI**

Which concept is described as: produces a 256-bit digest?

A. RSA
B. X.509
C. SHA-256
D. AES

Correct Answer: C

Concept: SHA-256 refers to the concept where produces a 256-bit digest.

Question 30 **Single Choice | Topic: Cryptography and PKI**

Which concept is described as: provides message authentication using a hash and key?

A. Symmetric encryption
B. CRL
C. HMAC
D. Root CA certificate

Correct Answer: C

Concept: HMAC refers to the concept where provides message authentication using a hash and key.

Question 31 **Single Choice | Topic: Cryptography and PKI**

Which concept is described as: verifies authenticity and integrity?

A. PKI
B. Digital signature
C. AES
D. Root CA certificate

Correct Answer: B

Concept: Digital signature refers to the concept where verifies authenticity and integrity.

Question 32 **Single Choice | Topic: Cryptography and PKI**

Which concept is described as: is used to verify a digital signature?

A. Symmetric encryption
B. ECB
C. Public key
D. DES

Correct Answer: C

Concept: Public key refers to the concept where is used to verify a digital signature.

Question 33 **Single Choice | Topic: Cryptography and PKI**

Which concept is described as: is used to create a digital signature?

A. DES
B. Private key
C. CA
D. Hash function

Correct Answer: B

Concept: Private key refers to the concept where is used to create a digital signature.

Question 34 **Single Choice | Topic: Cryptography and PKI**

Which concept is described as: manages public keys and certificates?

A. RSA
B. ECB
C. Symmetric encryption
D. PKI

Correct Answer: D

Concept: PKI refers to the concept where manages public keys and certificates.

Question 35 **Single Choice | Topic: Cryptography and PKI**

Which concept is described as: issues and signs digital certificates?

A. SHA-256
B. HMAC
C. CA
D. Symmetric encryption

Correct Answer: C

Concept: CA refers to the concept where issues and signs digital certificates.

Question 36 **Single Choice | Topic: Cryptography and PKI**

Which concept is described as: validates identity before certificate issuance?

A. Asymmetric encryption
B. MD5
C. RA
D. DES

Correct Answer: C

Concept: RA refers to the concept where validates identity before certificate issuance.

Question 37 **Single Choice | Topic: Cryptography and PKI**

Which concept is described as: lists revoked certificates?

A. Asymmetric encryption
B. CRL
C. MD5
D. ECB

Correct Answer: B

Concept: CRL refers to the concept where lists revoked certificates.

Question 38 **Single Choice | Topic: Cryptography and PKI**

Which concept is described as: provides online certificate status checking?

A. OCSP
B. Digital signature
C. Private key
D. HMAC

Correct Answer: A

Concept: OCSP refers to the concept where provides online certificate status checking.

Question 39 **Single Choice | Topic: Cryptography and PKI**

Which concept is described as: is the standard certificate format?

A. Public key
B. RSA
C. X.509
D. Root CA certificate

Correct Answer: C

Concept: X.509 refers to the concept where is the standard certificate format.

Question 40 **Single Choice | Topic: Cryptography and PKI**

Which concept is described as: is self-signed and acts as a trust anchor?

A. Root CA certificate
B. MD5
C. PKI
D. Public key

Correct Answer: A

Concept: Root CA certificate refers to the concept where is self-signed and acts as a trust anchor.

Question 41 **Single Choice | Topic: Cryptography and PKI**

Which concept is described as: provides authenticated encryption?

A. GCM
B. Root CA certificate
C. DES
D. Digital signature

Correct Answer: A

Concept: GCM refers to the concept where provides authenticated encryption.

Question 42 **Single Choice | Topic: Cryptography and PKI**

Which concept is described as: should be avoided because it reveals plaintext patterns?

A. CRL
B. ECB
C. GCM
D. RA

Correct Answer: B

Concept: ECB refers to the concept where should be avoided because it reveals plaintext patterns.

34 Cryptography Basics

Question 1 **Single Choice | Topic: Cryptography Basics**

Which encryption type uses the same key for encryption and decryption?

A. Symmetric encryption
B. Asymmetric encryption
C. Hashing
D. Steganography

Correct Answer: A

Concept: Symmetric encryption uses a single shared key for both encryption and decryption.

Question 2 **Single Choice | Topic: Cryptography Basics**

Which encryption type uses a key pair: public key and private key?

A. Symmetric encryption
B. Asymmetric encryption
C. Hashing
D. Encoding

Correct Answer: B

Concept: Asymmetric encryption uses a public key to encrypt and a private key to decrypt (or vice versa).

Question 3 **Single Choice | Topic: Cryptography Basics**

Which algorithm is a symmetric encryption standard?

A. RSA
B. AES
C. ECC
D. DSA

Correct Answer: B

Concept: AES (Advanced Encryption Standard) is a widely used symmetric encryption algorithm.

Question 4 **Single Choice | Topic: Cryptography Basics**

Which algorithm is an asymmetric encryption algorithm?

A. AES
B. DES
C. RSA
D. 3DES

Correct Answer: C

Concept: RSA is an asymmetric algorithm used for encryption, digital signatures, and key exchange.

35 Hash Algorithms

Question 1 **Single Choice | Topic: Hash Algorithms**

What is a key property of a cryptographic hash function?

A. Reversible output
B. Fixed-length output regardless of input size
C. Requires a public key
D. Uses the same key for hashing and verification

Correct Answer: B

Concept: Hash functions produce a fixed-length digest for any input size.

Question 2 **Single Choice | Topic: Hash Algorithms**

Which hash algorithm produces a 256-bit digest?

A. MD5
B. SHA-1
C. SHA-256
D. SHA-512

Correct Answer: C

Concept: SHA-256 produces a 256-bit hash digest.

Question 3 **Single Choice | Topic: Hash Algorithms**

Which hash algorithm is considered broken and unsuitable for security use?

A. SHA-256
B. MD5
C. SHA-3
D. BLAKE2

Correct Answer: B

Concept: MD5 is vulnerable to collision attacks and should not be used for security-sensitive applications.

36 Digital Signatures

Question 1 | **Single Choice | Topic: Digital Signatures**

What is the purpose of a digital signature?

A. Encrypt all data
B. Verify authenticity and integrity
C. Compress files
D. Hide sender identity

Correct Answer: B

Concept: Digital signatures verify the sender identity and ensure data has not been altered.

37 PKI

Question 1 | **Single Choice | Topic: PKI**

What does PKI stand for?

A. Public Key Infrastructure
B. Private Key Interface
C. Packet Knowledge Index
D. Policy Key Identifier

Correct Answer: A

Concept: PKI provides a framework for managing public keys, certificates, and trust relationships.

Question 2 | **Single Choice | Topic: PKI**

Which entity issues digital certificates in a PKI?

A. Registration Authority
B. Certificate Authority
C. End user
D. Repository

Correct Answer: B

Concept: A Certificate Authority (CA) issues and signs digital certificates.

Question 3 | **Single Choice | Topic: PKI**

Which PKI component validates an applicant identity before certificate issuance?

A. Certificate Authority
B. Registration Authority
C. Repository
D. CRL

Correct Answer: B

Concept: A Registration Authority (RA) verifies identities and forwards requests to the CA.

Question 4 **Single Choice | Topic: PKI**

What is a CRL used for?

A. Storing public keys
B. Listing revoked certificates
C. Encrypting emails
D. Generating hash values

Correct Answer: B

Concept: A Certificate Revocation List (CRL) contains certificates that have been revoked before expiration.

38 VPN

Question 1 **Single Choice | Topic: VPN**

Which VPN deployment connects two fixed sites?

A. Remote access VPN
B. Site-to-site VPN
C. Clientless VPN
D. Host-to-host VPN

Correct Answer: B

Concept: Site-to-site VPNs connect entire networks at different locations.

Question 2 **Single Choice | Topic: VPN**

What is a key advantage of GRE?

A. Built-in encryption
B. Can encapsulate multicast and non-IP protocols
C. Uses TCP
D. Automatic key exchange

Correct Answer: B

Concept: GRE can tunnel multicast and non-IP traffic, making it useful for routing protocols.

Question 3 **Single Choice | Topic: VPN**

Which protocol number is assigned to GRE?

A. 47
B. 50
C. 51
D. 89

Correct Answer: A

Concept: GRE is IP protocol 47.

Question 4 **Single Choice | Topic: VPN**

What does IKE Phase 1 establish?

A. IPsec SA
B. IKE SA
C. Routing table
D. DHCP lease

Correct Answer: B

Concept: IKE Phase 1 establishes a secure IKE SA for protected negotiation.

Question 5 **Single Choice | Topic: VPN**

What does IKE Phase 2 establish?

A. IKE SA
B. IPsec SA
C. SSL session
D. ARP entry

Correct Answer: B

Concept: IKE Phase 2 negotiates the IPsec SA that protects actual data traffic.

Question 6 **Single Choice | Topic: VPN**

Which authentication method does IKE support? (single best answer)

A. Pre-shared key
B. Digital signatures
C. Both pre-shared key and digital signatures
D. None

Correct Answer: C

Concept: IKE supports both pre-shared key and digital signature authentication.

Question 7 **Single Choice | Topic: VPN**

Which Diffie-Hellman group provides stronger key exchange?

A. Group 1
B. Group 2
C. Group 5
D. Group 14 or higher

Correct Answer: D

Concept: Higher Diffie-Hellman groups such as 14 or above provide stronger security.

Question 8 **Single Choice | Topic: VPN**

Which IKE version provides faster SA setup and improved reliability?

A. IKEv1
B. IKEv2
C. IKEv3
D. ISAKMP

Correct Answer: B

Concept: IKEv2 simplifies negotiation and improves resilience compared to IKEv1.

Question 9 **Single Choice | Topic: VPN**

Which UDP ports does L2TP use?

A. 500 and 4500
B. 1701
C. 443
D. 50 and 51

Correct Answer: B

Concept: L2TP uses UDP port 1701.

Question 10 **Single Choice | Topic: VPN**

Why is L2TP often combined with IPsec?

A. To provide encryption
B. To increase speed
C. To reduce overhead
D. To avoid tunnels

Correct Answer: A

Concept: L2TP lacks encryption, so IPsec is added for confidentiality and integrity.

Question 11 **Single Choice | Topic: VPN**

Which SSL VPN mode grants the remote host a virtual IP on the internal network?

A. Web proxy
B. Network extension
C. File sharing
D. Port forwarding

Correct Answer: B

Concept: Network extension mode assigns a virtual IP and routes traffic into the corporate network.

Question 12 **Single Choice | Topic: VPN**

Which SSL VPN mode is best for accessing internal web apps without installing a client?

A. Web proxy
B. Network extension
C. IPsec
D. L2TP

Correct Answer: A

Concept: Web proxy mode requires only a browser.

Question 13 **Single Choice | Topic: VPN**

Which encapsulation mode is typically used for site-to-site VPNs?

A. Transport mode
B. Tunnel mode
C. Bridge mode
D. Trunk mode

Correct Answer: B

Concept: Site-to-site VPNs typically use tunnel mode to protect the entire original packet.

Question 14 **Single Choice | Topic: VPN**

Which tunneling protocol is considered a Layer 2 VPN technology?

A. GRE
B. IPsec
C. L2TP
D. SSL VPN

Correct Answer: C

Concept: L2TP operates at Layer 2, while GRE, IPsec, and SSL VPN operate at higher layers.

Question 15 **Single Choice | Topic: VPN**

Which component defines the security parameters for an IPsec connection?

A. SPD
B. SA
C. NAT
D. ARP

Correct Answer: B

Concept: A Security Association (SA) defines algorithms, keys, and lifetimes for IPsec.

Question 16 **Single Choice | Topic: VPN**

The main goal of a VPN is to provide:

A. Secure communication over public networks
B. Faster Internet speed
C. Free public Wi-Fi
D. Unencrypted remote access

Correct Answer: A

Concept: VPNs create secure encrypted tunnels over untrusted networks.

Question 17 **Single Choice | Topic: VPN**

GRE protocol number is:

A. 47
B. 50
C. 51
D. 89

Correct Answer: A

Concept: GRE is encapsulated using IP protocol 47.

Question 18 **Single Choice | Topic: VPN**

IPsec ESP provides:

A. Encryption and authentication
B. Only routing
C. Only NAT
D. Only compression

Correct Answer: A

Concept: ESP provides confidentiality, integrity, and authentication.

Question 19 **Single Choice | Topic: VPN**

IPsec AH provides:

A. Authentication and integrity without encryption
B. Encryption only
C. Compression only
D. Key exchange

Correct Answer: A

Concept: AH authenticates and protects integrity but does not encrypt.

Question 20 **Single Choice | Topic: VPN**

IKE negotiates:

A. Security associations and keys for IPsec
B. DHCP leases
C. DNS records
D. MAC addresses

Correct Answer: A

Concept: IKE handles key exchange and SA negotiation.

Question 21 **Single Choice | Topic: VPN**

SSL VPN commonly uses port:

A. 80
B. 443
C. 22
D. 1701

Correct Answer: B

Concept: SSL VPN uses HTTPS/TCP 443.

Question 22 **Single Choice | Topic: VPN**

L2TP is commonly paired with which protocol for encryption?

A. IPsec
B. GRE
C. OSPF
D. BGP

Correct Answer: A

Concept: L2TP provides tunneling and is paired with IPsec for encryption.

Question 23 **Single Choice | Topic: VPN**

Which concept is described as: provides secure communication over public networks?

A. VPN
B. Tunnel mode
C. AH
D. Transport mode

Correct Answer: A

Concept: VPN refers to the concept where provides secure communication over public networks.

Question 24 **Single Choice | Topic: VPN**

Which concept is described as: connects entire networks at different locations?

A. AH
B. VPN
C. Site-to-site VPN
D. L2TP over IPsec

Correct Answer: C

Concept: Site-to-site VPN refers to the concept where connects entire networks at different locations.

Question 25 **Single Choice | Topic: VPN**

Which concept is described as: connects individual users to a corporate network?

A. Tunnel mode
B. IPsec
C. IKEv2
D. Remote-access VPN

Correct Answer: D

Concept: Remote-access VPN refers to the concept where connects individual users to a corporate network.

Question 26 **Single Choice | Topic: VPN**

Which concept is described as: is IP protocol 47 and lacks built-in encryption?

A. AH
B. L2TP
C. Site-to-site VPN
D. GRE

Correct Answer: D

Concept: GRE refers to the concept where is IP protocol 47 and lacks built-in encryption.

Question 27 **Single Choice | Topic: VPN**

Which concept is described as: can tunnel multicast and non-IP protocols?

A. L2TP over IPsec
B. Web proxy mode
C. IKE
D. GRE

Correct Answer: D

Concept: GRE refers to the concept where can tunnel multicast and non-IP protocols.

Question 28 **Single Choice | Topic: VPN**

Which concept is described as: provides confidentiality, integrity, and authentication?

A. Web proxy mode
B. Site-to-site VPN
C. IKE
D. IPsec

Correct Answer: D

Concept: IPsec refers to the concept where provides confidentiality, integrity, and authentication.

Question 29 **Single Choice | Topic: VPN**

Which concept is described as: provides encryption and authentication?

A. IKE
B. SSL VPN
C. VPN
D. ESP

Correct Answer: D

Concept: ESP refers to the concept where provides encryption and authentication.

Question 30 **Single Choice | Topic: VPN**

Which concept is described as: provides authentication without encryption?

A. AH
B. GRE
C. Tunnel mode
D. Transport mode

Correct Answer: A

Concept: AH refers to the concept where provides authentication without encryption.

Question 31 **Single Choice | Topic: VPN**

Which concept is described as: negotiates IPsec keys and security associations?

A. IPsec
B. Site-to-site VPN
C. IKE
D. AH

Correct Answer: C

Concept: IKE refers to the concept where negotiates IPsec keys and security associations.

Question 32 **Single Choice | Topic: VPN**

Which concept is described as: encrypts the entire original IP packet?

A. Tunnel mode
B. GRE
C. Remote-access VPN
D. Site-to-site VPN

Correct Answer: A

Concept: Tunnel mode refers to the concept where encrypts the entire original IP packet.

Question 33 **Single Choice | Topic: VPN**

Which concept is described as: encrypts only the IP payload?

A. AH
B. Remote-access VPN
C. Site-to-site VPN
D. Transport mode

Correct Answer: D

Concept: Transport mode refers to the concept where encrypts only the IP payload.

Question 34 **Single Choice | Topic: VPN**

Which concept is described as: uses UDP port 1701 and lacks encryption?

A. L2TP
B. Network extension mode
C. Site-to-site VPN
D. Tunnel mode

Correct Answer: A

Concept: L2TP refers to the concept where uses UDP port 1701 and lacks encryption.

Question 35 **Single Choice | Topic: VPN**

Which concept is described as: combines L2TP tunneling with IPsec encryption?

A. IKEv2
B. L2TP over IPsec
C. IKE
D. Network extension mode

Correct Answer: B

Concept: L2TP over IPsec refers to the concept where combines L2TP tunneling with IPsec encryption.

Question 36 Single Choice | Topic: VPN

Which concept is described as: commonly uses TCP port 443?

A. Remote-access VPN
B. AH
C. Tunnel mode
D. SSL VPN

Correct Answer: D

Concept: SSL VPN refers to the concept where commonly uses TCP port 443.

Question 37 Single Choice | Topic: VPN

Which concept is described as: provides browser-based access to web apps?

A. Transport mode
B. Web proxy mode
C. Tunnel mode
D. GRE

Correct Answer: B

Concept: Web proxy mode refers to the concept where provides browser-based access to web apps.

Question 38 Single Choice | Topic: VPN

Which concept is described as: assigns a virtual IP to the remote host?

A. Network extension mode
B. IKE
C. GRE
D. Site-to-site VPN

Correct Answer: A

Concept: Network extension mode refers to the concept where assigns a virtual IP to the remote host.

Question 39 Single Choice | Topic: VPN

Which concept is described as: is faster and more reliable than IKEv1?

A. Transport mode
B. IKEv2
C. GRE
D. VPN

Correct Answer: B

Concept: IKEv2 refers to the concept where is faster and more reliable than IKEv1.

39 VPN Overview

Question 1 Single Choice | Topic: VPN Overview

What does VPN stand for?

A. Virtual Private Network
B. Virtual Public Network
C. Verified Private Node
D. Virtual Protocol Network

Correct Answer: A

Concept: A VPN extends a private network across a public network using encryption and tunneling.

Question 2 Single Choice | Topic: VPN Overview

Which benefit does a VPN provide over the public Internet?

A. Higher physical speed
B. Confidentiality and secure remote access
C. Free bandwidth
D. No encryption needed

Correct Answer: B

Concept: VPNs provide confidentiality, integrity, and secure access over untrusted networks.

Question 3 Single Choice | Topic: VPN Overview

Which VPN type connects individual remote users to a corporate network?

A. Site-to-site VPN
B. Remote-access VPN
C. Intranet VPN
D. Extranet VPN

Correct Answer: B

Concept: Remote-access VPNs allow individual users to securely connect to a corporate network.

40 GRE VPN

Question 1 Single Choice | Topic: GRE VPN

Which protocol does GRE use as its transport protocol?

A. TCP
B. UDP
C. IP protocol 47
D. IP protocol 50

Correct Answer: C

Concept: GRE encapsulates packets using IP protocol number 47.

Question 2 **Single Choice | Topic: GRE VPN**

Which statement about GRE is correct?

A. GRE provides encryption by default
B. GRE is a simple tunneling protocol without built-in encryption
C. GRE cannot tunnel multicast
D. GRE uses UDP port 500

Correct Answer: B

Concept: GRE provides encapsulation and tunneling but does not provide encryption or authentication by itself.

41 IPsec VPN

Question 1 **Single Choice | Topic: IPsec VPN**

Which IPsec protocol provides encryption and confidentiality?

A. AH
B. ESP
C. IKE
D. GRE

Correct Answer: B

Concept: ESP (Encapsulating Security Payload) provides encryption, authentication, and integrity.

Question 2 **Single Choice | Topic: IPsec VPN**

Which IPsec protocol provides authentication and integrity but not encryption?

A. AH
B. ESP
C. IKE
D. L2TP

Correct Answer: A

Concept: AH (Authentication Header) provides source authentication and integrity but does not encrypt payloads.

Question 3 **Single Choice | Topic: IPsec VPN**

Which protocol is used to establish and manage IPsec security associations?

A. AH
B. ESP
C. IKE
D. GRE

Correct Answer: C

Concept: IKE (Internet Key Exchange) negotiates keys and Security Associations for IPsec.

Question 4 **Single Choice | Topic: IPsec VPN**

Which IPsec mode encrypts the entire original IP packet and adds a new IP header?

A. Transport mode
B. Tunnel mode
C. Gre mode
D. Bridge mode

Correct Answer: B

Concept: Tunnel mode encapsulates and encrypts the entire original packet, adding a new IP header.

Question 5 **Single Choice | Topic: IPsec VPN**

Which IPsec mode encrypts only the payload and is used for host-to-host communications?

A. Transport mode
B. Tunnel mode
C. Bridge mode
D. Trunk mode

Correct Answer: A

Concept: Transport mode encrypts only the IP payload, leaving the original IP header intact.

42 L2TP VPN

Question 1 **Single Choice | Topic: L2TP VPN**

Which layer-2 tunneling protocol is often combined with IPsec for encryption?

A. PPTP
B. L2TP
C. GRE
D. MPLS

Correct Answer: B

Concept: L2TP provides tunneling and is commonly paired with IPsec to provide encryption.

43 SSL VPN

Question 1 **Single Choice | Topic: SSL VPN**

Which VPN type is commonly accessed through a web browser without a dedicated client?

A. IPsec VPN
B. SSL VPN
C. GRE VPN
D. L2TP VPN

Correct Answer: B

Concept: SSL VPN can be accessed via a web browser, making it convenient for remote users.

Question 2 — **Single Choice | Topic: SSL VPN**

Which SSL VPN mode provides access to specific web applications through a browser?

A. Web proxy mode
B. Network extension mode
C. File sharing mode
D. Full tunnel mode

Correct Answer: A

Concept: Web proxy mode allows browser-based access to internal web applications.

PART II

Multiple-Answer Questions

44 Network Security Concepts

Question 1 — **Multiple Choice | Topic: Network Security Concepts**

Which of the following are characteristics of network security? (Choose all that apply)

A. Confidentiality
B. Integrity
C. Availability
D. Controllability
E. Non-repudiation

Correct Answer: A,B,C,D,E

Concept: Network security commonly encompasses confidentiality, integrity, availability, controllability, and non-repudiation.

Question 2 — **Multiple Choice | Topic: Network Security Concepts**

Which are stages in the development history of network security? (Choose all that apply)

A. Communication security period
B. Information security period
C. Information assurance period
D. Cyberspace security period

Correct Answer: A,B,C,D

Concept: Network security has evolved through communication security, information security, information assurance, and cyberspace security periods.

Question 3 **Multiple Choice | Topic: Network Security Concepts**

Which are core information security properties? (Choose all that apply)

A. Confidentiality
B. Integrity
C. Availability
D. Controllability
E. Non-repudiation

Correct Answer: A,B,C,D,E

Concept: Network security includes confidentiality, integrity, availability, controllability, and non-repudiation.

45 Network Security Standards

Question 1 **Multiple Choice | Topic: Network Security Standards**

Which are typical Classified Protection 2.0 lifecycle phases? (Choose all that apply)

A. Classification
B. Filing
C. Construction and rectification
D. Level assessment
E. Supervision and inspection

Correct Answer: A,B,C,D,E

Concept: The full lifecycle includes classification, filing, construction/rectification, level assessment, and supervision/inspection.

Question 2 **Multiple Choice | Topic: Network Security Standards**

Which are benefits of implementing ISO 27001? (Choose all that apply)

A. Systematic risk management
B. Improved stakeholder confidence
C. Legal/regulatory compliance
D. Guaranteed zero attacks

Correct Answer: A,B,C

Concept: ISO 27001 helps manage risk, build confidence, and comply with regulations, but cannot guarantee zero attacks.

Question 3 **Multiple Choice | Topic: Network Security Standards**

Which are typical Classified Protection 2.0 security extension requirements? (Choose all that apply)

A. Cloud computing
B. Mobile Internet
C. Internet of Things
D. Industrial control systems

Correct Answer: A,B,C,D

Concept: Classified Protection 2.0 extends coverage to cloud, mobile Internet, IoT, and industrial control scenarios.

Question 4 **Multiple Choice | Topic: Network Security Standards**

Which belong to the Classified Protection lifecycle? (Choose all that apply)

A. Classification
B. Filing
C. Construction and rectification
D. Level assessment
E. Supervision and inspection

Correct Answer: A,B,C,D,E

Concept: The full lifecycle includes classification, filing, construction, level assessment, and supervision.

Question 5 **Multiple Choice | Topic: Network Security Standards**

Which are ISO 27000 family standards? (Choose all that apply)

A. ISO 27001
B. ISO 27002
C. ISO 27005
D. ISO 9001

Correct Answer: A,B,C

Concept: ISO 27001, 27002, and 27005 are information security standards; ISO 9001 is quality management.

Question 6 **Multiple Choice | Topic: Network Security Standards**

Which of the following relate to ISO? (Choose all that apply)

A. ISO 27001
B. ISO 27002
C. ISO 27001:2022

Correct Answer: A,B,C

Concept: These concepts all relate to ISO in Network Security Standards.

Question 7 — Multiple Choice | Topic: Network Security Standards

Which of the following relate to Classified? (Choose all that apply)

A. Classified Protection 2.0
B. Classified Protection levels
C. Classified Protection lifecycle

Correct Answer: A,B,C

Concept: These concepts all relate to Classified in Network Security Standards.

46 Network Fundamentals

Question 1 — Multiple Choice | Topic: Network Fundamentals

Which layers are included in the TCP/IP equivalent model? (Choose all that apply)

A. Application
B. Transport
C. Network
D. Data link
E. Physical

Correct Answer: A,B,C,D,E

Concept: The TCP/IP equivalent model includes application, transport, network, data link, and physical layers.

Question 2 — Multiple Choice | Topic: Network Fundamentals

Which are connectionless protocols? (Choose all that apply)

A. UDP
B. IP
C. ICMP
D. TCP

Correct Answer: A,B,C

Concept: UDP, IP, and ICMP are connectionless; TCP is connection-oriented.

Question 3 — Multiple Choice | Topic: Network Fundamentals

Which fields are in a TCP header? (Choose all that apply)

A. Source port
B. Destination port
C. Sequence number
D. Acknowledgment number

Correct Answer: A,B,C,D

Concept: TCP headers include source/destination ports, sequence and acknowledgment numbers, flags, window size, etc.

Question 4 **Multiple Choice | Topic: Network Fundamentals**

Which devices operate primarily at Layer 1? (Choose all that apply)

A. Hub
B. Repeater
C. Switch
D. Router

Correct Answer: A,B

Concept: Hubs and repeaters operate at the physical layer.

Question 5 **Multiple Choice | Topic: Network Fundamentals**

Which protocols use TCP? (Choose all that apply)

A. HTTP
B. FTP
C. SMTP
D. DNS (sometimes)

Correct Answer: A,B,C,D

Concept: HTTP, FTP, and SMTP use TCP; DNS primarily uses UDP but can use TCP for zone transfers and large responses.

Question 6 **Multiple Choice | Topic: Network Fundamentals**

Which protocols use UDP? (Choose all that apply)

A. DNS
B. TFTP
C. SNMP
D. DHCP

Correct Answer: A,B,C,D

Concept: DNS, TFTP, SNMP, and DHCP commonly use UDP.

Question 7 **Multiple Choice | Topic: Network Fundamentals**

Which are valid IPv4 private address ranges? (Choose all that apply)

A. 10.0.0.0/8
B. 172.16.0.0/12
C. 192.168.0.0/16
D. 224.0.0.0/4

Correct Answer: A,B,C

Concept: RFC 1918 private ranges are 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16.

Question 8 **Multiple Choice | Topic: Network Fundamentals**

Which are OSI layers? (Choose all that apply)

A. Physical
B. Data link
C. Network
D. Transport
E. Application

Correct Answer: A,B,C,D,E

Concept: The OSI model includes these layers plus session and presentation.

Question 9 **Multiple Choice | Topic: Network Fundamentals**

Which are TCP/IP equivalent model layers? (Choose all that apply)

A. Application
B. Transport
C. Network
D. Data link
E. Physical

Correct Answer: A,B,C,D,E

Concept: The TCP/IP equivalent model includes all five layers.

Question 10 **Multiple Choice | Topic: Network Fundamentals**

Which protocols use TCP? (Choose all that apply)

A. HTTP
B. FTP
C. SMTP
D. SSH

Correct Answer: A,B,C,D

Concept: HTTP, FTP, SMTP, and SSH all use TCP.

Question 11 **Multiple Choice | Topic: Network Fundamentals**

Which devices operate at Layer 2? (Choose all that apply)

A. Switch
B. Bridge
C. Router
D. Hub

Correct Answer: A,B

Concept: Switches and bridges operate at the data link layer.

Question 12 — **Multiple Choice | Topic: Network Fundamentals**

Which of the following relate to TCP/IP? (Choose all that apply)

A. TCP/IP standard model
B. TCP/IP equivalent model

Correct Answer: A,B

Concept: These concepts all relate to TCP/IP in Network Fundamentals.

47 OSI and TCP/IP Models

Question 1 — **Multiple Choice | Topic: OSI and TCP/IP Models**

Which layers are part of the OSI model? (Choose all that apply)

A. Physical
B. Data link
C. Network
D. Transport
E. Application

Correct Answer: A,B,C,D,E

Concept: The OSI model includes physical, data link, network, transport, session, presentation, and application layers.

48 Network Protocols

Question 1 — **Multiple Choice | Topic: Network Protocols**

Which protocols operate at the application layer? (Choose all that apply)

A. HTTP
B. FTP
C. DNS
D. SMTP

Correct Answer: A,B,C,D

Concept: HTTP, FTP, DNS, and SMTP are all application-layer protocols.

49 Network Devices

Question 1 — Multiple Choice | Topic: Network Devices

Which devices can operate at Layer 2? (Choose all that apply)

A. Switch
B. Bridge
C. Router
D. Hub

Correct Answer: A,B

Concept: Switches and bridges operate at the data link layer; routers are Layer 3 and hubs are Layer 1.

50 Enterprise Security Threats

Question 1 — Multiple Choice | Topic: Enterprise Security Threats

Which are examples of malware? (Choose all that apply)

A. Virus
B. Worm
C. Trojan
D. Ransomware
E. Spyware

Correct Answer: A,B,C,D,E

Concept: All listed items are types of malware.

Question 2 — Multiple Choice | Topic: Enterprise Security Threats

Which are social engineering attacks? (Choose all that apply)

A. Phishing
B. Pretexting
C. Baiting
D. Tailgating

Correct Answer: A,B,C,D

Concept: Social engineering manipulates people through phishing, pretexting, baiting, tailgating, etc.

Question 3 — Multiple Choice | Topic: Enterprise Security Threats

Which protect communication confidentiality? (Choose all that apply)

A. Encryption
B. VPN
C. TLS/SSL
D. MAC filtering

Correct Answer: A,B,C

Concept: Encryption, VPNs, and TLS/SSL protect confidentiality; MAC filtering controls device access.

Question 4 | **Multiple Choice | Topic: Enterprise Security Threats**

Which are typical security zones? (Choose all that apply)

- **A.** Trust
- **B.** Untrust
- **C.** DMZ
- **D.** Local

Correct Answer: A,B,C,D

Concept: Huawei firewalls define Trust, Untrust, DMZ, and Local security zones by default.

Question 5 | **Multiple Choice | Topic: Enterprise Security Threats**

Which are endpoint security controls? (Choose all that apply)

- **A.** Antivirus
- **B.** Host firewall
- **C.** Application whitelisting
- **D.** Full-disk encryption

Correct Answer: A,B,C,D

Concept: Endpoint security includes antivirus, host firewalls, whitelisting, and encryption.

Question 6 | **Multiple Choice | Topic: Enterprise Security Threats**

Which are malware types? (Choose all that apply)

- **A.** Virus
- **B.** Worm
- **C.** Trojan
- **D.** Ransomware

Correct Answer: A,B,C,D

Concept: All listed are categories of malware.

Question 7 | **Multiple Choice | Topic: Enterprise Security Threats**

Which are denial-of-service attacks? (Choose all that apply)

- **A.** SYN flood
- **B.** DDoS
- **C.** Ping of death
- **D.** Smurf attack

Correct Answer: A,B,C,D

Concept: These attacks aim to disrupt service availability.

51 Zone Border Security

Question 1 Multiple Choice | Topic: Zone Border Security

Which controls are typically applied at zone borders? (Choose all that apply)

A. Firewall policies
B. NAT
C. IPS/IDS
D. VPN termination

Correct Answer: A,B,C,D

Concept: Zone borders commonly use firewalls, NAT, IPS/IDS, and VPN gateways for protection.

52 Firewall Basics

Question 1 Multiple Choice | Topic: Firewall Basics

Which are Huawei firewall default security zones? (Choose all that apply)

A. Trust
B. Untrust
C. DMZ
D. Local

Correct Answer: A,B,C,D

Concept: Huawei firewalls provide Trust, Untrust, DMZ, and Local zones by default.

Question 2 Multiple Choice | Topic: Firewall Basics

Which actions can a security policy perform? (Choose all that apply)

A. Permit
B. Deny
C. Log
D. Redirect

Correct Answer: A,B,C

Concept: Policies can permit, deny, and log traffic; redirect may be available in specific features.

Question 3 Multiple Choice | Topic: Firewall Basics

Which are common firewall deployment modes? (Choose all that apply)

A. Routing mode
B. Transparent mode
C. Hybrid mode
D. Bridge mode

Correct Answer: A,B,C

Concept: Firewalls commonly deploy in routing, transparent (bridge), and hybrid modes.

Question 4 **Multiple Choice | Topic: Firewall Basics**

Which conditions can be matched in a security policy? (Choose all that apply)

A. Source zone
B. Destination zone
C. Source IP
D. Destination port
E. User

Correct Answer: A,B,C,D,E

Concept: Policies can match zones, IPs, ports, users, applications, services, and time.

Question 5 **Multiple Choice | Topic: Firewall Basics**

Which are default Huawei security zones? (Choose all that apply)

A. Trust
B. Untrust
C. DMZ
D. Local

Correct Answer: A,B,C,D

Concept: Huawei firewalls provide Trust, Untrust, DMZ, and Local zones by default.

Question 6 **Multiple Choice | Topic: Firewall Basics**

Which can be matched in security policies? (Choose all that apply)

A. Source zone
B. Destination zone
C. Source IP
D. Destination port
E. User

Correct Answer: A,B,C,D,E

Concept: Policies can match zones, addresses, ports, users, services, applications, and time.

Question 7 **Multiple Choice | Topic: Firewall Basics**

Which of the following relate to Security? (Choose all that apply)

A. Security zones
B. Security policy logging

Correct Answer: A,B

Concept: These concepts all relate to Security in Firewall Basics.

53 Security Zones

Question 1 — Multiple Choice | Topic: Security Zones

Which are default security zones on Huawei firewalls? (Choose all that apply)

A. Trust
B. Untrust
C. DMZ
D. Local

Correct Answer: A,B,C,D

Concept: Huawei firewalls provide four default zones: Trust, Untrust, DMZ, and Local.

54 Security Policies

Question 1 — Multiple Choice | Topic: Security Policies

Which elements can be referenced in a security policy? (Choose all that apply)

A. Security zones
B. IP addresses/address groups
C. Services
D. Users and user groups
E. Time ranges

Correct Answer: A,B,C,D,E

Concept: Security policies can reference zones, addresses, services, users, applications, and time ranges.

55 Stateful Inspection

Question 1 — Multiple Choice | Topic: Stateful Inspection

Which are advantages of stateful inspection over simple packet filtering? (Choose all that apply)

A. Tracks connection state
B. Allows return traffic automatically
C. Better performance for established sessions
D. Can inspect application-layer payloads

Correct Answer: A,B,C

Concept: Stateful inspection tracks connections, permits return traffic, and improves performance; deep payload inspection is more advanced.

56 NAT

Question 1 **Multiple Choice | Topic: NAT**

Which are private IPv4 address ranges? (Choose all that apply)

A. 10.0.0.0/8
B. 172.16.0.0/12
C. 192.168.0.0/16
D. 127.0.0.0/8

Correct Answer: A,B,C

Concept: RFC 1918 private ranges are 10/8, 172.16/12, and 192.168/16.

Question 2 **Multiple Choice | Topic: NAT**

Which are valid uses of NAT Server? (Choose all that apply)

A. Publishing a web server
B. Publishing an email server
C. Remote access to an internal host
D. Hiding internal topology

Correct Answer: A,B,C,D

Concept: NAT Server can publish services, enable remote access, and obscure internal addressing.

Question 3 **Multiple Choice | Topic: NAT**

Which are NAT types? (Choose all that apply)

A. Source NAT
B. Destination NAT
C. Bidirectional NAT
D. NAT Server

Correct Answer: A,B,C,D

Concept: Huawei firewalls support all four NAT types.

Question 4 **Multiple Choice | Topic: NAT**

Which source NAT modes exist? (Choose all that apply)

A. NAT No-PAT
B. NAPT
C. Easy-IP
D. NAT Server

Correct Answer: A,B,C

Concept: NAT Server is not a source NAT mode.

Question 5 **Multiple Choice | Topic: NAT**

Which of the following relate to NAT? (Choose all that apply)

A. NAT
B. NAT Server
C. NAT No-PAT
D. NAT ALG

Correct Answer: A,B,C,D

Concept: These concepts all relate to NAT in NAT.

57 NAT Basics

Question 1 **Multiple Choice | Topic: NAT Basics**

Which are types of NAT supported by Huawei firewalls? (Choose all that apply)

A. Source NAT
B. Destination NAT
C. Bidirectional NAT
D. NAT Server

Correct Answer: A,B,C,D

Concept: Huawei firewalls support source NAT, destination NAT, bidirectional NAT, and NAT server.

58 Source NAT

Question 1 **Multiple Choice | Topic: Source NAT**

Which are common source NAT modes? (Choose all that apply)

A. NAT No-PAT
B. NAPT
C. Easy-IP
D. NAT Server

Correct Answer: A,B,C

Concept: Source NAT includes NAT No-PAT (one-to-one), NAPT (many-to-one), and Easy-IP (interface-based).

59 NAT ALG

Question 1 | **Multiple Choice | Topic: NAT ALG**

Which protocols may require NAT ALG? (Choose all that apply)

A. FTP
B. SIP
C. H.323
D. DNS

Correct Answer: A,B,C

Concept: Protocols embedding address/port information in payloads (FTP, SIP, H.323) need NAT ALG.

60 Firewall Hot Standby

Question 1 | **Multiple Choice | Topic: Firewall Hot Standby**

Which are VRRP components? (Choose all that apply)

A. Virtual IP
B. Virtual MAC
C. Master router
D. Backup router

Correct Answer: A,B,C,D

Concept: VRRP uses a virtual IP, virtual MAC, master, and backup routers.

Question 2 | **Multiple Choice | Topic: Firewall Hot Standby**

Which are HRP synchronization modes? (Choose all that apply)

A. Real-time
B. Fast
C. Batch
D. Auto

Correct Answer: A,B,C

Concept: HRP supports real-time, fast, and batch synchronization modes.

Question 3 | **Multiple Choice | Topic: Firewall Hot Standby**

Which are valid hot-standby modes? (Choose all that apply)

A. Active/standby
B. Load sharing
C. Cluster
D. Standalone

Correct Answer: A,B

Concept: Huawei firewall hot standby supports active/standby and load-sharing modes.

Question 4 **Multiple Choice | Topic: Firewall Hot Standby**

Which items should be checked when HRP is not synchronizing? (Choose all that apply)

A. Heartbeat link status
B. HRP enable status
C. VGMP priority
D. Configuration consistency

Correct Answer: A,B,C,D

Concept: Check heartbeat links, HRP status, priorities, and configuration consistency.

Question 5 **Multiple Choice | Topic: Firewall Hot Standby**

Which are HRP synchronization contents? (Choose all that apply)

A. Session table
B. NAT entries
C. Server-map entries
D. Configuration

Correct Answer: A,B,C,D

Concept: HRP synchronizes all these items between peers.

Question 6 **Multiple Choice | Topic: Firewall Hot Standby**

Which are VRRP states? (Choose all that apply)

A. Master
B. Backup
C. Initialize
D. Failover

Correct Answer: A,B,C

Concept: VRRP states include master, backup, and initialize.

Question 7 **Multiple Choice | Topic: Firewall Hot Standby**

Which of the following relate to VRRP? (Choose all that apply)

A. VRRP
B. VRRP master
C. VRRP backup

Correct Answer: A,B,C

Concept: These concepts all relate to VRRP in Firewall Hot Standby.

Question 8 — **Multiple Choice | Topic: Firewall Hot Standby**

Which of the following relate to HRP? (Choose all that apply)

A. HRP
B. HRP session backup

Correct Answer: A,B

Concept: These concepts all relate to HRP in Firewall Hot Standby.

61 VRRP

Question 1 — **Multiple Choice | Topic: VRRP**

Which are VRRP roles? (Choose all that apply)

A. Master
B. Backup
C. Initialize
D. Failover

Correct Answer: A,B,C

Concept: VRRP devices can be in master, backup, or initialize states.

62 HRP

Question 1 — **Multiple Choice | Topic: HRP**

Which data does HRP synchronize? (Choose all that apply)

A. Session table
B. NAT entries
C. Server-map entries
D. Configurations

Correct Answer: A,B,C,D

Concept: HRP synchronizes sessions, NAT entries, server-map entries, and configurations.

63 Intrusion Prevention and Antivirus

Question 1 **Multiple Choice | Topic: Intrusion Prevention and Antivirus**

Which are common attack phases? (Choose all that apply)

A. Reconnaissance
B. Exploitation
C. Lateral movement
D. Exfiltration

Correct Answer: A,B,C,D

Concept: Cyber kill chain phases include reconnaissance, exploitation, lateral movement, and exfiltration.

Question 2 **Multiple Choice | Topic: Intrusion Prevention and Antivirus**

Which are types of intrusion detection/prevention systems? (Choose all that apply)

A. NIDS
B. NIPS
C. HIDS
D. HIPS

Correct Answer: A,B,C,D

Concept: Network and host-based IDS/IPS exist for different monitoring scopes.

Question 3 **Multiple Choice | Topic: Intrusion Prevention and Antivirus**

Which factors should be considered when tuning IPS signatures? (Choose all that apply)

A. False positive rate
B. False negative rate
C. Performance impact
D. Business context

Correct Answer: A,B,C,D

Concept: Effective tuning balances detection accuracy, performance, and business needs.

Question 4 **Multiple Choice | Topic: Intrusion Prevention and Antivirus**

Which are methods to improve malware detection? (Choose all that apply)

A. Keep signatures updated
B. Use heuristics
C. Deploy sandboxing
D. User education

Correct Answer: A,B,C,D

Concept: Updated signatures, heuristics, sandboxing, and user awareness improve detection.

Question 5 **Multiple Choice | Topic: Intrusion Prevention and Antivirus**

Which are IPS deployment modes? (Choose all that apply)

A. Inline
B. Tap/SPAN
C. Out-of-band
D. Bridge

Correct Answer: A,B,C

Concept: IPS can be inline, tapped, or out-of-band.

Question 6 **Multiple Choice | Topic: Intrusion Prevention and Antivirus**

Which are antivirus detection methods? (Choose all that apply)

A. Signature-based
B. Heuristic
C. Sandboxing
D. Behavioral analysis

Correct Answer: A,B,C,D

Concept: Modern antivirus uses all these techniques.

Question 7 **Multiple Choice | Topic: Intrusion Prevention and Antivirus**

Which of the following relate to False? (Choose all that apply)

A. False positive
B. False negative

Correct Answer: A,B

Concept: These concepts all relate to False in Intrusion Prevention and Antivirus.

64 Intrusion Prevention

Question 1 **Multiple Choice | Topic: Intrusion Prevention**

Which are common IPS actions? (Choose all that apply)

A. Block
B. Permit
C. Reset connection
D. Alert/log

Correct Answer: A,C,D

Concept: IPS can block malicious traffic, reset connections, and alert/log events.

65 Antivirus

Question 1 | Multiple Choice | Topic: Antivirus

Which are antivirus detection techniques? (Choose all that apply)

A. Signature-based
B. Heuristic
C. Sandboxing
D. Behavioral analysis

Correct Answer: A,B,C,D

Concept: Modern antivirus uses signatures, heuristics, sandboxing, and behavioral analysis.

66 AAA and User Authentication

Question 1 | Multiple Choice | Topic: AAA and User Authentication

Which are AAA functions? (Choose all that apply)

A. Authentication
B. Authorization
C. Accounting
D. Alerting

Correct Answer: A,B,C

Concept: AAA provides authentication, authorization, and accounting.

Question 2 | Multiple Choice | Topic: AAA and User Authentication

Which can be authentication sources on Huawei firewalls? (Choose all that apply)

A. Local database
B. RADIUS server
C. HWTACACS server
D. LDAP server

Correct Answer: A,B,C,D

Concept: Firewalls can authenticate against local, RADIUS, HWTACACS, and LDAP sources.

Question 3 | Multiple Choice | Topic: AAA and User Authentication

Which are differences between RADIUS and TACACS+? (Choose all that apply)

A. RADIUS uses UDP; TACACS+ uses TCP
B. RADIUS combines auth/authz; TACACS+ separates them
C. RADIUS encrypts only password; TACACS+ encrypts entire payload
D. RADIUS is Cisco proprietary

Correct Answer: A,B,C

Concept: RADIUS uses UDP and combines auth/authz; TACACS+ uses TCP and separates AAA; only the password is encrypted in RADIUS.

Question 4 **Multiple Choice | Topic: AAA and User Authentication**

Which are AAA components? (Choose all that apply)

A. Authentication
B. Authorization
C. Accounting
D. Auditing

Correct Answer: A,B,C

Concept: AAA consists of authentication, authorization, and accounting.

Question 5 **Multiple Choice | Topic: AAA and User Authentication**

Which can authenticate users on a Huawei firewall? (Choose all that apply)

A. Local database
B. RADIUS
C. HWTACACS
D. LDAP

Correct Answer: A,B,C,D

Concept: Firewalls support all these authentication sources.

67 AAA Basics

Question 1 **Multiple Choice | Topic: AAA Basics**

Which are components of AAA? (Choose all that apply)

A. Authentication
B. Authorization
C. Accounting
D. Auditing

Correct Answer: A,B,C

Concept: AAA consists of authentication, authorization, and accounting.

68 Firewall User Authentication

Question 1 **Multiple Choice | Topic: Firewall User Authentication**

Which are common firewall authentication methods? (Choose all that apply)

A. Local authentication
B. RADIUS
C. LDAP
D. HWTACACS

Correct Answer: A,B,C,D

Concept: Firewalls support local, RADIUS, LDAP, and HWTACACS authentication.

69 Cryptography and PKI

Question 1 **Multiple Choice | Topic: Cryptography and PKI**

Which are asymmetric algorithms? (Choose all that apply)

A. RSA
B. ECC
C. DSA
D. AES

Correct Answer: A,B,C

Concept: RSA, ECC, and DSA are asymmetric; AES is symmetric.

Question 2 **Multiple Choice | Topic: Cryptography and PKI**

Which are block cipher modes? (Choose all that apply)

A. ECB
B. CBC
C. CTR
D. GCM

Correct Answer: A,B,C,D

Concept: ECB, CBC, CTR, and GCM are all AES block cipher modes.

Question 3 **Multiple Choice | Topic: Cryptography and PKI**

Which are secure hash algorithms? (Choose all that apply)

A. SHA-256
B. SHA-3
C. BLAKE2
D. MD5

Correct Answer: A,B,C

Concept: SHA-256, SHA-3, and BLAKE2 are considered secure; MD5 is broken.

Question 4 **Multiple Choice | Topic: Cryptography and PKI**

Which are PKI trust models? (Choose all that apply)

A. Single CA
B. Hierarchical CA
C. Cross-certification
D. Bridge CA

Correct Answer: A,B,C,D

Concept: PKI can use single, hierarchical, cross-certified, or bridge CA models.

Question 5 **Multiple Choice | Topic: Cryptography and PKI**

Which fields are typically found in an X.509 certificate? (Choose all that apply)

A. Subject
B. Issuer
C. Public key
D. Validity period

Correct Answer: A,B,C,D

Concept: X.509 certificates contain subject, issuer, public key, validity, serial number, and signature.

Question 6 **Multiple Choice | Topic: Cryptography and PKI**

Which are symmetric algorithms? (Choose all that apply)

A. AES
B. DES
C. 3DES
D. RSA

Correct Answer: A,B,C

Concept: AES, DES, and 3DES are symmetric; RSA is asymmetric.

Question 7 **Multiple Choice | Topic: Cryptography and PKI**

Which are hash algorithms? (Choose all that apply)

A. MD5
B. SHA-256
C. SHA-1
D. SHA-3

Correct Answer: A,B,C,D

Concept: These are all hash algorithms, though MD5 and SHA-1 are deprecated for security use.

Question 8 **Multiple Choice | Topic: Cryptography and PKI**

Which are PKI components? (Choose all that apply)

A. CA
B. RA
C. Repository
D. CRL/OCSP

Correct Answer: A,B,C,D

Concept: PKI includes certificate authorities, registration authorities, repositories, and revocation systems.

70 Cryptography Basics

Question 1 **Multiple Choice | Topic: Cryptography Basics**

Which are symmetric encryption algorithms? (Choose all that apply)

A. AES
B. DES
C. 3DES
D. RSA

Correct Answer: A,B,C

Concept: AES, DES, and 3DES are symmetric; RSA is asymmetric.

71 PKI

Question 1 **Multiple Choice | Topic: PKI**

Which are core PKI components? (Choose all that apply)

A. Certificate Authority
B. Registration Authority
C. Certificate repository
D. CRL/OCSP

Correct Answer: A,B,C,D

Concept: PKI includes CAs, RAs, repositories, and revocation mechanisms such as CRL and OCSP.

72 VPN

Question 1 **Multiple Choice | Topic: VPN**

Which are benefits of using a VPN? (Choose all that apply)

A. Confidentiality
B. Data integrity
C. Authentication
D. Secure remote access

Correct Answer: A,B,C,D

Concept: VPNs provide confidentiality, integrity, authentication, and secure remote access.

Question 2 **Multiple Choice | Topic: VPN**

Which are IPsec security services? (Choose all that apply)

A. Confidentiality
B. Integrity
C. Authentication
D. Anti-replay

Correct Answer: A,B,C,D

Concept: IPsec provides confidentiality, integrity, authentication, and anti-replay protection.

Question 3 — Multiple Choice | Topic: VPN

Which are phases of IKEv1? (Choose all that apply)

A. Phase 1
B. Phase 2
C. Phase 3
D. Phase 1.5

Correct Answer: A,B

Concept: IKEv1 has two phases: Phase 1 (IKE SA) and Phase 2 (IPsec SA).

Question 4 — Multiple Choice | Topic: VPN

Which protocols operate at the IP layer? (Choose all that apply)

A. AH
B. ESP
C. GRE
D. L2TP

Correct Answer: A,B

Concept: AH and ESP are IP-layer protocols; GRE is also IP-layer but L2TP is not.

Question 5 — Multiple Choice | Topic: VPN

Which are VPN types? (Choose all that apply)

A. Site-to-site
B. Remote-access
C. Intranet
D. Extranet

Correct Answer: A,B,C,D

Concept: VPNs can be categorized by these deployment models.

Question 6 — Multiple Choice | Topic: VPN

Which are IPsec protocols? (Choose all that apply)

A. AH
B. ESP
C. IKE
D. GRE

Correct Answer: A,B,C

Concept: AH, ESP, and IKE are IPsec protocols; GRE is separate.

Question 7 **Multiple Choice | Topic: VPN**

Which are SSL VPN access modes? (Choose all that apply)

A. Web proxy
B. File sharing
C. Network extension
D. Port forwarding

Correct Answer: A,B,C,D

Concept: SSL VPN commonly supports all these modes.

Question 8 **Multiple Choice | Topic: VPN**

Which of the following relate to GRE? (Choose all that apply)

A. GRE
B. GRE

Correct Answer: A,B

Concept: These concepts all relate to GRE in VPN.

Question 9 **Multiple Choice | Topic: VPN**

Which of the following relate to L2TP? (Choose all that apply)

A. L2TP
B. L2TP over IPsec

Correct Answer: A,B

Concept: These concepts all relate to L2TP in VPN.

73 VPN Overview

Question 1 **Multiple Choice | Topic: VPN Overview**

Which are common VPN types? (Choose all that apply)

A. Site-to-site VPN
B. Remote-access VPN
C. Intranet VPN
D. Extranet VPN

Correct Answer: A,B,C,D

Concept: VPNs can be site-to-site, remote-access, intranet, or extranet based on deployment.

74 IPsec VPN

Question 1 **Multiple Choice | Topic: IPsec VPN**

Which are IPsec protocols? (Choose all that apply)

A. AH
B. ESP
C. IKE
D. GRE

Correct Answer: A,B,C

Concept: AH, ESP, and IKE are part of IPsec; GRE is a separate tunneling protocol.

75 SSL VPN

Question 1 **Multiple Choice | Topic: SSL VPN**

Which are common SSL VPN access modes? (Choose all that apply)

A. Web proxy
B. File sharing
C. Network extension
D. Port forwarding

Correct Answer: A,B,C,D

Concept: SSL VPN can offer web proxy, file sharing, network extension, and port forwarding.

PART III

True / False Questions

76 Network Security Concepts

Question 1 **True / False | Topic: Network Security Concepts**

Network security in the narrow sense only refers to national cybersecurity laws.

A. True
B. False

Correct Answer: False

Concept: Narrow-sense network security covers vendor solutions and enterprise network protection, not only national laws.

Question 2 **True / False | Topic: Network Security Concepts**

The CIA triad includes confidentiality, integrity, and availability.

A. True
B. False

Correct Answer: True

Concept: Confidentiality, integrity, and availability are the three foundational goals of information security.

Question 3 **True / False | Topic: Network Security Concepts**

Building network security helps enterprises reduce losses from attacks.

A. True
B. False

Correct Answer: True

Concept: Security controls reduce the impact of breaches, downtime, and data loss, thereby lowering financial losses.

Question 4 **True / False | Topic: Network Security Concepts**

Non-repudiation ensures that senders or receivers cannot deny their actions.

A. True
B. False

Correct Answer: True

Concept: Non-repudiation provides evidence to prevent denial of participation.

Question 5 **True / False | Topic: Network Security Concepts**

Availability means anyone can access the system at any time.

A. True
B. False

Correct Answer: False

Concept: Availability means authorized users can access information when required, not unlimited access.

Question 6 **True / False | Topic: Network Security Concepts**

Zero trust means everything inside the network is trusted by default.

A. True
B. False

Correct Answer: False

Concept: Zero trust never assumes trust based on network location and verifies every request.

Question 7 **True / False | Topic: Network Security Concepts**

Integrity ensures that data is not tampered with during transmission.

A. True
B. False

Correct Answer: True

Concept: Integrity protects against unauthorized modification or destruction of data.

Question 8 **True / False | Topic: Network Security Concepts**

Network security broadly refers to cyberspace security at a national level.

A. True
B. False

Correct Answer: True

Concept: Broad network security covers national cyberspace protection.

Question 9 **True / False | Topic: Network Security Concepts**

The CIA triad excludes availability.

A. True
B. False

Correct Answer: False

Concept: The CIA triad includes confidentiality, integrity, and availability.

Question 10 **True / False | Topic: Network Security Concepts**

Risk is the combination of threat likelihood and impact.

A. True
B. False

Correct Answer: True

Concept: Risk assesses probability and consequence of security incidents.

Question 11 **True / False | Topic: Network Security Concepts**

Broad network security means that is synonymous with national cyberspace security.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Broad network security.

Question 12 **True / False | Topic: Network Security Concepts**

Broad network security is unrelated to is synonymous with national cyberspace security.

A. True
B. False

Correct Answer: False

Concept: Broad network security is directly related to is synonymous with national cyberspace security.

Question 13 **True / False | Topic: Network Security Concepts**

Narrow network security means that focuses on enterprise network protection solutions.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Narrow network security.

Question 14 **True / False | Topic: Network Security Concepts**

Narrow network security is unrelated to focuses on enterprise network protection solutions.

A. True
B. False

Correct Answer: False

Concept: Narrow network security is directly related to focuses on enterprise network protection solutions.

Question 15 **True / False | Topic: Network Security Concepts**

CIA triad means that consists of confidentiality, integrity, and availability.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes CIA triad.

Question 16 **True / False | Topic: Network Security Concepts**

CIA triad is unrelated to consists of confidentiality, integrity, and availability.

A. True
B. False

Correct Answer: False

Concept: CIA triad is directly related to consists of confidentiality, integrity, and availability.

Question 17 **True / False | Topic: Network Security Concepts**

Information assurance period means that added controllability and non-repudiation.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Information assurance period.

Question 18 **True / False | Topic: Network Security Concepts**

Information assurance period is unrelated to added controllability and non-repudiation.

A. True
B. False

Correct Answer: False

Concept: Information assurance period is directly related to added controllability and non-repudiation.

Question 19 **True / False | Topic: Network Security Concepts**

Cyberspace security means that protects facilities, data, users, and operations.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Cyberspace security.

Question 20 **True / False | Topic: Network Security Concepts**

Cyberspace security is unrelated to protects facilities, data, users, and operations.

A. True
B. False

Correct Answer: False

Concept: Cyberspace security is directly related to protects facilities, data, users, and operations.

Question 21 **True / False | Topic: Network Security Concepts**

Confidentiality means that ensures information is accessible only to authorized users.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Confidentiality.

Question 22 **True / False | Topic: Network Security Concepts**

Confidentiality is unrelated to ensures information is accessible only to authorized users.

A. True
B. False

Correct Answer: False

Concept: Confidentiality is directly related to ensures information is accessible only to authorized users.

Question 23 **True / False | Topic: Network Security Concepts**

Integrity means that ensures data is not tampered with during transmission or storage.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Integrity.

Question 24 **True / False | Topic: Network Security Concepts**

Integrity is unrelated to ensures data is not tampered with during transmission or storage.

A. True
B. False

Correct Answer: False

Concept: Integrity is directly related to ensures data is not tampered with during transmission or storage.

Question 25 **True / False | Topic: Network Security Concepts**

Availability means that ensures authorized users can access information when needed.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Availability.

Question 26 **True / False | Topic: Network Security Concepts**

Availability is unrelated to ensures authorized users can access information when needed.

A. True
B. False

Correct Answer: False

Concept: Availability is directly related to ensures authorized users can access information when needed.

Question 27 **True / False | Topic: Network Security Concepts**

Controllability means that enables monitoring and control of information and systems.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Controllability.

Question 28 **True / False | Topic: Network Security Concepts**

Controllability is unrelated to enables monitoring and control of information and systems.

A. True
B. False

Correct Answer: False

Concept: Controllability is directly related to enables monitoring and control of information and systems.

Question 29 **True / False | Topic: Network Security Concepts**

Non-repudiation means that prevents senders or receivers from denying actions.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Non-repudiation.

Question 30 **True / False | Topic: Network Security Concepts**

Non-repudiation is unrelated to prevents senders or receivers from denying actions.

A. True
B. False

Correct Answer: False

Concept: Non-repudiation is directly related to prevents senders or receivers from denying actions.

Question 31 **True / False | Topic: Network Security Concepts**

Risk means that is the combination of threat likelihood and impact.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Risk.

Question 32 **True / False | Topic: Network Security Concepts**

Risk is unrelated to is the combination of threat likelihood and impact.

A. True
B. False

Correct Answer: False

Concept: Risk is directly related to is the combination of threat likelihood and impact.

Question 33 **True / False | Topic: Network Security Concepts**

Vulnerability means that is a weakness that can be exploited by threats.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Vulnerability.

Question 34 **True / False | Topic: Network Security Concepts**

Vulnerability is unrelated to is a weakness that can be exploited by threats.

A. True
B. False

Correct Answer: False

Concept: Vulnerability is directly related to is a weakness that can be exploited by threats.

Question 35 **True / False | Topic: Network Security Concepts**

Threat means that is a potential danger to an asset.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Threat.

Question 36 **True / False | Topic: Network Security Concepts**

Threat is unrelated to is a potential danger to an asset.

A. True
B. False

Correct Answer: False

Concept: Threat is directly related to is a potential danger to an asset.

Question 37 **True / False | Topic: Network Security Concepts**

Exploit means that is a method that takes advantage of a vulnerability.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Exploit.

Question 38 **True / False | Topic: Network Security Concepts**

Exploit is unrelated to is a method that takes advantage of a vulnerability.

A. True
B. False

Correct Answer: False

Concept: Exploit is directly related to is a method that takes advantage of a vulnerability.

Question 39 **True / False | Topic: Network Security Concepts**

Future security trend means that includes zero-trust architecture.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Future security trend.

Question 40 **True / False | Topic: Network Security Concepts**

Future security trend is unrelated to includes zero-trust architecture.

A. True
B. False

Correct Answer: False

Concept: Future security trend is directly related to includes zero-trust architecture.

Question 41 **True / False | Topic: Network Security Concepts**

AI and machine learning means that help detect unknown threats and anomalies.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes AI and machine learning.

Question 42 **True / False | Topic: Network Security Concepts**

AI and machine learning is unrelated to help detect unknown threats and anomalies.

A. True
B. False

Correct Answer: False

Concept: AI and machine learning is directly related to help detect unknown threats and anomalies.

77 Network Security Standards

Question 1 **True / False | Topic: Network Security Standards**

ISO 27002 can be used to certify an enterprise information security system.

A. True
B. False

Correct Answer: False

Concept: ISO 27001 is the certification standard; ISO 27002 is a code of practice with controls.

Question 2 **True / False | Topic: Network Security Standards**

Classified Protection 2.0 defines protection levels from Level 1 to Level 5.

A. True
B. False

Correct Answer: True

Concept: Classified Protection 2.0 uses five levels, with higher levels requiring stronger controls and oversight.

Question 3 **True / False | Topic: Network Security Standards**

Network operators can always determine the final protection level on their own without review.

A. True
B. False

Correct Answer: False

Concept: For Level 2 and above, expert review and approval are required; only Level 1 can be determined by the operator.

Question 4 **True / False | Topic: Network Security Standards**

ISO 27001 provides a list of security controls but is not certifiable.

A. True
B. False

Correct Answer: False

Concept: ISO 27001 is the certifiable standard specifying ISMS requirements; ISO 27002 provides controls.

Question 5 **True / False | Topic: Network Security Standards**

Classified Protection 2.0 requires security and informatization to be planned together.

A. True
B. False

Correct Answer: True

Concept: The framework emphasizes synchronized planning, construction, and use of security.

Question 6 **True / False | Topic: Network Security Standards**

Level assessment is part of the Classified Protection lifecycle.

A. True
B. False

Correct Answer: True

Concept: After construction, a level assessment verifies that protection measures meet the required level.

Question 7 **True / False | Topic: Network Security Standards**

ISO 27001 is certifiable.

A. True
B. False

Correct Answer: True

Concept: Organizations can be certified against ISO 27001 requirements.

Question 8 **True / False | Topic: Network Security Standards**

Classified Protection 2.0 has only three levels.

A. True
B. False

Correct Answer: False

Concept: Classified Protection 2.0 defines five levels.

Question 9 **True / False | Topic: Network Security Standards**

ISO 27001 means that specifies certifiable ISMS requirements.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes ISO 27001.

Question 10 **True / False | Topic: Network Security Standards**

ISO 27001 is unrelated to specifies certifiable ISMS requirements.

A. True
B. False

Correct Answer: False

Concept: ISO 27001 is directly related to specifies certifiable ISMS requirements.

Question 11 **True / False | Topic: Network Security Standards**

ISO 27002 means that provides a code of practice for security controls.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes ISO 27002.

Question 12 **True / False | Topic: Network Security Standards**

ISO 27002 is unrelated to provides a code of practice for security controls.

A. True
B. False

Correct Answer: False

Concept: ISO 27002 is directly related to provides a code of practice for security controls.

Question 13 **True / False | Topic: Network Security Standards**

PDCA means that is the continuous improvement methodology used in ISO 27001.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes PDCA.

Question 14 **True / False | Topic: Network Security Standards**

PDCA is unrelated to is the continuous improvement methodology used in ISO 27001.

A. True
B. False

Correct Answer: False

Concept: PDCA is directly related to is the continuous improvement methodology used in ISO 27001.

Question 15 **True / False | Topic: Network Security Standards**

Classified Protection 2.0 means that is a Chinese national cybersecurity framework.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Classified Protection 2.0.

Question 16 **True / False | Topic: Network Security Standards**

Classified Protection 2.0 is unrelated to is a Chinese national cybersecurity framework.

A. True
B. False

Correct Answer: False

Concept: Classified Protection 2.0 is directly related to is a Chinese national cybersecurity framework.

Question 17 **True / False | Topic: Network Security Standards**

Classified Protection levels means that range from Level 1 to Level 5.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Classified Protection levels.

Question 18 **True / False | Topic: Network Security Standards**

Classified Protection levels is unrelated to range from Level 1 to Level 5.

A. True
B. False

Correct Answer: False

Concept: Classified Protection levels is directly related to range from Level 1 to Level 5.

Question 19 **True / False | Topic: Network Security Standards**

Level 2 and above means that require expert review and approval.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Level 2 and above.

Question 20 **True / False | Topic: Network Security Standards**

Level 2 and above is unrelated to require expert review and approval.

A. True
B. False

Correct Answer: False

Concept: Level 2 and above is directly related to require expert review and approval.

Question 21 **True / False | Topic: Network Security Standards**

Classified Protection lifecycle means that includes classification, filing, construction, assessment, and supervision.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Classified Protection lifecycle.

Question 22 **True / False | Topic: Network Security Standards**

Classified Protection lifecycle is unrelated to includes classification, filing, construction, assessment, and supervision.

A. True
B. False

Correct Answer: False

Concept: Classified Protection lifecycle is directly related to includes classification, filing, construction, assessment, and supervision.

Question 23 **True / False | Topic: Network Security Standards**

Security synchronization means that means planning security with information system construction.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Security synchronization.

Question 24 **True / False | Topic: Network Security Standards**

Security synchronization is unrelated to means planning security with information system construction.

A. True
B. False

Correct Answer: False

Concept: Security synchronization is directly related to means planning security with information system construction.

Question 25 True / False | Topic: Network Security Standards

ISO 27001:2022 means that organizes controls into four themes.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes ISO 27001:2022.

Question 26 True / False | Topic: Network Security Standards

ISO 27001:2022 is unrelated to organizes controls into four themes.

A. True
B. False

Correct Answer: False

Concept: ISO 27001:2022 is directly related to organizes controls into four themes.

Question 27 True / False | Topic: Network Security Standards

Cybersecurity Law means that underpins Classified Protection in China.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Cybersecurity Law.

Question 28 True / False | Topic: Network Security Standards

Cybersecurity Law is unrelated to underpins Classified Protection in China.

A. True
B. False

Correct Answer: False

Concept: Cybersecurity Law is directly related to underpins Classified Protection in China.

78 Network Fundamentals

Question 1 True / False | Topic: Network Fundamentals

The OSI model was developed before the TCP/IP model.

A. True
B. False

Correct Answer: False

Concept: TCP/IP was developed earlier and became the de facto standard; OSI was later as a reference model.

Question 2 **True / False | Topic: Network Fundamentals**

DNS always uses TCP port 53.

A. True
B. False

Correct Answer: False

Concept: DNS primarily uses UDP port 53; TCP port 53 is used for zone transfers and large responses.

Question 3 **True / False | Topic: Network Fundamentals**

A switch forwards packets based on IP addresses.

A. True
B. False

Correct Answer: False

Concept: A switch forwards frames based on MAC addresses; routers forward packets based on IP addresses.

Question 4 **True / False | Topic: Network Fundamentals**

IPv4 addresses are 32 bits long.

A. True
B. False

Correct Answer: True

Concept: IPv4 addresses are 32-bit binary numbers, usually written in dotted-decimal notation.

Question 5 **True / False | Topic: Network Fundamentals**

ICMP is a transport-layer protocol.

A. True
B. False

Correct Answer: False

Concept: ICMP is a network-layer protocol used for diagnostics and error reporting.

Question 6 **True / False | Topic: Network Fundamentals**

TCP provides connection-oriented reliable delivery.

A. True
B. False

Correct Answer: True

Concept: TCP establishes connections and ensures reliable, ordered delivery.

Question 7 **True / False | Topic: Network Fundamentals**

A hub creates separate collision domains per port.

A. True
B. False

Correct Answer: False

Concept: A hub shares a single collision domain; switches create separate collision domains.

Question 8 **True / False | Topic: Network Fundamentals**

IPv6 addresses are 128 bits long.

A. True
B. False

Correct Answer: True

Concept: IPv6 addresses are 128-bit binary numbers.

Question 9 **True / False | Topic: Network Fundamentals**

OSI model means that has seven layers.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes OSI model.

Question 10 **True / False | Topic: Network Fundamentals**

OSI model is unrelated to has seven layers.

A. True
B. False

Correct Answer: False

Concept: OSI model is directly related to has seven layers.

Question 11 **True / False | Topic: Network Fundamentals**

TCP/IP standard model means that has four layers.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes TCP/IP standard model.

Question 12 **True / False | Topic: Network Fundamentals**

TCP/IP standard model is unrelated to has four layers.

A. True
B. False

Correct Answer: False

Concept: TCP/IP standard model is directly related to has four layers.

Question 13 **True / False | Topic: Network Fundamentals**

TCP/IP equivalent model means that has five layers.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes TCP/IP equivalent model.

Question 14 **True / False | Topic: Network Fundamentals**

TCP/IP equivalent model is unrelated to has five layers.

A. True
B. False

Correct Answer: False

Concept: TCP/IP equivalent model is directly related to has five layers.

Question 15 **True / False | Topic: Network Fundamentals**

Physical layer means that transmits raw bits over a medium.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Physical layer.

Question 16 **True / False | Topic: Network Fundamentals**

Physical layer is unrelated to transmits raw bits over a medium.

A. True
B. False

Correct Answer: False

Concept: Physical layer is directly related to transmits raw bits over a medium.

Question 17 **True / False | Topic: Network Fundamentals**

Data link layer means that handles framing and MAC addressing.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Data link layer.

Question 18 **True / False | Topic: Network Fundamentals**

Data link layer is unrelated to handles framing and MAC addressing.

A. True
B. False

Correct Answer: False

Concept: Data link layer is directly related to handles framing and MAC addressing.

Question 19 **True / False | Topic: Network Fundamentals**

Network layer means that handles logical addressing and routing.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Network layer.

Question 20 **True / False | Topic: Network Fundamentals**

Network layer is unrelated to handles logical addressing and routing.

A. True
B. False

Correct Answer: False

Concept: Network layer is directly related to handles logical addressing and routing.

Question 21 **True / False | Topic: Network Fundamentals**

Transport layer means that provides end-to-end communication and reliability.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Transport layer.

Question 22 **True / False | Topic: Network Fundamentals**

Transport layer is unrelated to provides end-to-end communication and reliability.

A. True
B. False

Correct Answer: False

Concept: Transport layer is directly related to provides end-to-end communication and reliability.

Question 23 **True / False | Topic: Network Fundamentals**

Session layer means that manages application dialogs.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Session layer.

Question 24 **True / False | Topic: Network Fundamentals**

Session layer is unrelated to manages application dialogs.

A. True
B. False

Correct Answer: False

Concept: Session layer is directly related to manages application dialogs.

Question 25 **True / False | Topic: Network Fundamentals**

Presentation layer means that handles encryption and data format conversion.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Presentation layer.

Question 26 **True / False | Topic: Network Fundamentals**

Presentation layer is unrelated to handles encryption and data format conversion.

A. True
B. False

Correct Answer: False

Concept: Presentation layer is directly related to handles encryption and data format conversion.

Question 27 **True / False | Topic: Network Fundamentals**

Application layer means that provides network services to applications.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Application layer.

Question 28 **True / False | Topic: Network Fundamentals**

Application layer is unrelated to provides network services to applications.

A. True
B. False

Correct Answer: False

Concept: Application layer is directly related to provides network services to applications.

Question 29 **True / False | Topic: Network Fundamentals**

TCP means that is connection-oriented and reliable.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes TCP.

Question 30 **True / False | Topic: Network Fundamentals**

TCP is unrelated to is connection-oriented and reliable.

A. True
B. False

Correct Answer: False

Concept: TCP is directly related to is connection-oriented and reliable.

Question 31 **True / False | Topic: Network Fundamentals**

UDP means that is connectionless and has low overhead.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes UDP.

Question 32 **True / False | Topic: Network Fundamentals**

UDP is unrelated to is connectionless and has low overhead.

A. True
B. False

Correct Answer: False

Concept: UDP is directly related to is connectionless and has low overhead.

Question 33 **True / False | Topic: Network Fundamentals**

FTP means that uses TCP ports 20 and 21.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes FTP.

Question 34 **True / False | Topic: Network Fundamentals**

FTP is unrelated to uses TCP ports 20 and 21.

A. True
B. False

Correct Answer: False

Concept: FTP is directly related to uses TCP ports 20 and 21.

Question 35 **True / False | Topic: Network Fundamentals**

HTTP means that uses TCP port 80.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes HTTP.

Question 36 **True / False | Topic: Network Fundamentals**

HTTP is unrelated to uses TCP port 80.

A. True
B. False

Correct Answer: False

Concept: HTTP is directly related to uses TCP port 80.

Question 37 **True / False | Topic: Network Fundamentals**

HTTPS means that uses TCP port 443.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes HTTPS.

Question 38 **True / False | Topic: Network Fundamentals**

HTTPS is unrelated to uses TCP port 443.

A. True
B. False

Correct Answer: False

Concept: HTTPS is directly related to uses TCP port 443.

Question 39 **True / False | Topic: Network Fundamentals**

SSH means that uses TCP port 22.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes SSH.

Question 40 **True / False | Topic: Network Fundamentals**

SSH is unrelated to uses TCP port 22.

A. True
B. False

Correct Answer: False

Concept: SSH is directly related to uses TCP port 22.

Question 41 **True / False | Topic: Network Fundamentals**

Telnet means that uses TCP port 23 and is insecure.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Telnet.

Question 42 **True / False | Topic: Network Fundamentals**

Telnet is unrelated to uses TCP port 23 and is insecure.

A. True
B. False

Correct Answer: False

Concept: Telnet is directly related to uses TCP port 23 and is insecure.

Question 43 **True / False | Topic: Network Fundamentals**

DNS means that primarily uses UDP port 53.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes DNS.

Question 44 **True / False | Topic: Network Fundamentals**

DNS is unrelated to primarily uses UDP port 53.

A. True
B. False

Correct Answer: False

Concept: DNS is directly related to primarily uses UDP port 53.

Question 45 **True / False | Topic: Network Fundamentals**

SMTP means that uses TCP port 25.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes SMTP.

Question 46 **True / False | Topic: Network Fundamentals**

SMTP is unrelated to uses TCP port 25.

A. True
B. False

Correct Answer: False

Concept: SMTP is directly related to uses TCP port 25.

Question 47 **True / False | Topic: Network Fundamentals**

DHCP means that dynamically assigns IP addresses.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes DHCP.

Question 48 **True / False | Topic: Network Fundamentals**

DHCP is unrelated to dynamically assigns IP addresses.

A. True
B. False

Correct Answer: False

Concept: DHCP is directly related to dynamically assigns IP addresses.

Question 49 **True / False | Topic: Network Fundamentals**

ARP means that maps IP addresses to MAC addresses.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes ARP.

Question 50 **True / False | Topic: Network Fundamentals**

ARP is unrelated to maps IP addresses to MAC addresses.

A. True
B. False

Correct Answer: False

Concept: ARP is directly related to maps IP addresses to MAC addresses.

Question 51 **True / False | Topic: Network Fundamentals**

ICMP means that provides error reporting and diagnostics.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes ICMP.

Question 52 **True / False | Topic: Network Fundamentals**

ICMP is unrelated to provides error reporting and diagnostics.

A. True
B. False

Correct Answer: False

Concept: ICMP is directly related to provides error reporting and diagnostics.

Question 53 **True / False | Topic: Network Fundamentals**

IPv4 address means that is 32 bits long.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes IPv4 address.

Question 54 **True / False | Topic: Network Fundamentals**

IPv4 address is unrelated to is 32 bits long.

A. True
B. False

Correct Answer: False

Concept: IPv4 address is directly related to is 32 bits long.

Question 55 **True / False | Topic: Network Fundamentals**

IPv6 address means that is 128 bits long.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes IPv6 address.

Question 56 **True / False | Topic: Network Fundamentals**

IPv6 address is unrelated to is 128 bits long.

A. True
B. False

Correct Answer: False

Concept: IPv6 address is directly related to is 128 bits long.

Question 57 **True / False | Topic: Network Fundamentals**

MAC address means that is a Layer 2 hardware address.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes MAC address.

Question 58 **True / False | Topic: Network Fundamentals**

MAC address is unrelated to is a Layer 2 hardware address.

A. True
B. False

Correct Answer: False

Concept: MAC address is directly related to is a Layer 2 hardware address.

Question 59 **True / False | Topic: Network Fundamentals**

Switch means that operates at Layer 2 and forwards frames by MAC.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Switch.

Question 60 **True / False | Topic: Network Fundamentals**

Switch is unrelated to operates at Layer 2 and forwards frames by MAC.

A. True
B. False

Correct Answer: False

Concept: Switch is directly related to operates at Layer 2 and forwards frames by MAC.

Question 61 **True / False | Topic: Network Fundamentals**

Router means that operates at Layer 3 and forwards packets by IP.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Router.

Question 62 **True / False | Topic: Network Fundamentals**

Router is unrelated to operates at Layer 3 and forwards packets by IP.

A. True
B. False

Correct Answer: False

Concept: Router is directly related to operates at Layer 3 and forwards packets by IP.

Question 63 **True / False | Topic: Network Fundamentals**

Hub means that operates at Layer 1 and repeats signals.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Hub.

Question 64 **True / False | Topic: Network Fundamentals**

Hub is unrelated to operates at Layer 1 and repeats signals.

A. True
B. False

Correct Answer: False

Concept: Hub is directly related to operates at Layer 1 and repeats signals.

Question 65 **True / False | Topic: Network Fundamentals**

VLAN means that logically segments a network into broadcast domains.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes VLAN.

Question 66 **True / False | Topic: Network Fundamentals**

VLAN is unrelated to logically segments a network into broadcast domains.

A. True
B. False

Correct Answer: False

Concept: VLAN is directly related to logically segments a network into broadcast domains.

Question 67 **True / False | Topic: Network Fundamentals**

STP means that prevents Layer 2 loops.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes STP.

Question 68 **True / False | Topic: Network Fundamentals**

STP is unrelated to prevents Layer 2 loops.

A. True
B. False

Correct Answer: False

Concept: STP is directly related to prevents Layer 2 loops.

Question 69 **True / False | Topic: Network Fundamentals**

RFC 1918 means that defines private IPv4 address ranges.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes RFC 1918.

Question 70 **True / False | Topic: Network Fundamentals**

RFC 1918 is unrelated to defines private IPv4 address ranges.

A. True
B. False

Correct Answer: False

Concept: RFC 1918 is directly related to defines private IPv4 address ranges.

79 OSI and TCP/IP Models

Question 1 **True / False | Topic: OSI and TCP/IP Models**

The TCP/IP model combines the OSI session and presentation layers into the application layer.

A. True
B. False

Correct Answer: True

Concept: The TCP/IP model does not have separate session and presentation layers; their functions are included in the application layer.

80 Network Protocols

Question 1 — **True / False | Topic: Network Protocols**

UDP guarantees ordered delivery of packets.

A. True
B. False

Correct Answer: False

Concept: UDP is connectionless and does not guarantee delivery, ordering, or duplicate protection.

81 Network Devices

Question 1 — **True / False | Topic: Network Devices**

A router forwards frames based on MAC addresses.

A. True
B. False

Correct Answer: False

Concept: Routers forward packets based on IP addresses; switches forward frames based on MAC addresses.

Question 2 — **True / False | Topic: Network Devices**

A switch creates a separate collision domain for each port.

A. True
B. False

Correct Answer: True

Concept: Each switch port is its own collision domain, reducing collisions compared to a hub.

82 Enterprise Security Threats

Question 1 — **True / False | Topic: Enterprise Security Threats**

A worm requires a host program to spread.

A. True
B. False

Correct Answer: False

Concept: Worms are standalone malware that can self-replicate; viruses require host programs.

Question 2 **True / False | Topic: Enterprise Security Threats**

Phishing attacks target human psychology rather than software vulnerabilities.

A. True
B. False

Correct Answer: True

Concept: Phishing uses deception and social engineering to trick users.

Question 3 **True / False | Topic: Enterprise Security Threats**

Using HTTP instead of HTTPS improves confidentiality.

A. True
B. False

Correct Answer: False

Concept: HTTP sends data in plaintext; HTTPS encrypts data to protect confidentiality.

Question 4 **True / False | Topic: Enterprise Security Threats**

The DMZ zone is typically more trusted than the Untrust zone but less trusted than the Trust zone.

A. True
B. False

Correct Answer: True

Concept: DMZ has an intermediate trust level for public-facing servers.

Question 5 **True / False | Topic: Enterprise Security Threats**

Patch management helps eliminate known vulnerabilities on hosts.

A. True
B. False

Correct Answer: True

Concept: Applying patches closes security holes that attackers could exploit.

Question 6 **True / False | Topic: Enterprise Security Threats**

All security threats originate from external attackers.

A. True
B. False

Correct Answer: False

Concept: Threats can be external or internal, accidental or malicious.

Question 7 **True / False | Topic: Enterprise Security Threats**

Phishing is a type of social engineering.

A. True
B. False

Correct Answer: True

Concept: Phishing manipulates users into revealing information.

Question 8 **True / False | Topic: Enterprise Security Threats**

A firewall at the network border can enforce zone-based policies.

A. True
B. False

Correct Answer: True

Concept: Firewalls enforce policies between security zones.

Question 9 **True / False | Topic: Enterprise Security Threats**

Least privilege reduces the impact of compromised accounts.

A. True
B. False

Correct Answer: True

Concept: Limiting access limits damage from compromised credentials.

Question 10 **True / False | Topic: Enterprise Security Threats**

Passive attack means that monitors traffic without altering it.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Passive attack.

Question 11 **True / False | Topic: Enterprise Security Threats**

Passive attack is unrelated to monitors traffic without altering it.

A. True
B. False

Correct Answer: False

Concept: Passive attack is directly related to monitors traffic without altering it.

Question 12 **True / False | Topic: Enterprise Security Threats**

Active attack means that modifies or disrupts systems or data.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Active attack.

Question 13 **True / False | Topic: Enterprise Security Threats**

Active attack is unrelated to modifies or disrupts systems or data.

A. True
B. False

Correct Answer: False

Concept: Active attack is directly related to modifies or disrupts systems or data.

Question 14 **True / False | Topic: Enterprise Security Threats**

Malware means that includes viruses, worms, Trojans, and ransomware.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Malware.

Question 15 **True / False | Topic: Enterprise Security Threats**

Malware is unrelated to includes viruses, worms, Trojans, and ransomware.

A. True
B. False

Correct Answer: False

Concept: Malware is directly related to includes viruses, worms, Trojans, and ransomware.

Question 16 **True / False | Topic: Enterprise Security Threats**

Virus means that requires a host program to spread.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Virus.

Question 17 **True / False | Topic: Enterprise Security Threats**

Virus is unrelated to requires a host program to spread.

A. True
B. False

Correct Answer: False

Concept: Virus is directly related to requires a host program to spread.

Question 18 **True / False | Topic: Enterprise Security Threats**

Worm means that can self-replicate across networks.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Worm.

Question 19 **True / False | Topic: Enterprise Security Threats**

Worm is unrelated to can self-replicate across networks.

A. True
B. False

Correct Answer: False

Concept: Worm is directly related to can self-replicate across networks.

Question 20 **True / False | Topic: Enterprise Security Threats**

Trojan means that disguises itself as legitimate software.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Trojan.

Question 21 **True / False | Topic: Enterprise Security Threats**

Trojan is unrelated to disguises itself as legitimate software.

A. True
B. False

Correct Answer: False

Concept: Trojan is directly related to disguises itself as legitimate software.

Question 22 **True / False | Topic: Enterprise Security Threats**

Ransomware means that encrypts data and demands payment.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Ransomware.

Question 23 **True / False | Topic: Enterprise Security Threats**

Ransomware is unrelated to encrypts data and demands payment.

A. True
B. False

Correct Answer: False

Concept: Ransomware is directly related to encrypts data and demands payment.

Question 24 **True / False | Topic: Enterprise Security Threats**

Phishing means that uses deception to steal information.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Phishing.

Question 25 **True / False | Topic: Enterprise Security Threats**

Phishing is unrelated to uses deception to steal information.

A. True
B. False

Correct Answer: False

Concept: Phishing is directly related to uses deception to steal information.

Question 26 **True / False | Topic: Enterprise Security Threats**

Social engineering means that manipulates people into breaking security rules.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Social engineering.

Question 27 **True / False | Topic: Enterprise Security Threats**

Social engineering is unrelated to manipulates people into breaking security rules.

A. True
B. False

Correct Answer: False

Concept: Social engineering is directly related to manipulates people into breaking security rules.

Question 28 **True / False | Topic: Enterprise Security Threats**

DoS attack means that aims to make a service unavailable.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes DoS attack.

Question 29 **True / False | Topic: Enterprise Security Threats**

DoS attack is unrelated to aims to make a service unavailable.

A. True
B. False

Correct Answer: False

Concept: DoS attack is directly related to aims to make a service unavailable.

Question 30 **True / False | Topic: Enterprise Security Threats**

DDoS attack means that uses many distributed sources to flood a target.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes DDoS attack.

Question 31 **True / False | Topic: Enterprise Security Threats**

DDoS attack is unrelated to uses many distributed sources to flood a target.

A. True
B. False

Correct Answer: False

Concept: DDoS attack is directly related to uses many distributed sources to flood a target.

Question 32 — True / False | Topic: Enterprise Security Threats

Man-in-the-middle attack means that intercepts communication between two parties.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Man-in-the-middle attack.

Question 33 — True / False | Topic: Enterprise Security Threats

Man-in-the-middle attack is unrelated to intercepts communication between two parties.

A. True
B. False

Correct Answer: False

Concept: Man-in-the-middle attack is directly related to intercepts communication between two parties.

Question 34 — True / False | Topic: Enterprise Security Threats

ARP spoofing means that forges ARP replies to redirect traffic.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes ARP spoofing.

Question 35 — True / False | Topic: Enterprise Security Threats

ARP spoofing is unrelated to forges ARP replies to redirect traffic.

A. True
B. False

Correct Answer: False

Concept: ARP spoofing is directly related to forges ARP replies to redirect traffic.

Question 36 — True / False | Topic: Enterprise Security Threats

SYN flood means that sends many half-open TCP connections.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes SYN flood.

Question 37 **True / False | Topic: Enterprise Security Threats**

SYN flood is unrelated to sends many half-open TCP connections.

A. True
B. False

Correct Answer: False

Concept: SYN flood is directly related to sends many half-open TCP connections.

Question 38 **True / False | Topic: Enterprise Security Threats**

Ping of death means that sends oversized ICMP packets.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Ping of death.

Question 39 **True / False | Topic: Enterprise Security Threats**

Ping of death is unrelated to sends oversized ICMP packets.

A. True
B. False

Correct Answer: False

Concept: Ping of death is directly related to sends oversized ICMP packets.

Question 40 **True / False | Topic: Enterprise Security Threats**

Smurf attack means that uses broadcast pings to amplify traffic.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Smurf attack.

Question 41 **True / False | Topic: Enterprise Security Threats**

Smurf attack is unrelated to uses broadcast pings to amplify traffic.

A. True
B. False

Correct Answer: False

Concept: Smurf attack is directly related to uses broadcast pings to amplify traffic.

Question 42 **True / False | Topic: Enterprise Security Threats**

Least privilege means that limits user access to the minimum necessary.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Least privilege.

Question 43 **True / False | Topic: Enterprise Security Threats**

Least privilege is unrelated to limits user access to the minimum necessary.

A. True
B. False

Correct Answer: False

Concept: Least privilege is directly related to limits user access to the minimum necessary.

Question 44 **True / False | Topic: Enterprise Security Threats**

Patch management means that eliminates known vulnerabilities.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Patch management.

Question 45 **True / False | Topic: Enterprise Security Threats**

Patch management is unrelated to eliminates known vulnerabilities.

A. True
B. False

Correct Answer: False

Concept: Patch management is directly related to eliminates known vulnerabilities.

Question 46 **True / False | Topic: Enterprise Security Threats**

SIEM means that centralizes log collection and analysis.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes SIEM.

Question 47 **True / False | Topic: Enterprise Security Threats**

SIEM is unrelated to centralizes log collection and analysis.

A. True
B. False

Correct Answer: False

Concept: SIEM is directly related to centralizes log collection and analysis.

Question 48 **True / False | Topic: Enterprise Security Threats**

DMZ means that hosts public-facing servers between Internet and internal network.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes DMZ.

Question 49 **True / False | Topic: Enterprise Security Threats**

DMZ is unrelated to hosts public-facing servers between Internet and internal network.

A. True
B. False

Correct Answer: False

Concept: DMZ is directly related to hosts public-facing servers between Internet and internal network.

Question 50 **True / False | Topic: Enterprise Security Threats**

Insider threat means that originates from within the organization.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Insider threat.

Question 51 **True / False | Topic: Enterprise Security Threats**

Insider threat is unrelated to originates from within the organization.

A. True
B. False

Correct Answer: False

Concept: Insider threat is directly related to originates from within the organization.

83 Communication Network Security

Question 1 **True / False | Topic: Communication Network Security**

Encryption is an effective way to protect data confidentiality over public networks.

A. True
B. False

Correct Answer: True

Concept: Encryption ensures that intercepted data cannot be read without the proper key.

84 Zone Border Security

Question 1 **True / False | Topic: Zone Border Security**

A firewall at the zone border can enforce security policies between trusted and untrusted networks.

A. True
B. False

Correct Answer: True

Concept: Firewalls inspect and control traffic crossing security zone boundaries based on policy.

85 Firewall Basics

Question 1 **True / False | Topic: Firewall Basics**

Security policies on Huawei firewalls are matched from bottom to top by default.

A. True
B. False

Correct Answer: False

Concept: Policies are matched from top to bottom based on configured order/priority.

Question 2 **True / False | Topic: Firewall Basics**

A packet-filtering firewall tracks the state of every connection.

A. True
B. False

Correct Answer: False

Concept: Only stateful firewalls maintain connection state; packet filters inspect packets individually.

Question 3 **True / False | Topic: Firewall Basics**

ASPF can generate temporary server-map entries for dynamic protocols.

A. True
B. False

Correct Answer: True

Concept: ASPF creates server-map entries to permit negotiated secondary connections.

Question 4 **True / False | Topic: Firewall Basics**

Next-generation firewalls can identify applications regardless of port.

A. True
B. False

Correct Answer: True

Concept: NGFWs use application-layer inspection to identify applications beyond simple port numbers.

Question 5 **True / False | Topic: Firewall Basics**

Traffic between interfaces in the same security zone is always allowed.

A. True
B. False

Correct Answer: False

Concept: Even intra-zone traffic can be controlled by policies, though default behavior may vary by configuration.

Question 6 **True / False | Topic: Firewall Basics**

A deny action in a security policy can generate logs.

A. True
B. False

Correct Answer: True

Concept: Both permit and deny actions can be configured to log traffic.

Question 7 **True / False | Topic: Firewall Basics**

The default Huawei firewall action is permit.

A. True
B. False

Correct Answer: False

Concept: The default action is deny unless explicitly permitted.

Question 8 **True / False | Topic: Firewall Basics**

ASPF inspects application-layer negotiation.

A. True
B. False

Correct Answer: True

Concept: ASPF handles dynamic protocols such as FTP.

Question 9 **True / False | Topic: Firewall Basics**

Stateful firewalls use session tables to track connections.

A. True
B. False

Correct Answer: True

Concept: Session tables store connection state for return traffic.

Question 10 **True / False | Topic: Firewall Basics**

Trust zone priority means that is 85 by default.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Trust zone priority.

Question 11 **True / False | Topic: Firewall Basics**

Trust zone priority is unrelated to is 85 by default.

A. True
B. False

Correct Answer: False

Concept: Trust zone priority is directly related to is 85 by default.

Question 12 **True / False | Topic: Firewall Basics**

DMZ zone priority means that is 50 by default.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes DMZ zone priority.

Question 13 **True / False | Topic: Firewall Basics**

DMZ zone priority is unrelated to is 50 by default.

A. True
B. False

Correct Answer: False

Concept: DMZ zone priority is directly related to is 50 by default.

Question 14 **True / False | Topic: Firewall Basics**

Untrust zone priority means that is 5 by default.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Untrust zone priority.

Question 15 **True / False | Topic: Firewall Basics**

Untrust zone priority is unrelated to is 5 by default.

A. True
B. False

Correct Answer: False

Concept: Untrust zone priority is directly related to is 5 by default.

Question 16 **True / False | Topic: Firewall Basics**

Local zone priority means that is 100 by default.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Local zone priority.

Question 17 **True / False | Topic: Firewall Basics**

Local zone priority is unrelated to is 100 by default.

A. True
B. False

Correct Answer: False

Concept: Local zone priority is directly related to is 100 by default.

Question 18 **True / False | Topic: Firewall Basics**

Default security policy action means that is deny.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Default security policy action.

Question 19 **True / False | Topic: Firewall Basics**

Default security policy action is unrelated to is deny.

A. True
B. False

Correct Answer: False

Concept: Default security policy action is directly related to is deny.

Question 20 **True / False | Topic: Firewall Basics**

Stateful firewall means that tracks connections in a session table.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Stateful firewall.

Question 21 **True / False | Topic: Firewall Basics**

Stateful firewall is unrelated to tracks connections in a session table.

A. True
B. False

Correct Answer: False

Concept: Stateful firewall is directly related to tracks connections in a session table.

Question 22 **True / False | Topic: Firewall Basics**

Packet-filtering firewall means that inspects packets individually without state.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Packet-filtering firewall.

Question 23 **True / False | Topic: Firewall Basics**

Packet-filtering firewall is unrelated to inspects packets individually without state.

A. True
B. False

Correct Answer: False

Concept: Packet-filtering firewall is directly related to inspects packets individually without state.

Question 24 **True / False | Topic: Firewall Basics**

ASPF means that inspects multi-channel protocols like FTP.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes ASPF.

Question 25 **True / False | Topic: Firewall Basics**

ASPF is unrelated to inspects multi-channel protocols like FTP.

A. True
B. False

Correct Answer: False

Concept: ASPF is directly related to inspects multi-channel protocols like FTP.

Question 26 **True / False | Topic: Firewall Basics**

Next-generation firewall means that includes application awareness and IPS.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Next-generation firewall.

Question 27 **True / False | Topic: Firewall Basics**

Next-generation firewall is unrelated to includes application awareness and IPS.

A. True
B. False

Correct Answer: False

Concept: Next-generation firewall is directly related to includes application awareness and IPS.

Question 28 **True / False | Topic: Firewall Basics**

Security zones means that group interfaces with similar trust levels.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Security zones.

Question 29 **True / False | Topic: Firewall Basics**

Security zones is unrelated to group interfaces with similar trust levels.

A. True
B. False

Correct Answer: False

Concept: Security zones is directly related to group interfaces with similar trust levels.

Question 30 **True / False | Topic: Firewall Basics**

Intra-zone traffic means that can be controlled by policies.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Intra-zone traffic.

Question 31 **True / False | Topic: Firewall Basics**

Intra-zone traffic is unrelated to can be controlled by policies.

A. True
B. False

Correct Answer: False

Concept: Intra-zone traffic is directly related to can be controlled by policies.

Question 32 **True / False | Topic: Firewall Basics**

Policy hit count means that helps verify which rules are used.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Policy hit count.

Question 33 **True / False | Topic: Firewall Basics**

Policy hit count is unrelated to helps verify which rules are used.

A. True
B. False

Correct Answer: False

Concept: Policy hit count is directly related to helps verify which rules are used.

Question 34 **True / False | Topic: Firewall Basics**

Security policy logging means that can be enabled for permit and deny actions.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Security policy logging.

Question 35 **True / False | Topic: Firewall Basics**

Security policy logging is unrelated to can be enabled for permit and deny actions.

A. True
B. False

Correct Answer: False

Concept: Security policy logging is directly related to can be enabled for permit and deny actions.

Question 36 **True / False | Topic: Firewall Basics**

Routing mode means that is a firewall deployment mode.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Routing mode.

Question 37 **True / False | Topic: Firewall Basics**

Routing mode is unrelated to is a firewall deployment mode.

A. True
B. False

Correct Answer: False

Concept: Routing mode is directly related to is a firewall deployment mode.

Question 38 **True / False | Topic: Firewall Basics**

Transparent mode means that is a firewall deployment mode.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Transparent mode.

Question 39 **True / False | Topic: Firewall Basics**

Transparent mode is unrelated to is a firewall deployment mode.

A. True
B. False

Correct Answer: False

Concept: Transparent mode is directly related to is a firewall deployment mode.

Question 40 **True / False | Topic: Firewall Basics**

Hybrid mode means that is a firewall deployment mode.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Hybrid mode.

Question 41 **True / False | Topic: Firewall Basics**

Hybrid mode is unrelated to is a firewall deployment mode.

A. True
B. False

Correct Answer: False

Concept: Hybrid mode is directly related to is a firewall deployment mode.

86 Security Zones

Question 1 **True / False | Topic: Security Zones**

The Local zone represents traffic destined to or originated from the firewall itself.

A. True
B. False

Correct Answer: True

Concept: Local zone traffic involves the firewall services and management interfaces.

87 Security Policies

Question 1 **True / False | Topic: Security Policies**

By default, Huawei firewalls permit all traffic unless a policy explicitly denies it.

A. True
B. False

Correct Answer: False

Concept: The default action is to deny traffic unless a security policy explicitly permits it.

88 Stateful Inspection

Question 1 **True / False | Topic: Stateful Inspection**

A stateful firewall can allow return traffic for an established session without a separate policy.

A. True
B. False

Correct Answer: True

Concept: Return packets matching an existing session table entry are permitted automatically.

89 ASPF

Question 1 **True / False | Topic: ASPF**

ASPF is used to support protocols that open secondary or dynamic connections.

A. True
B. False

Correct Answer: True

Concept: ASPF inspects application-layer negotiation to dynamically permit related channels.

90 NAT

Question 1 **True / False | Topic: NAT**

NAT can hide internal network topology from external observers.

A. True
B. False

Correct Answer: True

Concept: NAT translates private addresses, making internal topology invisible from outside.

Question 2 **True / False | Topic: NAT**

Easy-IP uses a fixed public IP address pool.

A. True
B. False

Correct Answer: False

Concept: Easy-IP uses the firewall egress interface IP address dynamically.

Question 3 **True / False | Topic: NAT**

NAT ALG is required for HTTP because HTTP embeds IP addresses in payloads.

A. True
B. False

Correct Answer: False

Concept: HTTP generally does not embed IP addresses in payloads, so ALG is not required.

Question 4 **True / False | Topic: NAT**

NAPT conserves public IPv4 addresses by using port multiplexing.

A. True
B. False

Correct Answer: True

Concept: NAPT maps many private addresses to one public IP using unique port numbers.

Question 5 **True / False | Topic: NAT**

FTP data channel failure can be caused by disabled NAT ALG.

A. True
B. False

Correct Answer: True

Concept: Without ALG, the firewall cannot identify and permit the negotiated FTP data channel.

Question 6 **True / False | Topic: NAT**

NAPT translates both IP addresses and port numbers.

A. True
B. False

Correct Answer: True

Concept: NAPT uses port multiplexing for many-to-one translation.

Question 7 **True / False | Topic: NAT**

NAT Server is used for outbound Internet access.

A. True
B. False

Correct Answer: False

Concept: NAT Server publishes inbound services.

Question 8 **True / False | Topic: NAT**

Static NAT preserves a one-to-one public-private mapping.

A. True
B. False

Correct Answer: True

Concept: Static NAT maps one public IP to one private IP consistently.

Question 9 **True / False | Topic: NAT**

NAT means that translates IP addresses between private and public.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes NAT.

Question 10 **True / False | Topic: NAT**

NAT is unrelated to translates IP addresses between private and public.

A. True
B. False

Correct Answer: False

Concept: NAT is directly related to translates IP addresses between private and public.

Question 11 **True / False | Topic: NAT**

Source NAT means that translates the source IP of outbound packets.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Source NAT.

Question 12 True / False | Topic: NAT

Source NAT is unrelated to translates the source IP of outbound packets.

A. True
B. False

Correct Answer: False

Concept: Source NAT is directly related to translates the source IP of outbound packets.

Question 13 True / False | Topic: NAT

Destination NAT means that translates the destination IP of inbound packets.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Destination NAT.

Question 14 True / False | Topic: NAT

Destination NAT is unrelated to translates the destination IP of inbound packets.

A. True
B. False

Correct Answer: False

Concept: Destination NAT is directly related to translates the destination IP of inbound packets.

Question 15 True / False | Topic: NAT

NAT Server means that publishes internal services to the Internet.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes NAT Server.

Question 16 True / False | Topic: NAT

NAT Server is unrelated to publishes internal services to the Internet.

A. True
B. False

Correct Answer: False

Concept: NAT Server is directly related to publishes internal services to the Internet.

Question 17 **True / False | Topic: NAT**

Bidirectional NAT means that translates both source and destination addresses.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Bidirectional NAT.

Question 18 **True / False | Topic: NAT**

Bidirectional NAT is unrelated to translates both source and destination addresses.

A. True
B. False

Correct Answer: False

Concept: Bidirectional NAT is directly related to translates both source and destination addresses.

Question 19 **True / False | Topic: NAT**

NAPT means that translates many private addresses to one public IP using ports.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes NAPT.

Question 20 **True / False | Topic: NAT**

NAPT is unrelated to translates many private addresses to one public IP using ports.

A. True
B. False

Correct Answer: False

Concept: NAPT is directly related to translates many private addresses to one public IP using ports.

Question 21 **True / False | Topic: NAT**

NAT No-PAT means that performs one-to-one address translation preserving ports.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes NAT No-PAT.

Question 22 **True / False | Topic: NAT**

NAT No-PAT is unrelated to performs one-to-one address translation preserving ports.

A. True
B. False

Correct Answer: False

Concept: NAT No-PAT is directly related to performs one-to-one address translation preserving ports.

Question 23 **True / False | Topic: NAT**

Easy-IP means that uses the outbound interface IP for translation.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Easy-IP.

Question 24 **True / False | Topic: NAT**

Easy-IP is unrelated to uses the outbound interface IP for translation.

A. True
B. False

Correct Answer: False

Concept: Easy-IP is directly related to uses the outbound interface IP for translation.

Question 25 **True / False | Topic: NAT**

NAT ALG means that rewrites addresses inside application payloads.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes NAT ALG.

Question 26 **True / False | Topic: NAT**

NAT ALG is unrelated to rewrites addresses inside application payloads.

A. True
B. False

Correct Answer: False

Concept: NAT ALG is directly related to rewrites addresses inside application payloads.

Question 27 **True / False | Topic: NAT**

RFC 1918 means that defines private IPv4 address ranges.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes RFC 1918.

Question 28 **True / False | Topic: NAT**

RFC 1918 is unrelated to defines private IPv4 address ranges.

A. True
B. False

Correct Answer: False

Concept: RFC 1918 is directly related to defines private IPv4 address ranges.

Question 29 **True / False | Topic: NAT**

Hairpin NAT means that lets internal users access servers via public NAT address.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Hairpin NAT.

Question 30 **True / False | Topic: NAT**

Hairpin NAT is unrelated to lets internal users access servers via public NAT address.

A. True
B. False

Correct Answer: False

Concept: Hairpin NAT is directly related to lets internal users access servers via public NAT address.

Question 31 **True / False | Topic: NAT**

Static NAT means that provides a persistent one-to-one mapping.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Static NAT.

Question 32 — True / False | Topic: NAT

Static NAT is unrelated to provides a persistent one-to-one mapping.

A. True
B. False

Correct Answer: False

Concept: Static NAT is directly related to provides a persistent one-to-one mapping.

91 NAT Basics

Question 1 — True / False | Topic: NAT Basics

NAT can conserve public IPv4 addresses.

A. True
B. False

Correct Answer: True

Concept: NAT allows many private hosts to share a smaller number of public IP addresses.

92 Source NAT

Question 1 — True / False | Topic: Source NAT

NAPT translates both IP addresses and port numbers.

A. True
B. False

Correct Answer: True

Concept: NAPT multiplexes many private addresses onto one public IP by translating ports.

93 Destination NAT

Question 1 — True / False | Topic: Destination NAT

Destination NAT is typically used for outbound Internet access.

A. True
B. False

Correct Answer: False

Concept: Destination NAT is used for inbound traffic to expose internal services.

94 NAT Server

Question 1 — True / False | Topic: NAT Server

NAT Server maps public IP addresses to private server addresses.

A. True
B. False

Correct Answer: True

Concept: NAT Server publishes internal services by mapping public IPs/ports to private servers.

95 NAT ALG

Question 1 — True / False | Topic: NAT ALG

NAT ALG is needed when protocols embed IP addresses inside application payloads.

A. True
B. False

Correct Answer: True

Concept: ALG inspects application-layer data to update embedded addresses/ports.

96 Firewall Hot Standby

Question 1 — True / False | Topic: Firewall Hot Standby

In active/standby mode, both firewalls actively forward the same traffic.

A. True
B. False

Correct Answer: False

Concept: Only the active firewall forwards traffic; the standby takes over upon failure.

Question 2 — True / False | Topic: Firewall Hot Standby

VRRP provides default gateway redundancy for end hosts.

A. True
B. False

Correct Answer: True

Concept: VRRP allows multiple routers to act as a single virtual gateway.

Question 3 — True / False | Topic: Firewall Hot Standby

In VRRP, only the router with the physical IP matching the virtual IP can become master.

A. True
B. False

Correct Answer: False

Concept: Any router with sufficient priority can become master, not just the IP owner.

Question 4 — True / False | Topic: Firewall Hot Standby

VGMP prevents individual VRRP groups from becoming inconsistent during failover.

A. True
B. False

Correct Answer: True

Concept: VGMP manages multiple VRRP groups as a single entity.

Question 5 — True / False | Topic: Firewall Hot Standby

HRP can synchronize NAT sessions between active and standby firewalls.

A. True
B. False

Correct Answer: True

Concept: HRP keeps NAT/session state consistent across hot-standby peers.

Question 6 — True / False | Topic: Firewall Hot Standby

Load-sharing mode improves resource utilization compared to active/standby mode.

A. True
B. False

Correct Answer: True

Concept: Multiple firewalls forward traffic simultaneously in load-sharing mode.

Question 7 — True / False | Topic: Firewall Hot Standby

The HRP heartbeat link can share bandwidth with regular user traffic without risk.

A. True
B. False

Correct Answer: False

Concept: Sharing heartbeat and user traffic can cause false failover due to congestion.

Question 8 True / False | Topic: Firewall Hot Standby

VRRP provides a virtual gateway IP.

A. True
B. False

Correct Answer: True

Concept: VRRP allows multiple routers to share a virtual IP.

Question 9 True / False | Topic: Firewall Hot Standby

HRP synchronizes only configurations.

A. True
B. False

Correct Answer: False

Concept: HRP synchronizes configurations and runtime state.

Question 10 True / False | Topic: Firewall Hot Standby

Load-sharing mode allows multiple firewalls to forward traffic.

A. True
B. False

Correct Answer: True

Concept: Multiple active firewalls share the traffic load.

Question 11 True / False | Topic: Firewall Hot Standby

VRRP means that provides gateway redundancy with a virtual IP.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes VRRP.

Question 12 True / False | Topic: Firewall Hot Standby

VRRP is unrelated to provides gateway redundancy with a virtual IP.

A. True
B. False

Correct Answer: False

Concept: VRRP is directly related to provides gateway redundancy with a virtual IP.

Question 13 **True / False | Topic: Firewall Hot Standby**

VRRP master means that forwards traffic for the virtual IP.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes VRRP master.

Question 14 **True / False | Topic: Firewall Hot Standby**

VRRP master is unrelated to forwards traffic for the virtual IP.

A. True
B. False

Correct Answer: False

Concept: VRRP master is directly related to forwards traffic for the virtual IP.

Question 15 **True / False | Topic: Firewall Hot Standby**

VRRP backup means that takes over if the master fails.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes VRRP backup.

Question 16 **True / False | Topic: Firewall Hot Standby**

VRRP backup is unrelated to takes over if the master fails.

A. True
B. False

Correct Answer: False

Concept: VRRP backup is directly related to takes over if the master fails.

Question 17 **True / False | Topic: Firewall Hot Standby**

VGMP means that manages VRRP group states consistently.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes VGMP.

Question 18 True / False | Topic: Firewall Hot Standby

VGMP is unrelated to manages VRRP group states consistently.

A. True
B. False

Correct Answer: False

Concept: VGMP is directly related to manages VRRP group states consistently.

Question 19 True / False | Topic: Firewall Hot Standby

HRP means that synchronizes sessions and configurations.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes HRP.

Question 20 True / False | Topic: Firewall Hot Standby

HRP is unrelated to synchronizes sessions and configurations.

A. True
B. False

Correct Answer: False

Concept: HRP is directly related to synchronizes sessions and configurations.

Question 21 True / False | Topic: Firewall Hot Standby

Active/standby mode means that has one firewall forwarding traffic.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Active/standby mode.

Question 22 True / False | Topic: Firewall Hot Standby

Active/standby mode is unrelated to has one firewall forwarding traffic.

A. True
B. False

Correct Answer: False

Concept: Active/standby mode is directly related to has one firewall forwarding traffic.

Question 23 **True / False | Topic: Firewall Hot Standby**

Load-sharing mode means that has multiple firewalls forwarding traffic.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Load-sharing mode.

Question 24 **True / False | Topic: Firewall Hot Standby**

Load-sharing mode is unrelated to has multiple firewalls forwarding traffic.

A. True
B. False

Correct Answer: False

Concept: Load-sharing mode is directly related to has multiple firewalls forwarding traffic.

Question 25 **True / False | Topic: Firewall Hot Standby**

Heartbeat link means that should be dedicated and reliable.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Heartbeat link.

Question 26 **True / False | Topic: Firewall Hot Standby**

Heartbeat link is unrelated to should be dedicated and reliable.

A. True
B. False

Correct Answer: False

Concept: Heartbeat link is directly related to should be dedicated and reliable.

Question 27 **True / False | Topic: Firewall Hot Standby**

HRP session backup means that ensures connections survive failover.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes HRP session backup.

Question 28 **True / False | Topic: Firewall Hot Standby**

HRP session backup is unrelated to ensures connections survive failover.

A. True
B. False

Correct Answer: False

Concept: HRP session backup is directly related to ensures connections survive failover.

Question 29 **True / False | Topic: Firewall Hot Standby**

Preempt delay means that reduces rapid master-backup flapping.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Preempt delay.

Question 30 **True / False | Topic: Firewall Hot Standby**

Preempt delay is unrelated to reduces rapid master-backup flapping.

A. True
B. False

Correct Answer: False

Concept: Preempt delay is directly related to reduces rapid master-backup flapping.

97 VRRP

Question 1 **True / False | Topic: VRRP**

VRRP uses a virtual IP address shared between multiple physical routers.

A. True
B. False

Correct Answer: True

Concept: VRRP allows multiple routers to present a single virtual gateway IP to hosts.

98 VGMP

Question 1 **True / False | Topic: VGMP**

VGMP ensures that VRRP groups on a firewall switch state consistently.

A. True
B. False

Correct Answer: True

Concept: VGMP prevents split-brain scenarios by managing all VRRP groups as a single entity.

99 HRP

Question 1 **True / False | Topic: HRP**

HRP only synchronizes configuration files, not session tables.

A. True
B. False

Correct Answer: False

Concept: HRP synchronizes both configurations and runtime state such as session tables.

100 Intrusion Prevention and Antivirus

Question 1 **True / False | Topic: Intrusion Prevention and Antivirus**

IPS is typically deployed inline to block threats.

A. True
B. False

Correct Answer: True

Concept: Inline deployment allows IPS to actively block malicious traffic.

Question 2 **True / False | Topic: Intrusion Prevention and Antivirus**

Anomaly-based detection requires a database of known attack signatures.

A. True
B. False

Correct Answer: False

Concept: Anomaly detection uses baselines of normal behavior, not attack signatures.

Question 3 **True / False | Topic: Intrusion Prevention and Antivirus**

Heuristic analysis can detect previously unknown malware.

A. True
B. False

Correct Answer: True

Concept: Heuristics identify suspicious behavior even without a known signature.

Question 4 **True / False | Topic: Intrusion Prevention and Antivirus**

Signature-based antivirus is effective against all zero-day threats.

A. True
B. False

Correct Answer: False

Concept: Zero-day threats lack signatures, so signature-only detection cannot catch them.

Question 5 **True / False | Topic: Intrusion Prevention and Antivirus**

An IDS can block malicious traffic inline.

A. True
B. False

Correct Answer: False

Concept: IDS typically detects and alerts; IPS blocks inline.

Question 6 **True / False | Topic: Intrusion Prevention and Antivirus**

Heuristic detection can identify unknown threats.

A. True
B. False

Correct Answer: True

Concept: Heuristics analyze behavior to find unknown malware.

Question 7 **True / False | Topic: Intrusion Prevention and Antivirus**

Sandboxing executes files in an isolated environment.

A. True
B. False

Correct Answer: True

Concept: Sandboxing observes behavior without risking production systems.

Question 8 **True / False | Topic: Intrusion Prevention and Antivirus**

IDS means that detects intrusions and generates alerts.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes IDS.

Question 9 **True / False | Topic: Intrusion Prevention and Antivirus**

IDS is unrelated to detects intrusions and generates alerts.

A. True
B. False

Correct Answer: False

Concept: IDS is directly related to detects intrusions and generates alerts.

Question 10 **True / False | Topic: Intrusion Prevention and Antivirus**

IPS means that detects and blocks intrusions inline.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes IPS.

Question 11 **True / False | Topic: Intrusion Prevention and Antivirus**

IPS is unrelated to detects and blocks intrusions inline.

A. True
B. False

Correct Answer: False

Concept: IPS is directly related to detects and blocks intrusions inline.

Question 12 **True / False | Topic: Intrusion Prevention and Antivirus**

Signature detection means that matches known attack patterns.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Signature detection.

Question 13 **True / False | Topic: Intrusion Prevention and Antivirus**

Signature detection is unrelated to matches known attack patterns.

A. True
B. False

Correct Answer: False

Concept: Signature detection is directly related to matches known attack patterns.

Question 14 **True / False | Topic: Intrusion Prevention and Antivirus**

Anomaly detection means that flags deviations from normal behavior.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Anomaly detection.

Question 15 **True / False | Topic: Intrusion Prevention and Antivirus**

Anomaly detection is unrelated to flags deviations from normal behavior.

A. True
B. False

Correct Answer: False

Concept: Anomaly detection is directly related to flags deviations from normal behavior.

Question 16 **True / False | Topic: Intrusion Prevention and Antivirus**

False positive means that incorrectly blocks legitimate traffic.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes False positive.

Question 17 **True / False | Topic: Intrusion Prevention and Antivirus**

False positive is unrelated to incorrectly blocks legitimate traffic.

A. True
B. False

Correct Answer: False

Concept: False positive is directly related to incorrectly blocks legitimate traffic.

Question 18 **True / False | Topic: Intrusion Prevention and Antivirus**

False negative means that misses an actual attack.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes False negative.

Question 19 **True / False | Topic: Intrusion Prevention and Antivirus**

False negative is unrelated to misses an actual attack.

A. True
B. False

Correct Answer: False

Concept: False negative is directly related to misses an actual attack.

Question 20 **True / False | Topic: Intrusion Prevention and Antivirus**

Sandboxing means that observes file behavior in an isolated environment.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Sandboxing.

Question 21 **True / False | Topic: Intrusion Prevention and Antivirus**

Sandboxing is unrelated to observes file behavior in an isolated environment.

A. True
B. False

Correct Answer: False

Concept: Sandboxing is directly related to observes file behavior in an isolated environment.

Question 22 **True / False | Topic: Intrusion Prevention and Antivirus**

Heuristic analysis means that detects malware by examining behavior.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Heuristic analysis.

Question 23 **True / False | Topic: Intrusion Prevention and Antivirus**

Heuristic analysis is unrelated to detects malware by examining behavior.

A. True
B. False

Correct Answer: False

Concept: Heuristic analysis is directly related to detects malware by examining behavior.

Question 24 **True / False | Topic: Intrusion Prevention and Antivirus**

Whitelisting means that allows only approved applications.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Whitelisting.

Question 25 **True / False | Topic: Intrusion Prevention and Antivirus**

Whitelisting is unrelated to allows only approved applications.

A. True
B. False

Correct Answer: False

Concept: Whitelisting is directly related to allows only approved applications.

Question 26 **True / False | Topic: Intrusion Prevention and Antivirus**

Antivirus signatures means that must be updated regularly.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Antivirus signatures.

Question 27 **True / False | Topic: Intrusion Prevention and Antivirus**

Antivirus signatures is unrelated to must be updated regularly.

A. True
B. False

Correct Answer: False

Concept: Antivirus signatures is directly related to must be updated regularly.

Question 28 **True / False | Topic: Intrusion Prevention and Antivirus**

NIDS means that monitors network traffic for intrusions.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes NIDS.

Question 29 **True / False | Topic: Intrusion Prevention and Antivirus**

NIDS is unrelated to monitors network traffic for intrusions.

A. True
B. False

Correct Answer: False

Concept: NIDS is directly related to monitors network traffic for intrusions.

Question 30 **True / False | Topic: Intrusion Prevention and Antivirus**

HIDS means that monitors host activity for intrusions.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes HIDS.

Question 31 **True / False | Topic: Intrusion Prevention and Antivirus**

HIDS is unrelated to monitors host activity for intrusions.

A. True
B. False

Correct Answer: False

Concept: HIDS is directly related to monitors host activity for intrusions.

Question 32 **True / False | Topic: Intrusion Prevention and Antivirus**

Reconnaissance means that is an early attack phase.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Reconnaissance.

Question 33 **True / False | Topic: Intrusion Prevention and Antivirus**

Reconnaissance is unrelated to is an early attack phase.

A. True
B. False

Correct Answer: False

Concept: Reconnaissance is directly related to is an early attack phase.

Question 34 **True / False | Topic: Intrusion Prevention and Antivirus**

Exploitation means that gains unauthorized access using vulnerabilities.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Exploitation.

Question 35 — True / False | Topic: Intrusion Prevention and Antivirus

Exploitation is unrelated to gains unauthorized access using vulnerabilities.

A. True
B. False

Correct Answer: False

Concept: Exploitation is directly related to gains unauthorized access using vulnerabilities.

101 Intrusion Prevention

Question 1 — True / False | Topic: Intrusion Prevention

An IDS is typically deployed inline and can block attacks.

A. True
B. False

Correct Answer: False

Concept: IDS is usually passive/out-of-band and only detects and alerts; IPS blocks inline.

Question 2 — True / False | Topic: Intrusion Prevention

Signature-based IPS can detect zero-day attacks effectively.

A. True
B. False

Correct Answer: False

Concept: Signatures require known attack patterns; zero-day attacks are unknown and often missed.

102 Antivirus

Question 1 — True / False | Topic: Antivirus

Antivirus signature databases must be updated regularly to detect new malware.

A. True
B. False

Correct Answer: True

Concept: New malware variants require updated signatures and detection engines.

Question 2 — True / False | Topic: Antivirus

Sandboxing can detect unknown malware by observing its runtime behavior.

A. True
B. False

Correct Answer: True

Concept: Sandboxing reveals malicious behavior of unknown samples in an isolated environment.

103 AAA and User Authentication

Question 1 **True / False | Topic: AAA and User Authentication**

Authentication verifies who a user is.

A. True
B. False

Correct Answer: True

Concept: Authentication confirms identity before granting access.

Question 2 **True / False | Topic: AAA and User Authentication**

TACACS+ separates authentication, authorization, and accounting.

A. True
B. False

Correct Answer: True

Concept: TACACS+ provides independent AAA functions.

Question 3 **True / False | Topic: AAA and User Authentication**

RADIUS encrypts the entire authentication request.

A. True
B. False

Correct Answer: False

Concept: RADIUS encrypts only the password attribute.

Question 4 **True / False | Topic: AAA and User Authentication**

Local authentication does not require an external server.

A. True
B. False

Correct Answer: True

Concept: Local authentication uses the firewall own user database.

Question 5 **True / False | Topic: AAA and User Authentication**

Authorization happens after authentication.

A. True
B. False

Correct Answer: True

Concept: Authorization defines what an authenticated user may do.

Question 6 **True / False | Topic: AAA and User Authentication**

TACACS+ uses UDP.

A. True
B. False

Correct Answer: False

Concept: TACACS+ uses TCP.

Question 7 **True / False | Topic: AAA and User Authentication**

Portal authentication requires a web browser.

A. True
B. False

Correct Answer: True

Concept: Portal presents a web login page to users.

Question 8 **True / False | Topic: AAA and User Authentication**

Authentication means that verifies user identity.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Authentication.

Question 9 **True / False | Topic: AAA and User Authentication**

Authentication is unrelated to verifies user identity.

A. True
B. False

Correct Answer: False

Concept: Authentication is directly related to verifies user identity.

Question 10 **True / False | Topic: AAA and User Authentication**

Authorization means that determines what a user can access.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Authorization.

Question 11 **True / False | Topic: AAA and User Authentication**

Authorization is unrelated to determines what a user can access.

A. True
B. False

Correct Answer: False

Concept: Authorization is directly related to determines what a user can access.

Question 12 **True / False | Topic: AAA and User Authentication**

Accounting means that records user activities and resource usage.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Accounting.

Question 13 **True / False | Topic: AAA and User Authentication**

Accounting is unrelated to records user activities and resource usage.

A. True
B. False

Correct Answer: False

Concept: Accounting is directly related to records user activities and resource usage.

Question 14 **True / False | Topic: AAA and User Authentication**

RADIUS means that uses UDP ports 1812 and 1813.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes RADIUS.

Question 15 **True / False | Topic: AAA and User Authentication**

RADIUS is unrelated to uses UDP ports 1812 and 1813.

A. True
B. False

Correct Answer: False

Concept: RADIUS is directly related to uses UDP ports 1812 and 1813.

Question 16 **True / False | Topic: AAA and User Authentication**

TACACS+ means that uses TCP port 49.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes TACACS+.

Question 17 **True / False | Topic: AAA and User Authentication**

TACACS+ is unrelated to uses TCP port 49.

A. True
B. False

Correct Answer: False

Concept: TACACS+ is directly related to uses TCP port 49.

Question 18 **True / False | Topic: AAA and User Authentication**

Portal authentication means that uses a web login page.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Portal authentication.

Question 19 **True / False | Topic: AAA and User Authentication**

Portal authentication is unrelated to uses a web login page.

A. True
B. False

Correct Answer: False

Concept: Portal authentication is directly related to uses a web login page.

Question 20 **True / False | Topic: AAA and User Authentication**

802.1X means that provides port-based network access control.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes 802.1X.

Question 21 **True / False | Topic: AAA and User Authentication**

802.1X is unrelated to provides port-based network access control.

A. True
B. False

Correct Answer: False

Concept: 802.1X is directly related to provides port-based network access control.

Question 22 **True / False | Topic: AAA and User Authentication**

Local authentication means that uses the device user database.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Local authentication.

Question 23 **True / False | Topic: AAA and User Authentication**

Local authentication is unrelated to uses the device user database.

A. True
B. False

Correct Answer: False

Concept: Local authentication is directly related to uses the device user database.

Question 24 **True / False | Topic: AAA and User Authentication**

LDAP means that queries a directory server for authentication.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes LDAP.

Question 25 **True / False | Topic: AAA and User Authentication**

LDAP is unrelated to queries a directory server for authentication.

A. True
B. False

Correct Answer: False

Concept: LDAP is directly related to queries a directory server for authentication.

Question 26 **True / False | Topic: AAA and User Authentication**

User group means that applies policies to multiple users collectively.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes User group.

Question 27 **True / False | Topic: AAA and User Authentication**

User group is unrelated to applies policies to multiple users collectively.

A. True
B. False

Correct Answer: False

Concept: User group is directly related to applies policies to multiple users collectively.

Question 28 **True / False | Topic: AAA and User Authentication**

SSO means that allows one authentication for multiple services.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes SSO.

Question 29 **True / False | Topic: AAA and User Authentication**

SSO is unrelated to allows one authentication for multiple services.

A. True
B. False

Correct Answer: False

Concept: SSO is directly related to allows one authentication for multiple services.

104 AAA Basics

Question 1 **True / False | Topic: AAA Basics**

Authorization determines whether an authenticated user is allowed to perform an action.

A. True
B. False

Correct Answer: True

Concept: Authorization enforces access control decisions based on identity and policy.

Question 2 — True / False | Topic: AAA Basics

Accounting in AAA records user activities and resource usage.

A. True
B. False

Correct Answer: True

Concept: Accounting provides logs for auditing, billing, and forensic analysis.

105 Firewall User Authentication

Question 1 — True / False | Topic: Firewall User Authentication

Portal authentication is commonly used for guest access and web-based login.

A. True
B. False

Correct Answer: True

Concept: Portal authentication provides a convenient web-based login experience.

Question 2 — True / False | Topic: Firewall User Authentication

RADIUS encrypts the entire authentication packet including the username.

A. True
B. False

Correct Answer: False

Concept: RADIUS encrypts only the password field; the rest of the packet is not encrypted.

106 Cryptography and PKI

Question 1 — True / False | Topic: Cryptography and PKI

AES is a symmetric encryption algorithm.

A. True
B. False

Correct Answer: True

Concept: AES uses the same key for encryption and decryption.

Question 2 — True / False | Topic: Cryptography and PKI

RSA is faster than AES for bulk data encryption.

A. True
B. False

Correct Answer: False

Concept: RSA is slower and typically used for key exchange or signatures, not bulk encryption.

Question 3 **True / False | Topic: Cryptography and PKI**

SHA-256 produces a 256-bit digest.

A. True
B. False

Correct Answer: True

Concept: SHA-256 generates a fixed 256-bit output.

Question 4 **True / False | Topic: Cryptography and PKI**

Hash functions are reversible.

A. True
B. False

Correct Answer: False

Concept: Hash functions are designed to be one-way.

Question 5 **True / False | Topic: Cryptography and PKI**

A digital signature provides non-repudiation.

A. True
B. False

Correct Answer: True

Concept: Digital signatures prove the signer created the message and cannot easily deny it.

Question 6 **True / False | Topic: Cryptography and PKI**

A root CA certificate is typically signed by another CA.

A. True
B. False

Correct Answer: False

Concept: Root CA certificates are self-signed and act as trust anchors.

Question 7 **True / False | Topic: Cryptography and PKI**

OCSP provides real-time certificate revocation checking.

A. True
B. False

Correct Answer: True

Concept: OCSP allows clients to query revocation status online.

Question 8 — **True / False | Topic: Cryptography and PKI**

A hash function output can be reversed to find the input.

A. True
B. False

Correct Answer: False

Concept: Hash functions are one-way.

Question 9 — **True / False | Topic: Cryptography and PKI**

A digital signature verifies integrity and authenticity.

A. True
B. False

Correct Answer: True

Concept: Signatures prove the message came from the signer and was not altered.

Question 10 — **True / False | Topic: Cryptography and PKI**

RSA is a symmetric encryption algorithm.

A. True
B. False

Correct Answer: False

Concept: RSA is asymmetric.

Question 11 — **True / False | Topic: Cryptography and PKI**

Symmetric encryption means that uses the same key for encryption and decryption.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Symmetric encryption.

Question 12 — **True / False | Topic: Cryptography and PKI**

Symmetric encryption is unrelated to uses the same key for encryption and decryption.

A. True
B. False

Correct Answer: False

Concept: Symmetric encryption is directly related to uses the same key for encryption and decryption.

Question 13 **True / False | Topic: Cryptography and PKI**

Asymmetric encryption means that uses public and private key pairs.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Asymmetric encryption.

Question 14 **True / False | Topic: Cryptography and PKI**

Asymmetric encryption is unrelated to uses public and private key pairs.

A. True
B. False

Correct Answer: False

Concept: Asymmetric encryption is directly related to uses public and private key pairs.

Question 15 **True / False | Topic: Cryptography and PKI**

AES means that is a symmetric encryption standard.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes AES.

Question 16 **True / False | Topic: Cryptography and PKI**

AES is unrelated to is a symmetric encryption standard.

A. True
B. False

Correct Answer: False

Concept: AES is directly related to is a symmetric encryption standard.

Question 17 **True / False | Topic: Cryptography and PKI**

RSA means that is an asymmetric encryption algorithm.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes RSA.

Question 18 **True / False | Topic: Cryptography and PKI**

RSA is unrelated to is an asymmetric encryption algorithm.

A. True
B. False

Correct Answer: False

Concept: RSA is directly related to is an asymmetric encryption algorithm.

Question 19 **True / False | Topic: Cryptography and PKI**

DES means that uses a 56-bit effective key.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes DES.

Question 20 **True / False | Topic: Cryptography and PKI**

DES is unrelated to uses a 56-bit effective key.

A. True
B. False

Correct Answer: False

Concept: DES is directly related to uses a 56-bit effective key.

Question 21 **True / False | Topic: Cryptography and PKI**

Hash function means that produces a fixed-length digest.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Hash function.

Question 22 **True / False | Topic: Cryptography and PKI**

Hash function is unrelated to produces a fixed-length digest.

A. True
B. False

Correct Answer: False

Concept: Hash function is directly related to produces a fixed-length digest.

Question 23 **True / False | Topic: Cryptography and PKI**

MD5 means that is considered broken due to collision attacks.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes MD5.

Question 24 **True / False | Topic: Cryptography and PKI**

MD5 is unrelated to is considered broken due to collision attacks.

A. True
B. False

Correct Answer: False

Concept: MD5 is directly related to is considered broken due to collision attacks.

Question 25 **True / False | Topic: Cryptography and PKI**

SHA-256 means that produces a 256-bit digest.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes SHA-256.

Question 26 **True / False | Topic: Cryptography and PKI**

SHA-256 is unrelated to produces a 256-bit digest.

A. True
B. False

Correct Answer: False

Concept: SHA-256 is directly related to produces a 256-bit digest.

Question 27 **True / False | Topic: Cryptography and PKI**

HMAC means that provides message authentication using a hash and key.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes HMAC.

Question 28 **True / False | Topic: Cryptography and PKI**

HMAC is unrelated to provides message authentication using a hash and key.

A. True
B. False

Correct Answer: False

Concept: HMAC is directly related to provides message authentication using a hash and key.

Question 29 **True / False | Topic: Cryptography and PKI**

Digital signature means that verifies authenticity and integrity.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Digital signature.

Question 30 **True / False | Topic: Cryptography and PKI**

Digital signature is unrelated to verifies authenticity and integrity.

A. True
B. False

Correct Answer: False

Concept: Digital signature is directly related to verifies authenticity and integrity.

Question 31 **True / False | Topic: Cryptography and PKI**

Public key means that is used to verify a digital signature.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Public key.

Question 32 **True / False | Topic: Cryptography and PKI**

Public key is unrelated to is used to verify a digital signature.

A. True
B. False

Correct Answer: False

Concept: Public key is directly related to is used to verify a digital signature.

Question 33 **True / False | Topic: Cryptography and PKI**

Private key means that is used to create a digital signature.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Private key.

Question 34 **True / False | Topic: Cryptography and PKI**

Private key is unrelated to is used to create a digital signature.

A. True
B. False

Correct Answer: False

Concept: Private key is directly related to is used to create a digital signature.

Question 35 **True / False | Topic: Cryptography and PKI**

PKI means that manages public keys and certificates.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes PKI.

Question 36 **True / False | Topic: Cryptography and PKI**

PKI is unrelated to manages public keys and certificates.

A. True
B. False

Correct Answer: False

Concept: PKI is directly related to manages public keys and certificates.

Question 37 **True / False | Topic: Cryptography and PKI**

CA means that issues and signs digital certificates.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes CA.

Question 38 **True / False | Topic: Cryptography and PKI**

CA is unrelated to issues and signs digital certificates.

A. True
B. False

Correct Answer: False

Concept: CA is directly related to issues and signs digital certificates.

Question 39 **True / False | Topic: Cryptography and PKI**

RA means that validates identity before certificate issuance.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes RA.

Question 40 **True / False | Topic: Cryptography and PKI**

RA is unrelated to validates identity before certificate issuance.

A. True
B. False

Correct Answer: False

Concept: RA is directly related to validates identity before certificate issuance.

Question 41 **True / False | Topic: Cryptography and PKI**

CRL means that lists revoked certificates.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes CRL.

Question 42 **True / False | Topic: Cryptography and PKI**

CRL is unrelated to lists revoked certificates.

A. True
B. False

Correct Answer: False

Concept: CRL is directly related to lists revoked certificates.

Question 43 **True / False | Topic: Cryptography and PKI**

OCSP means that provides online certificate status checking.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes OCSP.

Question 44 **True / False | Topic: Cryptography and PKI**

OCSP is unrelated to provides online certificate status checking.

A. True
B. False

Correct Answer: False

Concept: OCSP is directly related to provides online certificate status checking.

Question 45 **True / False | Topic: Cryptography and PKI**

X.509 means that is the standard certificate format.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes X.509.

Question 46 **True / False | Topic: Cryptography and PKI**

X.509 is unrelated to is the standard certificate format.

A. True
B. False

Correct Answer: False

Concept: X.509 is directly related to is the standard certificate format.

Question 47 **True / False | Topic: Cryptography and PKI**

Root CA certificate means that is self-signed and acts as a trust anchor.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Root CA certificate.

Question 48 **True / False | Topic: Cryptography and PKI**

Root CA certificate is unrelated to is self-signed and acts as a trust anchor.

A. True
B. False

Correct Answer: False

Concept: Root CA certificate is directly related to is self-signed and acts as a trust anchor.

Question 49 **True / False | Topic: Cryptography and PKI**

GCM means that provides authenticated encryption.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes GCM.

Question 50 **True / False | Topic: Cryptography and PKI**

GCM is unrelated to provides authenticated encryption.

A. True
B. False

Correct Answer: False

Concept: GCM is directly related to provides authenticated encryption.

Question 51 **True / False | Topic: Cryptography and PKI**

ECB means that should be avoided because it reveals plaintext patterns.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes ECB.

Question 52 **True / False | Topic: Cryptography and PKI**

ECB is unrelated to should be avoided because it reveals plaintext patterns.

A. True
B. False

Correct Answer: False

Concept: ECB is directly related to should be avoided because it reveals plaintext patterns.

107 Cryptography Basics

Question 1 — **True / False | Topic: Cryptography Basics**

Symmetric encryption is generally faster than asymmetric encryption.

A. True
B. False

Correct Answer: True

Concept: Symmetric algorithms are faster and suitable for bulk data encryption.

Question 2 — **True / False | Topic: Cryptography Basics**

Asymmetric encryption is ideal for encrypting large amounts of data directly.

A. True
B. False

Correct Answer: False

Concept: Asymmetric encryption is slower; it is often used for key exchange and signatures, while symmetric encryption handles bulk data.

108 Hash Algorithms

Question 1 — **True / False | Topic: Hash Algorithms**

A hash function can be reversed to recover the original input.

A. True
B. False

Correct Answer: False

Concept: Hash functions are one-way; the original input cannot be derived from the digest.

109 PKI

Question 1 — **True / False | Topic: PKI**

A digital certificate binds a public key to an identity.

A. True
B. False

Correct Answer: True

Concept: Certificates contain a public key and identity information, signed by a trusted CA.

Question 2 — True / False | Topic: PKI

A CRL lists certificates that are still valid.

A. True
B. False

Correct Answer: False

Concept: A CRL lists certificates that have been revoked and should no longer be trusted.

110 VPN

Question 1 — True / False | Topic: VPN

A VPN allows secure communication over an untrusted public network.

A. True
B. False

Correct Answer: True

Concept: VPNs create encrypted tunnels over public networks.

Question 2 — True / False | Topic: VPN

GRE can tunnel multicast traffic.

A. True
B. False

Correct Answer: True

Concept: GRE supports multicast tunneling, which is useful for routing protocols.

Question 3 — True / False | Topic: VPN

IPsec transport mode encrypts the entire original IP packet.

A. True
B. False

Correct Answer: False

Concept: Transport mode encrypts only the payload; tunnel mode encrypts the entire packet.

Question 4 — True / False | Topic: VPN

IKE is responsible for negotiating IPsec keys and security associations.

A. True
B. False

Correct Answer: True

Concept: IKE automates key exchange and SA negotiation.

Question 5 **True / False | Topic: VPN**

L2TP provides encryption by itself.

A. True
B. False

Correct Answer: False

Concept: L2TP does not encrypt traffic; it is usually combined with IPsec.

Question 6 **True / False | Topic: VPN**

SSL VPN can be accessed through a standard web browser.

A. True
B. False

Correct Answer: True

Concept: SSL VPN supports browser-based access via HTTPS.

Question 7 **True / False | Topic: VPN**

GRE provides encryption by default.

A. True
B. False

Correct Answer: False

Concept: GRE encapsulates but does not encrypt traffic.

Question 8 **True / False | Topic: VPN**

SSL VPN commonly uses TCP port 443.

A. True
B. False

Correct Answer: True

Concept: SSL VPN uses HTTPS/TCP 443.

Question 9 **True / False | Topic: VPN**

IPsec tunnel mode encrypts the entire original IP packet.

A. True
B. False

Correct Answer: True

Concept: Tunnel mode encapsulates the whole original packet.

Question 10 **True / False | Topic: VPN**

VPN means that provides secure communication over public networks.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes VPN.

Question 11 **True / False | Topic: VPN**

VPN is unrelated to provides secure communication over public networks.

A. True
B. False

Correct Answer: False

Concept: VPN is directly related to provides secure communication over public networks.

Question 12 **True / False | Topic: VPN**

Site-to-site VPN means that connects entire networks at different locations.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Site-to-site VPN.

Question 13 **True / False | Topic: VPN**

Site-to-site VPN is unrelated to connects entire networks at different locations.

A. True
B. False

Correct Answer: False

Concept: Site-to-site VPN is directly related to connects entire networks at different locations.

Question 14 **True / False | Topic: VPN**

Remote-access VPN means that connects individual users to a corporate network.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Remote-access VPN.

Question 15 **True / False | Topic: VPN**

Remote-access VPN is unrelated to connects individual users to a corporate network.

A. True
B. False

Correct Answer: False

Concept: Remote-access VPN is directly related to connects individual users to a corporate network.

Question 16 **True / False | Topic: VPN**

GRE means that is IP protocol 47 and lacks built-in encryption.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes GRE.

Question 17 **True / False | Topic: VPN**

GRE is unrelated to is IP protocol 47 and lacks built-in encryption.

A. True
B. False

Correct Answer: False

Concept: GRE is directly related to is IP protocol 47 and lacks built-in encryption.

Question 18 **True / False | Topic: VPN**

GRE means that can tunnel multicast and non-IP protocols.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes GRE.

Question 19 **True / False | Topic: VPN**

GRE is unrelated to can tunnel multicast and non-IP protocols.

A. True
B. False

Correct Answer: False

Concept: GRE is directly related to can tunnel multicast and non-IP protocols.

Question 20 **True / False | Topic: VPN**

IPsec means that provides confidentiality, integrity, and authentication.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes IPsec.

Question 21 **True / False | Topic: VPN**

IPsec is unrelated to provides confidentiality, integrity, and authentication.

A. True
B. False

Correct Answer: False

Concept: IPsec is directly related to provides confidentiality, integrity, and authentication.

Question 22 **True / False | Topic: VPN**

ESP means that provides encryption and authentication.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes ESP.

Question 23 **True / False | Topic: VPN**

ESP is unrelated to provides encryption and authentication.

A. True
B. False

Correct Answer: False

Concept: ESP is directly related to provides encryption and authentication.

Question 24 **True / False | Topic: VPN**

AH means that provides authentication without encryption.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes AH.

Question 25 **True / False | Topic: VPN**

AH is unrelated to provides authentication without encryption.

A. True
B. False

Correct Answer: False

Concept: AH is directly related to provides authentication without encryption.

Question 26 **True / False | Topic: VPN**

IKE means that negotiates IPsec keys and security associations.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes IKE.

Question 27 **True / False | Topic: VPN**

IKE is unrelated to negotiates IPsec keys and security associations.

A. True
B. False

Correct Answer: False

Concept: IKE is directly related to negotiates IPsec keys and security associations.

Question 28 **True / False | Topic: VPN**

Tunnel mode means that encrypts the entire original IP packet.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Tunnel mode.

Question 29 **True / False | Topic: VPN**

Tunnel mode is unrelated to encrypts the entire original IP packet.

A. True
B. False

Correct Answer: False

Concept: Tunnel mode is directly related to encrypts the entire original IP packet.

Question 30 **True / False | Topic: VPN**

Transport mode means that encrypts only the IP payload.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Transport mode.

Question 31 **True / False | Topic: VPN**

Transport mode is unrelated to encrypts only the IP payload.

A. True
B. False

Correct Answer: False

Concept: Transport mode is directly related to encrypts only the IP payload.

Question 32 **True / False | Topic: VPN**

L2TP means that uses UDP port 1701 and lacks encryption.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes L2TP.

Question 33 **True / False | Topic: VPN**

L2TP is unrelated to uses UDP port 1701 and lacks encryption.

A. True
B. False

Correct Answer: False

Concept: L2TP is directly related to uses UDP port 1701 and lacks encryption.

Question 34 **True / False | Topic: VPN**

L2TP over IPsec means that combines L2TP tunneling with IPsec encryption.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes L2TP over IPsec.

Question 35 **True / False | Topic: VPN**

L2TP over IPsec is unrelated to combines L2TP tunneling with IPsec encryption.

A. True
B. False

Correct Answer: False

Concept: L2TP over IPsec is directly related to combines L2TP tunneling with IPsec encryption.

Question 36 **True / False | Topic: VPN**

SSL VPN means that commonly uses TCP port 443.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes SSL VPN.

Question 37 **True / False | Topic: VPN**

SSL VPN is unrelated to commonly uses TCP port 443.

A. True
B. False

Correct Answer: False

Concept: SSL VPN is directly related to commonly uses TCP port 443.

Question 38 **True / False | Topic: VPN**

Web proxy mode means that provides browser-based access to web apps.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Web proxy mode.

Question 39 **True / False | Topic: VPN**

Web proxy mode is unrelated to provides browser-based access to web apps.

A. True
B. False

Correct Answer: False

Concept: Web proxy mode is directly related to provides browser-based access to web apps.

Question 40 **True / False | Topic: VPN**

Network extension mode means that assigns a virtual IP to the remote host.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes Network extension mode.

Question 41 **True / False | Topic: VPN**

Network extension mode is unrelated to assigns a virtual IP to the remote host.

A. True
B. False

Correct Answer: False

Concept: Network extension mode is directly related to assigns a virtual IP to the remote host.

Question 42 **True / False | Topic: VPN**

IKEv2 means that is faster and more reliable than IKEv1.

A. True
B. False

Correct Answer: True

Concept: This statement correctly describes IKEv2.

Question 43 **True / False | Topic: VPN**

IKEv2 is unrelated to is faster and more reliable than IKEv1.

A. True
B. False

Correct Answer: False

Concept: IKEv2 is directly related to is faster and more reliable than IKEv1.

111 GRE VPN

Question 1 **True / False | Topic: GRE VPN**

GRE provides strong encryption by default.

A. True
B. False

Correct Answer: False

Concept: GRE only encapsulates traffic; encryption must be added separately, often with IPsec.

112 IPsec VPN

Question 1 **True / False | Topic: IPsec VPN**

ESP provides both encryption and authentication.

A. True
B. False

Correct Answer: True

Concept: ESP can encrypt payloads and authenticate packets.

Question 2 **True / False | Topic: IPsec VPN**

AH provides payload encryption.

A. True
B. False

Correct Answer: False

Concept: AH provides authentication and integrity but does not encrypt the payload.

113 L2TP VPN

Question 1 **True / False | Topic: L2TP VPN**

L2TP alone does not provide encryption.

A. True
B. False

Correct Answer: True

Concept: L2TP provides tunneling but relies on IPsec or other mechanisms for encryption.

114 SSL VPN

Question 1 **True / False | Topic: SSL VPN**

SSL VPN typically uses TCP port 443.

A. True
B. False

Correct Answer: True

Concept: SSL/TLS VPNs commonly use HTTPS/TCP 443, which is usually allowed through firewalls.